

PERU AS IT IS:

A RESIDENCE IN LIMA,
AND OTHER PARTS OF THE PERUVIAN REPUBLIC,

COMPRISING

AN ACCOUNT OF THE SOCIAL AND PHYSICAL FEATURES
OF THAT COUNTRY.

BY ARCHIBALD SMITH, M.D.

IN TWO VOLUMES.

VOL. II.

LONDON:

RICHARD BENTLEY, NEW BURLINGTON STREET,
Publisher in Ordinary to Her Majesty.

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ERRATA.

Page 13, line 9, *for* pulverilla and massisa *read* polvorilla and maciza.
128, " 17, and in all other instances, *for* Pozuro *read* Pozuzo.
187, " 2, *for* realise *read* realize.
239, " 6, *for* the aborigines *read* those.

Transcriber's Note: The errata have been corrected but otherwise the original spelling (in both English and Spanish) has been preserved.

PERU AS IT IS.

CHAPTER I.

Site, population, and climate of Cerro Pasco.—Houses.—Coal, and other kinds of fuel.—Timber for use of the mines, &c.—Where brought from.—Fruit and provisions.—Mines.—Mantadas.—Boliches.—Habilitador.—Mint.—Returns of the mines.—Banks of Rescate.—Pasco foundery.

The town of Cerro Pasco, about fourteen thousand feet above the level of the sea, has its site in an irregular hollow on the northern side of a group of small hills, which commence at Old Pasco on the north-east limit of the high table-land of Bombon.

Cerro Pasco is thus situated at nearly equal distances, or about twenty leagues, from Tarma on the south and Huanaco on the north, both after their kind fertile and productive. It has the fine lake of Chinchaycocha, near old Pasco on its south; and, on the north, the outskirts of the town almost reach to a funnel-mouthed gullet which leads with a rapid descent to the village of Quinoa, three leagues distant. Its eastern and western aspects are bounded in the view by the respective ridges of the eastern and western Cordillera; and the intervening spaces between this bed of Peruvian treasure, and the stupendous barriers presented by these commanding summits, forming a grand amphitheatre, are enlivened throughout much of their extent by the innumerable herds of sheep and folds of cattle that roam and flourish upon them. Here and there are seen groups of the tame llama and shy vicuña; whilst the whole landscape is variegated with lakes, rivulets, and marshes, whose surfaces are ever rippled by the fluttering flocks of geese, ducks, snipes, plovers, water-hens, herons, yanavicas, flamingos, &c. which at their proper and appropriate seasons animate and adorn this wide expanse. Nor should we omit to mention that far towards the west, and skirting the limits of the great plains, are seen from the surrounding heights strange fragments of rock, as in the neighbourhood of Huallay, that assume to the distant eye the appearance of dark pine-trees rising under the shade of the adjacent mountains.

The waters of this mineral district are partly carried off by the famous Adit of Quiullacocha, and a considerable portion of these naturally percolate northward into the hollow of Rumillana near to Cerro, from whence starts the spring Puceoyaco, the source of the river Huallaga.

The population of Cerro Pasco is in a great degree migratory, for it increases and diminishes according as the mines are highly productive, or in

a state of poverty and inundation for want of proper drainage: were the drainage perfect, the treasure that might be extracted would be incalculable. The number of inhabitants is never, perhaps, under four or five thousand, and it has been known to swell up to thrice this amount,—the most active hands happily finding accommodation under ground. When the mines were thus productive, the abode of the master-miner rang with the clink of hard dollars, as the die was kept in constant motion; and the fair sex crowded from the more genial vales, and enlivened the miners' home with the song, guitar, and dance.

The climate of Cerro Pasco is for nearly one half the year, or from the end of November to May, exceedingly gloomy and variable. In the course of a few hours, the wind is often observed to take the round of the compass; and in the same time it changes from fair to rain, from rain to sleet, snow, hail, and rain again. The lanes, for streets they merit not to be termed, are during the greater part of these months wet and miry. The thermometer of Fahrenheit, during this period, rarely rises above 44° in the shade, and seldom falls so low as the freezing point.

But during the dry season, which reigns from May to November, it is much otherwise; and then, though the sun at noon shines forth with great power in the face of a cloudless canopy, the frosts at night are intense, and the evenings and the mornings are keenly piercing and cold. In the course of the month of August the air is so remarkably dry that the nose and fauces become parched and painful. The writer suffered so much from this troublesome affection as to find it necessary to seek a more temperate air a few leagues off, when the ailment disappeared immediately.

The severity of the climate of Cerro Pasco had little to mitigate its effects in the manner wherein houses were constructed in the time of the Spaniards. The dwellings are covered with thatch, and this is the unfortunate cause of frequent and destructive fires breaking out in the town. To avoid such accidents, one or two houses have been lately covered with lead.

It was not until the arrival of the Peruvian Mining Company, in December 1825, that the inhabitants were taught how to mitigate the evils of their inclement home by the construction of chimneys and proper fire-places, as well as glazed windows; and for the introduction of these comforts to the dwellings and firesides of the miners we have heard the

company blessed, long after its agents had to forsake those regions of subterranean wealth. Though this rich district has not the natural advantages of a favourable climate, yet it possesses that by which its rigour may be softened and its effects meliorated; it has abundance of coal.

Within five miles or thereabouts of Cerro Pasco, is a coal-mine of rather inferior quality, from which Captain Hodge, a distinguished miner from Cornwall, used to supply his customers. The coal burnt at the steam-engine was conveyed about five or six miles from the coal-mine, near the pueblo of Rancas, called “La mina de las Maquinas,” and is of very superior quality. The fuel, however, which is most common in Cerro Pasco, as well as all over the frigid districts of the Sierra, is “*champa*” a turf (not peat) cut from the surface of the marsh-land. Charcoal becomes very expensive; and the large *braseros* or pans, in which the rich miners once used to burn it, (though not always with impunity, because of the deleterious effect of the carbonic acid gas evolved,) are fortunately out of fashion since the advantages of the chimneys and grates have become known.

For smelting purposes, at different mining situations in the Sierra, the ordure of quadrupeds is collected on the plains in the dry season, and used as fuel with rushes and long grass which grow on the pasture-grounds.

Heavy timber for the use of the mines and mineral haciendas, and lighter timber for house-building in Cerro Pasco, is dragged a distance of sixty miles, and over bad and uneven road, by men and oxen, from the woods of Paucartambo, at the entrance into the Montaña, to the south-east of Pasco.

Fodder is sometimes exceedingly scarce and dear in Cerro Pasco. It frequently takes six reals, or one dollar (3s. or 4s.), daily, to feed a mule or horse with “alcaser” or barley-straw cut down when green, and conducted on beasts’ backs from the small villages in adjoining glens; and it is therefore customary for those who come to Pasco on business, and have several mules or horses, to send them away to the nearest convenient pastures until required to renew their journey. Indeed, in Cerro Pasco itself are to be seen in the wet season small patches of barley which never ripens, and is cut down green; but the quantity grown is too trifling to deserve notice, except in so far as it goes to show the sort of climate in this locality. Potatoes and “alcaser” are the principal productions of Quinoa; for, though its pastures are good, the temperature of the place is too cold to produce

maize: but a league or two lower down, at a village called Cajamarquilla, wheat may be grown, but in small quantity for want of sufficient arable land. At this place are numerous little gardens carefully cultivated, whence onions, cabbages, lettuces, and flowers for the use of churches and chapels, &c. are taken, and sold in Pasco market-place, which throughout the year is well supplied with a variety of fruits, plenty of good fresh meat and other provisions in abundance, from the warm and temperate valleys beneath, and lakes and plains around the mines.^[1]

“No one is ignorant,” says the intelligent and active prefect, Don Francisco Quiros, in his report to the departmental junta of his jurisdiction of Junin, assembled in Huanuco in 1833,—“No one is ignorant that the richest fountain of our national wealth this day is concentrated in the immense treasures of the mineral of Pasco. Its works, conducted with intelligence and managed with economy, would be more than enough to spread abroad abundance in all the republic, enough to draw towards us the productions of the whole universe, and to increase incalculably the delights and comforts of life. But the fatality which too often persecutes what is good, has plunged us in misery. Avaricious hands, desiring to enrich themselves in a moment, have for years back paralysed our best exertions; and by the indiscreet liberality of our mineral statute, ‘ordenanza,’ permitting the labouring miners to be paid in ore, and thereby violating more and more the principles of subterranean architecture, it would seem that instead of supporting the ample domes with solid pillars, pains had been taken to bring them down. There, as in a sepulchre, our most flattering hopes will be interred, if with a strong hand abuses so enormous shall not be checked,—if a wholesome severity may not be able to restrain the scandalous practice of thieving, as well as the irregular mode of subterranean labour.”

The mine labourer can choose for himself, by the laws of the mineral district, one of two sorts of payment. He can have four reals, or two shillings, daily as a fixed hire; or he may choose to retain a certain proportion of the ore, which he breaks down from the mine and carries, (panting loudly under his burden, contained in a leathern bag or *capacho*,) to the surface, where the division takes place by established measure; and the women, with a pot of chicha in hand, eagerly grasped at by the overheated and half-exhausted *capachero* or carrier, commonly stand by the

mouth of the mine to carry home the miner's share,—a bundle of ore called “mantada.” A common workman's daily share of metal may be worth a great deal of money, or it may be worth little or nothing. When the former is the case, the mine is said to be in “boya” or “bolla,” namely, a state of rich production, when the common labourer naturally insists upon being paid in metal; and again, when the mine does not produce good ore, or such as pays well, the labourer, who throws away his all on the pageantry of religious festivals and processions, claims his four reals per day's work, and will have no share in his employer's bad bargain.

At the mouth of the great mine, called the King's mine,—“La mina del Rey,”—which rendered the family of Yjura so famous and so wealthy,—a labourer has been known in our own day to refuse eighty dollars for his mantada, which abounded in pieces of polvorilla and maciza, or ore rich in native and nearly pure silver. Native and massive silver is, however, necessarily rare, and only occurs in small and scattered portions among other metals of good quality. The better the quality of the ore, so much greater is the damage and loss from robbery sustained by the mine and master miner; and it often happens that the cost of raising the ore, extracting the silver, paying government and local duties, the repair of the underground works, supply of salt and quicksilver, together with all the expense of major-domos, and wear and tear of mining utensils, loss on cargo-mules and llamas, &c. exceed the whole returns of the mine. Hence, we have known a most intelligent, active, and distinguished master miner of Pasco, four or five of whose mines yielded mostly all the rich ore extracted from that mineral in the years 1827 and 1828, declare that, after having put about two millions of dollars in circulation from the produce of these mines, he himself was rather a loser than a gainer, notwithstanding such abundant returns.

The number of mines at Cerro usually at work since they have fallen into the hands of the Patriots are comparatively few: though in the different districts or sections of this place, known under the names Santa Rosa, Yauricocha, Caya, Yanacancha, Cheupimarca, and Matagente, there are several hundred well-known mines from which silver has been, and yet may be, plentifully dug out, provided a perfect drainage should ever be effected. The mines of late actually productive in Cerro Pasco may be said generally to amount to about thirty in number, and to be at work for about eight

months in the year. Some of these are of course of inferior quality; but the metals which, by assay or experiment on the small scale, only yield six or seven *marcs* per *cajon* or box, (the marc being eight ounces, and the cajon twenty-five mules' load of ten arrobes, or two hundred and fifty pounds each,) are found to be worth the working, provided the ore be not very difficult to extract, either on account of the character of the vein, or the depth of the workings. The metals of Santa Rosa, when they yield ten marcs of silver per cajon, and when quicksilver is at a moderate price, pay the miner better than richer ores, because they do not tempt the cupidity of the labourers, who are therefore contented with the fixed sum of four reals wages per day, instead of the mantadas or bundles of metal already mentioned.

These mantadas are purchased by a class of men called bolicheros, or owners of boliches. This boliche is, to the common grinding-mills on the mine-estates or haciendas, what the hand-mill of the Israelites was to the modern corn-mills moved by machinery: it is a kind of rocking-stone, placed on the concave surface of a larger stone well accommodated underneath. Metal, in comparatively small quantities, is ground between these two stones by a man who, with the help of a long pole, balances himself on the upper roundish and heavy stone, which by the weight and motion of his own body he keeps rocking incessantly. The metal, or ore, thus ground, is the very richest; poor or ordinary ore could not pay on this small scale: but the ore bought of the labouring miners usually enriches the bolichero, who, tempted by the prospects of a ready fortune, does not hesitate to encourage the thieving practices complained of in the departmental report of our friend the prefect of Junin, himself a native of Cerro Pasco.

The town of Cerro Pasco—of which the very “adobes,” or unburnt bricks, partly used in some of the houses, contain silver—is itself so burrowed under, that one is really in no small danger of inadvertently, and especially at night, falling into old mines, or rather pits,—sometimes superficial, sometimes deep and fathomless, and half filled with water. The mines are irregularly wrought under ground; and the experienced hands burrow like rabbits through holes not commonly known, and so come at rich metals by stealth, to be immediately exchanged for dollars at the bolicheros. The best way to prevent such plunder would be to prohibit

boliches. While this species of robbery goes on, Cerro Pasco, though removed from the sphere of the earthquakes that infest the coast, is in risk of being swallowed up by the falling-in of the arches of the mines, supported on pillars frequently consisting of rich ore. The thieves pilfer from these pillars, and so weaken the supports of the whole underground fabric, that now and then entire arches fall in, sometimes producing a sacrifice of lives or other disastrous consequences.

We see the Pasco miners always in the midst of riches, and always embarrassed: they are kept in a state of continued tantalization. The miner, it is true, sometimes has immense and rapid gains, in spite of rogues and plunderers everywhere about him, at comparatively little expense of time or money; and this occasional success leads others to indulge in a hope of similar good fortune, which hurries the majority of speculators in this channel into pecuniary difficulties: for, as we have seen, the necessary outlay is often great without any compensation; and when the capital is too limited, though in the main the undertaking be a good one, ruin is near. Shopkeepers and dealers in plata-piña are tempted, by prospects of commercial advantage, to lend money to the harassed mine-owner to enable him to forward his works, and to repay the loan in piña^[2] at so much per marc. Such a lender is called “habilitador:” but it unluckily happens for this capitalist that, by the custom and usage of the miner, the last “habilitador” has a claim to be first paid, which leads to the worst practical results. The miner is generally a reckless gambler, who spends money as fast as it comes to him, not in improving his mines, but in indulging his vices; and in this manner the interest of the first habilitadors may be successively postponed to the claims of the most recent, who frequently is disappointed in his turn; while the difficulties of the miner are not removed, but merely prolonged, and he is involved in everlasting disputes and litigation.

The risk, expense, and delay occasioned at all times, and more particularly in days of civil broil, by the necessity of forwarding the bars of silver from Pasco to Lima for the purpose of coinage, are felt as so many real grievances by the miner; and it is known that these causes, with the desire to avoid the payment of the established duties, have led to a contraband trade across the by-ways of the mountains to the coast, which no number of custom-house officers could prevent, even on the extravagant supposition of their being proof against bribery and corruption. The evils

attendant on the existing arrangements led the legislature to pass a law for the establishment of a mint at the mines of Pasco; but this desirable object has not yet been carried into effect in a proper and efficient manner, though we understood that the prefect Quiros employed a native tradesman to erect some rude machinery by which a few hundred dollars were thrown off daily.

An extract from a memoir presented in the year 1832 to the congress in Lima by Mr. Tudela, the Peruvian minister of “Hacienda” or home affairs, may give an idea of the returns of the mines. “To animate mining industry,” says he, “one most essential thing is the convenient supply of quicksilver, with which our metals are generally extracted from their ores; because smelting is not suited for the greater number of these, neither is it used for those ores to the refining of which it is adapted.”^[3] The price of the quicksilver determines the profit or loss on the poorer metals; and neither exemption from duties of *cobos*^[4] and tithes, nor any other protection which the law dispenses to the mining corporation, compensate for the expense of this article.

“In Huancavelica, Peru possesses one of the richest quicksilver mines on earth,—a mine which comprehends forty-one hills, examined and found intersected with veins, of which one part alone, that of Saint Barbara, called the “Great,” yielded five thousand quintals of quicksilver per annum for two centuries. It was, therefore, a matter of importance to inquire if it could be conveniently worked; and it has been found that, with moderate support and certain arrangements, quicksilver may be procured at sixty-five dollars per quintal.”^[5]

“The operations of the mints concur in proving the necessity of banks: for, granting the mint of Lima has stamped in the last three years 4,902,762 dollars, and the mint at Cuzco, in the years 1829 and 1831, 969,939 dollars, the augmentation of coinage does not correspond to that which should result from the abolition of the duties of *cobos* and tithes since the 26th February 1830, and the increase in the coinage at Cuzco proceeds from other causes. When these duties” (viz. *cobos* and tithes) “were yet exacted, in the Lima mint alone, in the year 1827, more than 2,700,000 dollars were coined; but in that year, in addition to the beneficial drainage of the Pasco mines, the contraband trade had not extended itself as afterwards it did. To diminish

the evils produced by contraband trade in mining places or in their immediate environs, funds, in addition to banks of ‘Rescate,’ are necessary in the houses of coinage; with which being provided, neither the holders of bullion shall be deterred from presenting it because of the delay they experience in being paid its value, nor, if this delay be shortened, shall the treasury suffer the severe losses to which it is actually subjected. One hundred thousand dollars in the Lima mint, and fifty thousand in the mint of Cuzco, should prove sufficient to meet all difficulties.”

We may remark, that the want of such mint deposits as are alluded to by Mr. Tudela is one of the principal sources of mistrust in revolutionary times; as the possessor of bullion will rather run the risk of smuggling, than the chance either of losing all his capital, or of being long deprived of its value in hard dollars, if he carries it in the regular channel to the Lima mint. The banks of Rescate to which Mr. Tudela refers, now so much desired in Peru, are only funds deposited in certain situations, and under proper superintendence, for the miner to be thereby enabled to exchange his piña at a fixed and just value in current money, by which he is put in possession of dollars as soon as his piña is ready for the market. And this, we may venture to say, is the only sort of bank calculated to be of real service to the dissolute miner; as it encourages his industry, without putting it in his power to outrun his credit with the bank, or of ruining himself and family, and kindling the worst of passions in consequence of forfeiting his mines, which would frequently be the case if these were accepted in security for cash advances, to be spent probably in feasts, frolics, cards, and dicing, instead of being applied to the professed purpose of working his mines or improving his property.

The number of marcs^[6] of silver reduced to bars, in the foundery at Cerro Pasco, from the year 1825 to 1836 inclusive, is, according to the best information, as follows:—

<i>Years.</i>	<i>Marcs.</i>
1825	56,971
1826	163,852
1827	221,707
1828	201,330
1829	82,031

1830	95,265
1831	135,134
1832	219,378
1833	257,669
1834	272,558
1835	246,820
1836	237,840
	<u>2,190,555</u>

CHAPTER II.

Descent from Pasco to Huanuco.—Succession of works for grinding and amalgamating silver ore.—Quinoa.—Cajamarquilla.—Huariaca.—San Rafael.—Ambo.—Vale of Huanuco; its beauties and advantages.—State of agriculture in this vale, and traffic with Pasco.—The College named La Virtud Peruana.—Steam navigation on the river Huallaga, and civilization of the wild Indians of the Montaña.—Natural productions of the Montaña.

Some of the valleys in Peru, like that by Obrajillo and Canta, extend from the coast to the Cordillera: some are only a few leagues of rapid descent from the puna or lofty table-land, as Tarma, for example, from the heights of Junin; but others sink deeply into the bosom of the central Andes, or dip under the brow of the Montaña, as, for example, Guarrigancha and Huanuco, of the latter of which we purpose to offer a more particular account.

Huanuco is not to be confounded with the ancient town called Leon de Guanuco, of which the remarkable remains are still well worth visiting on the high pasture-land of Huamalies: for the city now called Huanuco, or, as some write it, Guanuco, is in a delightful valley, twenty-two leagues in a north-easterly direction from the mines of Cerro Pasco, with a descent of about seven thousand feet; thus situated, as nearly as may be, half-way in respect to altitude between Cerro Pasco and the ocean.

In the first three leagues of our descent from the mines to the vale, we pass by a number of mills for grinding metal, preparatory to its being mixed with salt and quicksilver for the purpose of amalgamation. These are situated in a narrow rocky glen; the rugged road through it lying often along the bed of the stream that wanders down it, putting a great many mills successively in motion as it is directed into troughs or canals leading to the clumsy machinery of the haciendas, to which the ore is conducted at great trouble and expense, on the backs of mules, donkeys, and llamas.

From the village of Quinoa, only three leagues from Cerro, and once celebrated for its gold mine, to the village of Cajamarquilla, two leagues lower down, the road is furrowed, deep, and miry during the wet season; but the pasture-grounds are good, and upon these the cattle of the miners are sent to feed at small expense. From two to three leagues below Cajamarquilla, of which we took notice in our account of Cerro Pasco, is Huariaca, a small town with a large plaza, or square, and very good houses.

This town is the centre of a curacy and seat of a governor, with a climate analogous to that of Obrajillo on the road between Lima and Cerro, or Cerro Pasco, formerly noticed. Its artificial productions are also the same as we formerly mentioned, viz. maize, wheat, beans, potatoes, &c.; but here natural vegetation is more luxuriant, and the air exceedingly benign: the frosts are seldom so keen as to blight or wither the parks of lucern, and troublesome heat is unknown. Huariaca is endeared to the memory of many a Cornish miner, who lost his health in Cerro Pasco, and at this rendezvous for convalescence rejoiced in the smiling aspect of nature, and enjoyed the delightful feeling of returning health. The writer, in common with several of his countrymen, has to lament the premature death of the curate of this place, Dr. Don Pablo de Marticurena; whose intelligence, hospitality, and amiable disposition rendered him an object of love and respect, while his house was the home of the traveller, and the abode of charity, without distinction of creed or country. A league below Huariaca, we cross a bridge placed over the small river of Cono or Pallanchacra, a short distance above which is the famous tepid mineral well of Cono; to which, as it is in a temperate little glen, the sick have frequent recourse. On the banks of this stream we have peaches in perfection and plenty; and as we approach towards the village of Saint Rafael, a few leagues lower down, we are amused by looking up at heights topped with Indian hamlets, and at little flats and declivities under crop of wheat and potatoes, &c. and, near the river, maize. The temperature of Saint Rafael is delicious, and this locality is free from any endemic disease.

From Saint Rafael to Ambo is a distance of several leagues of hard road, sometimes running close to the river's edge, often running along the steep, and with its rocky staircases and narrow passes subjected in time of rains to be blocked up by large stones and small trees, carried down by the mountain torrents. Where the glen expands towards the hill-tops, but closes so narrowly below as only to give room for the channel of the river, we find the road at certain narrows carried along the face of the rock; and here the craggy projections serve as supporters for poles or rafters extending along the intervening gaps, and covered with flags or brushwood laid on and coated with a little earth, thus forming an extremely awkward and narrow bridgeway suspended over the stream. At Ambo, nine leagues lower down than Huariaca, the aspect of the country is changed. Here the loud chirping (for it cannot be called croaking) of little frogs heard by night—the

granadilla in elegant flowering festoons seen by day on the pacay and lucuma tree, tell the warm and thirsty traveller that he has come to the land of “guarapo,”^[7] where he may enjoy the cool of the corridor, and cast off the load of his Sierra ponchos and heavy clothing.

From Ambo to the city of Huanuco we have five leagues of a charming ride; and from Ambo downwards, the Vale of Huanuco may be said to commence. In this vale the writer resided for three years. The year is, as usual, divided into the wet and dry seasons, observing the same periods of change as we have already noticed to belong to the seasons on the high Sierra. In this valley, however, snow never falls, except on the summits of the highest hills; and the thermometer of Fahrenheit is seldom seen to rise above 72° in the shade of the veranda, or wide-spreading fig-tree. In the hottest day, when every little stone on the surface of the newly-turned field glistens in the sunbeams, so as to torment the sight, the thermometer rises very high on being exposed in the open air to the direct rays of the sun; but, upon being removed into the shade, it again falls to a very few degrees above 70°; and scarcely ever throughout the whole year is it seen to sink under 66° of the night thermometer placed within doors,—thus manifesting an equability of atmospherical temperature altogether as extraordinary as it is benignant. So small, then, is the range of the thermometer in this fine locality, that the state of the internal circulation of our frame is but little disturbed by sudden changes induced by vicissitudes of temperature. To the uniform mildness of its atmosphere it may be principally owing that pulmonary consumption is as little a disease of this favoured locality as ague;^[8] for we never, during the period of three years that we resided here, had occasion to know of a single instance in which this disease originated in the valley; but those who, by residing in other situations, had their lungs nearly wasted by consumption and spitting of blood, have, in different parts of this valley, found a temporary asylum which afforded a prolongation of life when entire restoration to health was physically impossible. The climate is sometimes complained of as too dry, it being only during the rainy months that the perspiration commonly becomes sensible on moderate exertion. During the greater part of the year the reflected rays of the sun on the sides of the valley would render it intolerably hot, were it not for the daily breeze that, from about 11 A.M. to 3 P.M. comes with uniform

regularity from the Montaña, through the aperture in the mountains along which the river of Huanuco rolls towards the Huallaga and great Marañon.

In August and September, no perceptible dew falls; but during these months, when vegetation among the small neighbouring dales becomes scanty, the deer often steal in herds to the thickets near the river; and we have stalked them at midnight in the midst of the fields, without discovering a trace of moisture on the alfalfa leaf. The nights are always delightful; and the sky, when it does not rain, is pure, bright, and beautiful. The hills on the eastern side of the valley are clothed with pastures, have perennial springs and wood in their dingles and corries, and are capable of grazing cattle all the year: but opposite to these, on the western side, the hills, like those of the coast, are dull arid masses for nine months in the year, only furnishing a sparse growth of flowering shrubs and weeds on their sides; whilst their elevated tops alone throw forth a denser crop of sweet herbage, on which folds of cattle regale themselves in the months of January, February, and March,—at a season, as we have seen, when the uncultivated heights near the coast are scorched, and stripped of all vegetation except cacti and some bulbous plants. But the plains that spread round the base of the hills and mountains that go to form the Vale of Huanuco, are never allowed to take upon them the withered face of winter. By the aid of rivulets from the mountains, sometimes diverted from their natural channels by art, and carried, by circuitous aqueducts of many miles in extent, the numerous flats among the recesses of the heights and slopes, frequently elevated much above the lower plains, are kept ever verdant and productive, in like manner as the fields and enclosures in the bottom of the vale are fertilized by canals from the river. The best sugar-cane comes to maturity in about eighteen months or two years, and yields several cuttings of after-growth. The lucern or alfalfa, without the aid of top-dressing, gives six crops annually for an indefinite number of years; and in some favoured spots it yields a cutting in six weeks, and therefore gives eight crops yearly. The writer had a plot that yielded, at this rate, alfalfa of about a yard in height, and in good flower. The plantain, both long and short, and the richest tuna, or Indian fig, grow in abundance; the finest pineapples are brought from the neighbouring Montaña, where vegetation is much more rapid and vigorous than in the Vale of Huanuco. In this vale, however, the palta and cheremoya mellow on the branches in their native soil. The maguey, coffee, cotton, and vine, the pomegranate and orange, the citron, lemon, and lime, &c. flourish here; and

the meanest villager, as well as the humblest lodger under a cane-roofed shed, inhales with every breath the odours of never-failing blossoms. As the morning sun gilds the high ridges of this happy valley, its inhabitants are animated to the daily labours of the field by the cheerful voice of the prettily-plumaged inmates of their well-shaded bowers. Such, then, are some of the more prominent beauties and natural advantages of the Vale of Huanuco: and we may here mention, that the city of Huanuco is the principal seat of recreation for him who wastes his strength and frets his temper in the too often delusive pursuit of wealth in Cerro Pasco, and other inclement mining localities in the neighbourhood. In spite of their vexations and misfortunes, few can have invested themselves with a mood so sad or so cynic as not to enjoy and partake of the enthusiastic glee and antiquated gambols of a carnival feast in Huanuco.

The agriculture of Huanuco,—though alluring to the eye of the ordinary traveller, who only glances at its rich and waving fields, enclosed within tapias or fences of mud, and hedges of the Indian fig, and aloe or maguey plants,—is in every way defective as a branch of industry. The fields owe their luxuriance to nature rather than to man, except in the single advantage of water, which he often directs and supplies to them. Manure is a thing never thought of; and the ground seldom requires it, though we see the same spot year after year under crop: but much of the soil which is considered poor might be rendered fertile, in so favourable a climate, if the people would only take the trouble of cleaning out their large cattle-pens once a year; but this would be to diverge from their accustomed routine, which they dislike to forsake. The implements of husbandry are of the rudest kind. The plough, which is slight and single-handed, is constructed merely of wood, without mould-board, which we have seen a one-handed person manage with perfect dexterity. The ploughshare is a thick iron blade, only tied when required for use by a piece of thong, or lasso, on the point of the plough, which divides the earth very superficially. Where the iron is not at hand, as frequently happens, we understand that the poor peasant uses, instead, a share made of hard iron-wood that grows in the Montaña. Harrows they have, properly speaking, none: if we remember well, they sometimes use, instead, large clumsy rakes; and we have seen them use a green bough of a tree dragged over the sown ground, with a weight upon it to make it scratch the soil. In room of the roller, of which they never experienced the advantage, they break down the earth in the field intended

for cane-plants, after it has got eight or ten ploughings and cross-ploughings, with the heel of a short-handled hoe which they call “lampa;” a tool which they use with great dexterity in weeding the cane-fields and clearing aqueducts. For smoothing down the clods of earth, we have seen some Indians use a more antiquated instrument. It consisted of a soft, flat, and round stone, about the size of a small cheese, which had a hole beaten through its centre by dint of blows with a harder and pointed stone. To the stone thus perforated they fixed a long handle, and, as they swung it about, they did great execution in the work of “cuspiando” or field-levelling.

Lucern or alfalfa is daily cut down, and used green, as scores of cattle and the working oxen for the plough and sugar-mills are to be fed by it; yet the scythe is not in use among the great planters, who find it necessary to keep two or three individuals at the sickle to cut down food for herds, in the daytime fed on irrigated pastures, but at night fed in corrals or pens.

Potato-ground they are accustomed to break up on the face of steeps with deep narrow spades, to which long handles are attached, that afford good leverage. In the same manner the soil is turned up by those who have neither plough nor oxen, but who yet sow maize on the temperate flats on the hill-sides, and in the midst of thickets by mountain streams, where the soil is usually fertile, and materials for fencing are at hand. People thus circumstanced make holes in the ground with a sharp-pointed stick, where they bury the seed secure, that it may not be taken up by the fowls of the air; and that, when dropped in virgin soil, it may yield a luxuriant crop and plentiful harvest. The Indian sows the white-grained maize in preference to the yellow, (morocho,) as he considers that when toasted it makes the best “cancha,” which the poor Indian everywhere uses instead of bread; and that when boiled it makes the blandest “mote,” for so they call the simply boiled maize: it has moreover the credit of making the most savoury chicha, or beer, which they home-brew whenever they have a little surplus grain at their command. They also, as we were given to understand, make a kind of beer from the fermented juice of the maize-stalks compressed between small rollers of wood moved with the hand. The usual application of dry maize-leaves and stubble is to feed cattle, and for this purpose it is considered more fattening than either alfalfa or sugar-cane tops.

Agi, or pimento, is cultivated around the little Indian houses and gardens in the Vale of Huanuco; and without this condiment the natives hardly relish

any kind of food.

The sugar-mills in this valley are, the greater number of them, made of wood, and moved by oxen. On the larger estates small brass rollers are used; but with a single exception, on the estate of Andaguaylla, where we were concerned in erecting a water-mill for the purpose of grinding sugar-cane, the proprietors adhere to the old practice of working with oxen by day and by night throughout the year, barring accidents, and feasts or holy-days.

The beautiful hacienda or estate of Quicacan, Colonel Lucar's, is a model of industry and method after the fashion of the country; and the most distinguished family of Echegoyen have, in Colpa-grande, the finest cane-estate, as far as we know, in the interior of Peru. It extends for nine or ten miles along the fertile banks of the river, from the city of Huanuco towards the ascents that lead into the Montaña.

Respecting Huanuco, although the principal city or capital of the department to which it belongs, we have to observe, that the consumption of its agricultural produce, as well as its own internal prosperity, depends on the mineral seat of Cerro Pasco. When the population of the Cerro rises to ten or twelve thousand, every article of Huanuco produce is in high demand; but when, from any cause, the mines are not wrought, or when these are inundated from defective drainage, and the hands employed in working them are fewer in number, the Huanuqueños and other neighbouring agriculturists are greatly discouraged or actually ruined; because, deprived of this outlet for their produce, they cannot undertake the expense of sending sugar and spirits on mules to the coast. The consequence is, that they are frequently poor in the midst of plenty; like the owners of extensive herds of sheep on the high pasture-lands, whose wool is of little value to them, as it cannot pay for mule or llama carriage to the coast; and the scanty produce of the looms of the interior have little estimation, as the ruined "obrages," or manufactories, now amply testify. The shuttle is, moreover, nearly put at rest by the cheaper articles of warm woollen as well as cotton clothing continually introduced from the stores of our English manufacturers.

A staple article of Huanuco commerce with Cerro Pasco is the coca-leaf, from their Montaña, only distant about fifteen leagues from the city; an article of which they have several crops yearly. The indigo growers in the

contiguous Montaña have, we believe, forsaken their enterprise, for want of funds to proceed with the manufacture of what, from the samples produced, was considered a good article.

Much of the fruit of the Huanuco orchards is eaten at the tables of the Pasqueños, or inhabitants of Cerro; and in the convents are made excellent sweet preserves, which are highly valued, and circulated in the surrounding country as nice and most welcome presents rather than as formal articles of commerce.

Several lands formerly belonging to convents are now appropriated as endowments of the college of Huanuco, named *La Virtud Peruana*, which is the only school of its kind at present open in the department of Junin. This temple of Peruvian virtue, for so the Lyceum, which was formerly a convent, has been emphatically called, was installed in May 1829, under the rectorship of Doctor Don Gregorio de Cartagena; and the writer would now desire from his native country to offer to this acute and enlightened gentleman his grateful acknowledgments for the generous hospitality of which he was himself the object, when, pilgrim and stranger as he was, he knocked at the gates of the “Temple of Peruvian Virtue.” Doctor Don Gregorio de Cartagena, jointly with his distinguished relative Doctor Don Manuel Antonio de Valdizan, has the honour of being considered the founder of this college in his native city, as we learn from the speech of Doctor Don Buena Ventura Lopez, delivered in the college chapel on the day of installation, and published, with other harangues made on the same occasion, in the periodical then commenced as the first-fruits of the Huanuco press, under the very happy title of “The Echo of the Montaña.”

In the speech of Dr. Lopez, he encourages the rising generation to take the best advantage of the new path to knowledge, virtue, and honourable distinction now freely opened to them by the meritorious exertions of two of the most eminent natives of Huanuco. He exhorts his young hearers never to forget how much they owe to these patriots and benefactors:

“And you,” says he, “fortunate young men, in possession of advantages which were denied to your forefathers, let the names of your indulgent friends Valdizan and Cartagena, coupled with the obligations of this day, sink deep into your hearts: warmed as they are with feelings of the purest

delight, they will readily receive the generous impression, ay, and retain it for ever.”

The kind and affable inhabitants of this city in the bosom of the Andes have their imaginations excited with the hopes of their rising glories, and their own happy valley is too narrow for their expanding desires. So full are their literati of the flattering idea that an English colony on the river Huallaga may extend its industry and enterprise to the cultivation of the great pampa del Sacramento, that they already fancy proper depôts and harbours selected, docks prepared, and ships building from the timber of their own Montaña, to carry them a voyage of pleasure and profit round the world. They imagine little steamers up to *Playa-grande*, or even to the falls at Casapi, or the port of Cuchero on the river Chinchao, within a couple of days’ journey of their city; and, when their wishes are realized, they calculate that their now useless and neglected copper mines shall be more precious, and draw in upon them more wealth than ever did brilliants or diamonds on their distant neighbours of Brazil. And no wonder that the natives of this Elysian valley should be overjoyed at such prospects; since their long-continued communication with the canoe-men of the Huallaga on the one side, and in former times with those of the missionaries at the port or settlement of Mayro on the other, familiarise them with the notion of navigating the Huallaga and Ucayali; while the intervening plains of Sacramento they consider to be naturally the richest and most capable of improvement of any in the world. Even the miner of Cerro Pasco finds his fancy warmed when he reflects on the prospect of a steam navigation on the Marañon. Don Jose Lago y Lemus, one of the most distinguished of the veteran miners of Pasco, published in 1831 a pamphlet in illustration of the advantages that might accrue to the republic from this navigation. In this pamphlet he endeavours to show that the portions of Peruvian territory hitherto occupied, and consisting of arid coasts and rugged mountainous districts, are not to be compared, in point of natural interest or national importance, with the immense plains and fertile Montaña or wooded deserts on the eastern frontier; and he manifests a laudable and patriotic zeal in endeavouring to arouse the attention of his countrymen to this most momentous subject.

Don Jose expresses himself thus:

“The undersigned, being convinced of the truths he lays before the public, and at the same time anxiously desiring, in virtue of his appointment, both the welfare of the department and the province which he represents, he proposes to the most honourable Junta,” (viz. the departmental Junta of Junin, assembled in the city of Huanuco,) “a project of the grandest magnitude, capable of making the entire republic prosper, and of placing her in the rank and circumstances to compete with, and be the envy of, the most powerful states in the world. It will be said truly, that we were not heretofore ignorant of the treasures and riches of the actual productions in the Montañas of the Peruvian territory; it is equally certain that the want of hands, capital, and men of enterprise, have been powerful causes why we were unable to enjoy these natural advantages. If this be our state of weakness—if its commencement be traceable to our colonial condition, and that Providence has reserved the remedy till the epoch of our freedom and an age of intellectual light, let us make every effort to reap such incalculable benefits. Commercial relations are those that enlighten the people; by this powerful magic friendships are acquired, and with the most remote inhabitants of the globe bonds of brotherhood are established. Let, then, the grand canal of the Marañon be rendered navigable for steam vessels; so that, by the diverse and lesser streams that form this great river, we may procure them entrance to the immediate environs of our cities, towns, and villages, situated on the banks of the Huallaga.

“Ah, gentlemen! What a sudden and extraordinary emotion this idea excites in my mind! My imagination already combines the ideas that suggest themselves respecting this privileged city of Huanuco. Now its spacious fields are held worthy of higher cultivation and care; its abandoned streets I see crowded with useful citizens; the banks of its ample river Huallaga present a varied and charming perspective of shipping, newly elevated towns, open tracts of woodland, and cultivated lawns. Allured by the novelty of this scene, innumerable tribes of the wild Indians will unite themselves with us; they are our brethren, and, when thus intimately brought into contact with us, they may frankly discover to our knowledge those hidden treasures of our forests which their ignorance and barbarism hitherto concealed; and, as integral parts of Peru, they will conduce to its grandeur and respectability. Gentlemen, the most vivid imagination is lost in this contemplation, and finds itself overwhelmed by the number and vastness of the objects which crowd into its thoughts.”

The above patriotic effusion, very worthy of a departmental deputy of Junin, may appear to the reader to paint in too glowing terms the capabilities and importance of the Montaña on the confines of Huanuco. But, considering the extent and fertility of the territory, the navigable nature of its principal rivers, and the generally salubrious character of its climate, we believe that he who attempts to depict its various superiorities and advantages is more likely to come short of his object, than to overrate the reality which in imagination he may desire to trace.

Those regions in the Montaña which are watered by the Huallaga, Ucayali, and Marañon, with various subsidiary rivers interspersed among the intervening grounds, have as yet been but inadequately explored, and therefore only a very imperfect account can be offered of their aspect and natural productions.

From May to November the sun shines very powerfully in the Montaña, and consequently the soil, where it is cleared of wood,—for example, in the valley of Chinchao—becomes so parched that its surface opens in chinks; but underneath it always preserves humidity, and therefore needs no irrigation. From November to May it rains much, sometimes for six or seven days without intermission.

In the rivers, alligators, tortoises, and a variety of fish are found; and these also swarm in the ponds or lakes formed during the inundations of the rainy season. The most remarkable inhabitant of these waters is the manati, sometimes called pexebuey,^[9] from its supposed resemblance to the cow or ox. Like the cetaceous family to which it belongs, it suckles its young, and also feeds among the grass on the banks of the rivers.

The trees of the forest are inhabited by parrots, tanagers, and a surprising variety of birds, whose exquisite plumage vies with butterflies and flowers in the beauty, delicacy, and combination of their tints. Monkeys are so numerous as to form a chief article of animal food for the Indian hunter, dexterous in the use of the bow and arrow, or of the cerbatána, a long and hollow piece of wood through which he blows a small arrow, and hits his mark, at short distances, with fatal precision. There are very many venomous serpents. Wild-boars, deer, pumas, bears, tigers, and tapirs, frequent these forests, and are objects of the chase.

The vegetable productions of the Montaña, here considered as articles of commerce, or adopted for economical uses, are numerous. Among the valuable woods are cedar, and chonta or ebony, mahogany, walnut, and almond-tree. Edible herbs and roots, except the potato and yuca, are little cultivated; but coffee, plantains, and sugar-cane, of which a variety called the blue or azul is very luxuriant, are reared with some care, where nature indeed requires but little aid from the hand of man. The sugar-cane comes to maturity earlier than in other parts of Peru, and yields an annual crop at very little cost of production.

In the fertile vale of Chinchao, famous for its coca plantations, a few proprietors of Huanuco cultivate frijoles, or beans, for the use of the coca-gatherers: rice is also grown by the humid banks of the great rivers, and maize is everywhere sown as a necessary of life.

In the Montaña, chicha is made from maize, as in other parts of Peru; but the natives here make a drink called *masata*, not known in more civilized parts of the country, produced by chewing the yuca or maize, &c. and then leaving it to ferment, when, according to the quantity of water added to it, the fermented juice will be found of greater or less intoxicating power.

Indigo, as we have before noticed, is of Montaña growth, as is also tobacco.

Cotton, spun and wove into cloths of various texture by the Indians, requires no artificial assistance for its luxuriant growth. Lemons, limes, oranges, citrons, and other cooling fruit, are also productions of those parts.

The pine-apple is very abundant, as well as of delicious flavour, though it grows wild: and among the articles of spontaneous growth in the Montaña contiguous to Huanuco we may enumerate cacao or cocoa, cinnamon, guaiacum, vanilla, black wax, storax, dragon's blood, oil of Maria, gum caraña, balsam of copaiba, copal, and many other gums, balsams, and resins. Cinchona and sarsaparilla abound in great quantity.

For the following account of medicinal plants, collected during a journey down the river Huallaga, and through part of Maynas, we are indebted to the kindness of Mr. Mathews, an English botanist, formerly mentioned.

1. *Machagui huasca* is a *bejuco* or climber, the trunk and branches of which are intensely bitter. It grows at Tarapoto, and is used as a febrifuge.

2. *Diabolo huasca*^[10] grows at Tarapoto, and is used medicinally as a purgative.

3. *Uchu sanango*.—This is a species of *taberna-montana*, which grows at Tarapoto and Moyobamba. It is very piquant; produces a sensible degree of heat, and is used as a remedy in colds and rheumatic affections of the joints.

It is also said to be used in the preparation of the *pucuna* poison.—(See Humboldt.)

4. *Chiri sanango*.—Said to be contrary in its effects to the above; the natives hold it in some dread.

5. *Calentura huasca*, or *shiyintu*.—This is violent in its effects: it swells the gullet; produces quick, full pulse, and high fever; and in twenty-four hours after the fever the skin begins to peel off.

This remedy is taken for various complaints. The patient generally retires to his *chacra*, or country-house, to take medicine, where he is not liable to be molested; generally keeps his bed for eight days, and on the fifteenth day bathes. For four months it is necessary for those who take this remedy to diet themselves. On some men it produces no sensible effects.

The part of this plant used medicinally is the stem, which is roasted, pounded, and then taken in warm water or guarapo.

6. *Zuquilla*.—This is a thick-rooted variety of sarsaparilla.

7. *Guaco* grows about Tarapoto.

8. *Piñon*, or *croton tiglium*.—Three seeds of it eaten act as a drastic medicine.

9. *Carpuña*.—A few leaves (two or three) of this plant, put into warm spirit and water, act as a sudorific, which is employed in colds and rheumatic pains.

10. *Huyusa*.—The leaves of the huyusa are also used in small quantity, in form of infusion; and this remedy has the same virtues with the carpuña.

11. *Tapia bark*.—This is pounded into powder, and taken in cold water. It acts as a powerful emetic.

12. *Yerba de San Martin*.—The infusion from this plant is used for the same purpose as cubebs, or balsam of copaiba.

CHAPTER III.

The Department of Junin.—The river Marañon.—General sketch of the form of internal Government of Peru.—Particular account of the Prefecture or Department of Junin.—Mines.—Agriculture.—Manufactures.—Public Instruction.—Hospitals and Charitable Asylums.—Vaccination.—Junta of Health.—Public Baths.—Police.—Pantheons.—Roads.—Posts.—Public Treasury at Pasco.—Administration of Justice.—National Militia.

Of the three inland departments of Peru, namely, Cuzco, Ayacucho, and Junin, the latter is peculiarly distinguished by its mineral riches, and the rise of the great river Marañon, in the lake of Lauricocha, in the neighbourhood of Cerro Pasco. The length of this river, all its windings included, has been reckoned not less than one thousand one hundred leagues, of which nine hundred have been found to be navigable; and, at the distance of several hundred miles before it reaches the ocean, (where its mouth is one hundred and seventy or eighty miles wide,) the effect of the tides may be distinctly marked on its banks. For a very long way—some say two hundred miles or more—after it has entered the sea, it still continues fresh to the taste, or, at least, to a great degree unmingled with the retiring waters of the ocean, which it rolls back before the unsubdued force of its own mighty stream.

The provinces of this department are Jauja, Tarma, Pasco, Cajatambo, Huari, Huaylas, Huamalies, Conchucos, and Huanuco. Besides the precious metals, (and quicksilver, which for some time back has been regularly extracted from the mines of Jonta in Huamalies,) these provinces yield a great variety of cattle and vegetable produce. Huanuco, the principal city of the province of the same name, though no longer the seat of opulence or aristocracy, was once one of the chief cities of Peru under the ancient conquerors, and is at present chiefly distinguished as the capital of the whole department of Junin. The prefect of this jurisdiction extends his authority as far as the country of Maynas on the north, and to the banks of the river Paro, or Beni, on the east, passing through the intervening wilds of the Pajonal and pampa del Sacramento, &c. along windings of the forest best known to holy fathers and half-converted Indians. These wilds are inhabited on the south, and in the environs of the rivers Apurimac, Mantaro, Pangoa, Perene, Camar, and Sampoya, &c. by Campas, Piros, Mochobos, Ruanaguas, and other Indian tribes no longer invited to share of the blessings of Christianity; and to the north-east of the plain, or pampa of Sacramento, is the very important missionary settlement of Sarayacu, still

annexed to the department of Junin. Neither of these outskirts of an ill-sustained civil jurisdiction, nor the territories which thus lie very wide and uncultivated to the east of the main provinces, ever appear to have constituted a part of the ancient empire of the Incas. And not only by the rugged barriers of the eastern Cordillera, but by a difference of language, the untutored Indians of the Montaña are to this day separated from the true children of the Sun, whose common language, as the reader knows, is the Quichua, while the savages hitherto discovered speak almost as many tongues as there have appeared distinct tribes among them, except on the banks of the Ucayali, and in the vicinity of the chief missionary settlement there, where the Pano is the general or prevailing language of the somewhat christianized natives.

The government of Peru is, by its constitution, pronounced to be a popular, representative government; and in theory at least, though not in fact, the sovereignty emanates from the people, who are supposed to delegate its exercise to the legislative, executive, and judicial powers of the republic. It is not, however, now our intention to enter upon an account of the general government, as we only desire to enumerate some of the more important functions of the internal government.

OF THE OFFICE OF PREFECTS AND GOVERNORS, &C.

The superior political government of every *department* is vested in a prefect, under immediate subordination to the president of the republic; that of every province is entrusted to a sub-prefect, who is immediately subordinate to the prefect; that of the districts to a governor, who acknowledges the sub-prefect as his superior; and in every town, or Indian village, there is a still humbler officer called *alcalde*, who acts under the orders of the governor of his district, and is entrusted with the ordinary routine of local police.

To fill the appointment of prefect, sub-prefect, or governor, it is required that the candidate should be an active citizen, not under thirty years of age, and a man eminent for his probity.

The duties of such functionaries are,

1. To maintain public security and order in their respective territories.

2. To cause the articles of the political constitution, the laws enacted by congress, and the decrees and commands of the executive power, to be duly carried into effect.

3. To enforce the completion of sentences pronounced by the different tribunals and courts of justice.

4. To take care that the functionaries subordinate to each of them shall faithfully discharge their proper duties.

The prefects are also charged with the economical administration of the affairs of state within their respective departments. But they are restrained,

1. From interfering with, or in any degree interrupting, the course of popular elections.

2. From preventing the reunion of the departmental juntas, or interfering with these in the free exercise of their functions.

3. From taking any cognizance in judicial cases; but, should public tranquillity urgently require that any individual should be taken up, a prefect may command his immediate arrest,—transferring the delinquent, accompanied with the grounds of having taken him into custody, to the judicial magistrate or judge, within the precise term of forty-eight hours.

DEPARTMENTAL JUNTAS.

In the capital of every department there is a junta, composed of two members from each province. The object of these juntas is to promote the interest of the provinces in particular, and of the departments in general. The members are elected after the same manner with those of the Congress or Chamber of Deputies. [\[11\]](#)

The prefects of the departments have to open the annual sessions of the juntas, to report to them in writing on the state of the public affairs of their respective jurisdictions, and to suggest such measures as appear to them calculated to promote the general advantage of the departments.

Among the functions of this political body we may enumerate—

1. To propose, discuss, and agree about the best means of promoting the agricultural, mining, and other branches of industry in their respective

provinces.—2. To forward public education and instruction according to the method approved by Congress.—3. To watch over and promote charitable institutions; and, in general, all that relates to the interior police of the departments, except that of public security.—4. To proportion among the provinces the amount of the assessments of each department; and to ascertain, in case of complaint, the exact amount raised in the particular towns by their respective municipal authorities.—5. To determine the number of recruits for the service of the army and navy which each province and district should provide.—6. To take care that the chiefs of the national militia keep up good discipline in their corps, and have them always ready for service.—7. To see that the municipal corporations discharge their duties, and to inform the prefects of any abuses they may detect in them.—8. To examine the accounts which it is incumbent on the municipalities to render annually respecting the particular funds of the towns and villages.—9. To draw up every five years a statistical report of the departments.—10. To provide for the reduction and civilization of the indigenous tribes on the frontiers of any particular department, and to make it an object of special concern to allure them within the pale of civilized society by soothing and persuasive means.—11. To take cognizance of the imports and exports of the departments, and to transmit their observations thereon to the Minister of the Home Department, or *Hacienda*.—12. To apprise the Congress of any infraction of the constitution; and further, to elect *senators* from the lists presented by the provincial electoral colleges. But, to mention no more of the peculiar functions of the departmental juntas, we may conclude by observing, that such of their proceedings as hinge on the exercise of the powers or functions now enumerated, are transmitted through the medium and with the remarks of the prefects to the executive power, by whom again they are forwarded to the Congress as matters of legislative deliberation. ^[12]

Having premised the above articles relative to the internal government of the country in general, we shall now make some observations on the state of the very important department of Junin in particular; being able to do so, partly from personal knowledge, and partly from the report presented by the prefect of Junin, upon opening, in 1833, the session of the departmental junta in Huanuco; an official document whence we consider it incumbent upon us to draw much of the substance of the following observations, since

they refer to topics with which none can be supposed to be more conversant than the head of the local government.

MINES.

We shall here pass over the subject of mines, regarding which we have said enough under the head of Cerro Pasco; though there is no province in the whole department in which silver and even gold mines are not to be found; but the chief source of production at present is Cerro Pasco.

AGRICULTURE.

Of the agriculture of the Vale of Huanuco we have already treated; and, from what has been said, enough may be conceived of the general state of agriculture in the Sierra. We have also alluded to Jauja, in the preceding pages, as most productive in wheat; and abounding, as it does, in maize, lucern, peas, beans, &c. it is considered not only as the granary of the department to which it belongs, but also of all the central Sierra of Peru between the two great chains of the Andes. In the Vale of Jauja, as on the plains of Cajamarca, ^[13] vegetation is subject to blight from hoar-frost, which in some years is much more destructive to the crops than in others; but, upon the whole, the average wheat crops are very good and abundant. Huaylas, again, like Huanuco, produces excellent sugar; nay, that of Huaylas appears to be the finer-grained and purer of the two; for, though the sugar-refiners in Huanuco are generally brought from Huaylas, yet, in the hands of the same workmen, the sugar of the former locality does not equal that of the latter in the beauty of its crystallization.

MANUFACTURES.

The manufactures, we may cursorily remark, are in a very backward state; and though the natives, especially of Huanuco, have shown no small share of ingenuity in some of their mechanical contrivances, yet they want proper masters; and, however we may admire the progress they have made with such slender means of instruction, we cannot compare their performances with those of Europeans in the same line. In Tarma they make *ponchos*, or loose cloaks, of great beauty and fineness; and, on the colder

table-lands, warm but coarse blankets and ponchos, &c. are still made by the Indians. In the valleys, goat-skins are made into cordovans; cow-hide is made into saddle-bags, and almofrezes, or travelling-cases for bed and bedding; mats too are manufactured from rushes, and are very generally used as carpeting under the name of *esteras*. But the work of silversmiths is generally in a rude state even in Pasco; for the fine filigree work, for which inland Peru is celebrated, is made, not in the department of Junin, but at Guamanga, in the department of Ayacucho,—where the natives have also shown a decided talent for sculpture, though their works cannot be said to exhibit, as yet, much elegance or expression.

PUBLIC INSTRUCTION.

Without the aid of science, the arts of active life cannot duly advance in the career of improvement, and thus society remains a stranger to the higher refinements of civilization: hence, as the prefect of Junin well observes, the education of youth becomes a leading object of national interest and desire. But though in the department of Junin there are three colleges, that of Huanuco alone fulfils its purpose of instruction to a limited number of pupils; and there is reason to fear that it will soon share the fate of the other colleges in Huaras and Jauja, and be shut up for want of the necessary funds to defray its very moderate expenses. We are assured that the masters and professors of this recent institution, upon the foundation of which such bright hopes have been raised, are badly remunerated for their services; and have to contend against the perverseness of young persons whose early habits of self-indulgence while under the parental roof, ill prepared them to submit to the necessary restraints of a college, where the conscientiousness of the teacher, less pliant than parental fondness, will not sacrifice useful education to idle or vicious amusement, by imprudently winking at the errors and misdemeanors of the scholar.

Such being the inauspicious account of the only college of the department at present in exercise, the schools of elementary instruction do not seem to be in a more flattering position, though official measures have been taken to diffuse instruction among every class of the community. To the credit of the chief of the department, commands were issued by him for the erection of schools of elementary instruction in all the provinces, with

orders that proper reports concerning their condition should be regularly forwarded to the prefectorate at Pasco.

To these important subjects the prefect in the *mensaje*, or report, to which we have already alluded, endeavoured to awaken the attention of the departmental junta assembled in the city of Huanuco; when it was declared that the method of *mutual instruction*, adopted in that capital of local jurisdiction, in no way corresponded in its advantages or results to the time devoted to it by the pupils, or to the hopes at first entertained of its higher practical utility as a system.

The failure implied on this occasion may possibly have been less the fault of the system than of those who offered to apply it; for it was remarked as very worthy the consideration of the honourable junta, that, in reference to many of the schools intended for the improvement of the indigenous or Indian race, wherein they were merely taught a jargon of Spanish which they could not comprehend, it were better for them to be left in an untutored state of mind than to be placed under the melancholy influence of such teachers as presided over them. These were represented to be so imbecile, and so unacquainted with the merest rudiments of reading, or so abandoned and drowned in vice, as to be persons utterly unfit to guide the mind of infancy and innocence into a proper path. The junta were therefore called upon by their prefect to appoint some better means of instruction, which might at once serve to improve the virtuous feelings of the individual, and promote the national cause of civilization.

HOSPITALS AND CHARITABLE ASYLUMS.

It is affecting to think that, notwithstanding the wealth of which this department is the seat, the sick and invalid in general cannot find a home or place of assistance under their sufferings.

In Huaras, as well as in Huanuco, there were formerly well-endowed hospitals, but these are now fallen into such decay for want of funds for their support, that few indeed are the sick who can be accommodated or relieved in them; and, consequently, those in charge of these once useful establishments are daily put to the pain of refusing admission to a great many distressed persons, who are induced to seek their protection in the

hope of being cured of their ailments, or, if not, at least to breathe their last in the bosom of kindness and charity.

We are told by the prefect that an asylum or house of relief for the distressed poor never existed in the department: but, in his report to the departmental junta, he urges with an earnestness honourable to his feelings, that humanity calls for the immediate institution of establishments of this kind on behalf of the wretched victims of misfortune, whose very misery plunges them into despair. He also holds it to be a matter of public expediency to find a fixed home, and steady occupation, for those abandoned objects of compassion, who make traffic of their degradation and a boast of their debasement.

VACCINATION.

In the year 1832, it was found that the small-pox had just left dismal traces of its ravages in the department: fathers mourned their children now dead, or so disfigured and mutilated as to become unfit for the active business of life; the widow, too, wept for her lost husband, and the offspring of a mutual affection were left to feel the want of a father's care.

Curates, and municipal bodies, most particularly intrusted with the frequently repeated charge of preserving the vaccine fluid, unhappily neglected a trust so important; and the heads of families, who joined in the same carelessness, did not consider, until the fatal epidemic swept their children from their arms, that they were ever to taste the bitter fruit of their own improvident indifference. But to avoid the recurrence, on any future occasion, of so dreadful and destructive a malady, the prefect caused a supply of the precious vaccine matter to be procured from Lima; which, if carefully propagated, may yet save victims without number from adding to that depopulation which incessant warfare has, of late years, caused among his fellow countrymen.

JUNTAS OF HEALTH.

It has been proposed by the same active and intelligent prefect, Don Francisco Quiros, that juntas of health should be established in the capitals or principal towns of the provinces of his jurisdiction, with a view to

prevent the spread of contagious diseases;—to ascertain, and if possible correct, those physical causes and sources of disorder which are hostile to the healthy operations of the vital functions, and destructive to the growth of population.

It is, as we have seen, the duty of the departmental deputies to create these institutions, to frame rules for their regulation, and appoint fit persons for their management; while it would be the proper business of the prefect to see the resolutions of the junta carried into execution.

We only point to such proposals as the present, to show the reader how much such institutions are really wanted in Peru: not at all to mislead his judgment by inducing him to believe that there is the least appearance of their being established for a long time to come, unless, indeed, public tranquillity be soon restored; but people must perceive their wants before they desire to remove them, and the agitation of questions of civil amendment may ultimately lead to real improvement in their social condition.

PUBLIC BATHS.

In the dry and equable climate of Huanuco, bathing is not so necessary either to cool or to refresh the body as it is found to be in more humid and warm situations; for there is a bracing property in the dry air, which carries off the natural perspiration almost as rapidly as it is produced, and prevents that languor and discomfort experienced in a sultry atmosphere, where one perspires more than the air readily absorbs.

The inhabitants of this interesting province, and especially of the town of Huanuco, feel so little desire for the cold bath, that it is proverbial among them, that they only bathe in the river, or the canals of their delightful orchards, once in every year,—that is, on the day of San Juan, the 24th of June, the same on which the inhabitants of Lima celebrate their annual festival of Amencaes.

In the jurisdiction of Junin, however, natural warm and hot springs are exceedingly common, as well on the mountain plains, (which are in many places, as at Hualliy, covered with a saline incrustation,) as in the warmer valleys; and of these none are more resorted to, by invalids and

convalescents, than the ferruginous and tepid waters of the well-known baths of Cono, near Huariaca, and the still more celebrated sulphurous waters of Villo, in the district of Yanahuanca. Here there are two streams, of which the one is cold and the other hot; and being received into a reservoir in due proportions, baths may be always provided easily and cheaply, of any degree of temperature desired.

To make the medicinal waters of Villo—situated in a mild climate about one day's journey from Cerro Pasco on the one hand, and the city of Huanuco on the other,—as available as possible to the public, the patriotic prefect has recently taken measures to fit up convenient baths at this place. The well-known efficacy of the sulphurous waters in numberless instances of impaired health, the benignity of the climate in which nature has placed them, and the vicinity of this favoured spot to Cerro Pasco, have been the chief inducements to undertake this public work; which must prove of the utmost importance to the neighbourhood at large, and especially to miners and residents in the rigorous climate of Cerro, where health is more easily lost than regained, and where good medical attendance is rarely found.

POLICE.

Few of the municipalities of the department possess public rents and revenues calculated to answer the purpose to which they should be applied. But, in the absence of adequate funds and resources to forward all the objects of a general and well-regulated municipal police, there exists a valuable decree, which is very worthy of proper observance; for, in virtue of it, blasphemers, and those who, by their habitual indulgence in vice and vicious language, insult the better feelings of the community, are consigned to labour at public works, or compelled to sweep the streets, as the penalty of their infamous conduct. With further view to public order, the prefect has resolved to stigmatize, when he cannot hope at once to remove, the vice of drunkenness, in which the lower orders in general freely indulge in days of religious processions which are consecrated to sanctifying ends; but which the poor miner and uneducated villager think well observed by hearing mass in a morning, and contributing to the decoration of the saints clothed in tinselled and showy dresses, and surrounded by waxen lights without number.

On great religious days pavilions are not unfrequently erected in convenient situations for the reception of the effigies of the Virgin, our Saviour, and Cross, around which all sorts of silver and other ornaments are placed in fanciful confusion. The entrances into the churches and chapels, even in the rigid climate of Cerro and the adjacent *haciendas*, are lavishly adorned with beautiful lilies conveyed from the valleys; and wreaths and festoons of flowers hang over and around the doors of the pavilions and churches, which, when good metals are abundant, display a richness which only a mining country can be supposed to put within reach of the very humblest of the people. On such occasions the labouring miner often exhibits his person bedecked in the most gorgeous and expensive fashion; while he farcically dances, ankle-deep (if it happen to be the wet season) in mud, as gay and as merry as a London chimney-sweep on a May morning.

So marked is the taste for flowers among the poorest tenants of a mud-and-cane booth in the Vale of Huanuco, that on the festival of Corpus Christi,—a day of joy to the agricultural Indian, who always eats meat on this day, even should he have passed the rest of the year, like an anchorite, on vegetable diet,—the poor women and children on the sugar estates approach the house of their *patron* with hats, hands, and mantles full of the sweetest blossoms, which they strew before his door, and along his hall and corridor, as a sign of respect and rejoicing. To such expressions of good feeling he courteously responds by ordering the *tinaja*, or large jar of guarapo, to be placed at their disposal, under the direction of a major-domo or corporal of the field; and then with guitar and harp they engage in festive frolics.

But however desirous of enforcing a stricter observance of the days devoted to the service of the church, it has not been the aim of the prefect to check or discontinue the more innocent amusements of music and dance, or those of bull-fights and fire-works, in which the Indian also delights. He has struck at the principal cause of alienation from the house of God, namely, drunkenness, by condemning all those who are convicted of rioting or breaking the peace to sweep the streets for three successive days; or, should this be considered too light a correction, to labour for the same number of days in some other work of public utility.^[14]

Idle vagabonds, without useful occupation or property, and even without country, are pronounced by the civil authorities to be found in all parts

disseminating immorality and disorder; and seeing that to temporise with obnoxious characters of this sort is, in effect, to promote the cause of libertinism and idleness, it has been resolved at the prefectorate of Junin to persecute and exterminate, if they cannot amend, all such vicious intruders on society.

PANTHEONS OR CEMETERIES.

It has been long an established practice in Peru to bury the dead within the churches; a practice which, on the coast more especially, gave rise to a heavy exhalation, which very naturally rendered the incense burnt on the altar, independently of its mystical virtue, an agreeable and seasonable corrective for the sepulchral vapours of the rich and well-adorned temples of the metropolis.

This very unwholesome and improper custom has ceased in Lima since the erection of its Pantheon, and the example of the great capital has been followed in the remote departments. With few, if any, exceptions, cemeteries are now formed in all the provinces of Junin. But in Cerro Pasco, however, the burying-place was so very circumscribed and neglected, that, on the arrival of Don Francisco Quiros as chief of the department in the year 1832, there was not earth enough to cover the dead within the Pantheon walls, which altogether presented a very loathsome appearance. He caused the cemetery to be sufficiently enlarged, so that there should be nothing to render this place of rest offensive. Indeed, he expresses himself strongly on the urgency there was for the execution of this work: and though the stunted flowers of Pasco common do not always furnish a supply of fresh blossoms to be daily renewed over the graves of the departed,—and though no acacia, cypress, nor willow, no yew nor myrtle, can endure its elevated site,—Mr. Quiros enjoys the praise and the pleasure of having raised, in this inclement region of silver beds, a place of rest for his countrymen, which not even avarice can disturb; and glad would he be to see the children of the deceased steal to the graves of their fathers, there to pray over the remains of their kindred, and thus habitually cherish feelings of piety, humility, and hope. ^[15]

ROADS OF JUNIN.

Regarding the roads of the Sierra in general, enough has been said in the preceding pages; but of Junin, in particular, it remains for us to observe that very laudable efforts have been lately made for improving the roads of this department: yet no regular post-houses, with suitable accommodations for the traveller, are anywhere established; and the communication between the more remote provinces and Pasco is exceedingly bad. This is a great hinderance to commerce, and leads to inevitable delay and inconvenience in the transport of goods. However discreditable the fact may be to the corporation of miners, so little enterprise have they shown for the improvement of a place from which so much specie has been sent all over the world, that it is not without great difficulty, and loss of time and cattle, that they are able to convey the ore from the mines to the mills in the environs of Cerro; and all because of the miserable tracks which they use as roads. By the stream of Sacrafamilia alone, in the immediate neighbourhood of the mines, there are no fewer than eighty-eight ingenios or mills for grinding ore, some of which during the dry season are at a stand-still on account of the scarcity of water, and others at all seasons are interrupted in their work from the irregular supply of ore consequent on the bad means of conveyance. To obviate these great drawbacks on the industry of the miners, and general resources of the department, the prefect, some time since, commenced a cart-road, over which the ore might be conducted by oxen from the mines to the mills specified, through a tract neither extensive nor precipitous; but the undertaking was a great and a novel one for that part of the world in the year 1833, so little had such works of general advantage hitherto called up the attention and energies of the inhabitants.

POSTS.

The inhabitants of Cerro Pasco have the advantage of a weekly post between their town and the capital of the republic; and a direct correspondence twice a month with Huanuco, the capital of the department. By these arrangements an immediate communication is also held with the government, and the spirit of mercantile enterprise is thereby increased; Cerro being, as the reader may readily imagine, when the mines are highly productive, a most stirring place, visited by men of all climes, and full of traffic and speculation.

PUBLIC TREASURY AT PASCO.

The prefects in general are, as we have seen, not only entrusted with the maintenance of public order and security, but they are also at the head of financial affairs in their respective departments. In times of intestine warfare it has always happened that the Patriot government has exceeded the natural resources of the country, crippled as they are in all their branches by want of security, and consequently of capital. Thus there were, at the commencement of the year 1834 particularly, heavy arrears owing to the army, navy, and civil list. The supplies from the mint and custom-house were deeply pledged for sums advanced to the government; demands for payments, beyond what they could liquidate, were made upon the local treasuries of all the departments.^[16] and the treasury of Junin had to bear its share of all the demands of a needy government. By the report of the prefect to the departmental junta, of session 1833, it appears that, while in office for the previous year, he removed several abuses, regulated the accounts, and struck a fair balance of the ingress and egress of the Pasco treasury. He does not, however, state the amount of the departmental funds in this report, or present any data by which we are to form an estimate of the separate or aggregate revenue arising from the different provinces. All data of this sort the Patriot government is deficient in; and the real rental of the state can hardly at any time be clearly ascertained. A natural result of this fundamental defect in their statistics is, that, not knowing the precise extent of the population or pecuniary resources of the departments, the annual contributions cannot be laid in a well-regulated and just proportion to the means of each town, parish, and province. The difficulties and obscurities in which every branch of the public revenue, and especially of his own department, was involved, led the prefect of Junin publicly to declare his doubts concerning the integrity of the officers of the executive intrusted with the collection of imposts; and he broadly hints, that, in gratifying their self-love and interest, they forget the higher duties of the citizen. He therefore exhorts the honourable junta to discountenance all favouritism, to exercise a stern patriotism, and by fair inquiry to resolve the important questions,—namely, Whether or not more be annually exacted of the provinces than they, without injury to themselves or the state, have the power to contribute? Whether or not they do really pay more than can be legitimately required of them?

OF THE ADMINISTRATION OF JUSTICE.

Justice, in all the departments, is administered in the name of the republic; and in every town there are justices of the peace, whose business it is to hear both sides of the question at issue, and to endeavour to bring about an amicable termination without going formally to law: no demand, civil or criminal, save fiscal cases and others excepted by law, being admitted, unless this essential preliminary attempt at reconciliation has been put into practice.

In some of the provinces of the Junin department, as those of Huaylas and Huamalies, they have not judges duly learned and qualified in judicial proceedings; and consequently, in those parts, the office of the judge devolves upon the sub-prefects, who are alike ignorant of the law and its forms of application. Hence we may suppose they must be very unfit surrogates in such delicate matters as affect the person and property of individuals, and the good order of society.

Justice in the civil department is ill administered in Cerro Pasco, for which the prefect assigns a good reason; namely, that here criminal suits continually occur to engross the time and attention of the judge, so that it is impossible for him without assistance to attend to the ready despatch of merely civil causes, which are less urgent. The public are great sufferers from this imperfect judicial arrangement; and not only an additional judge, but several more notaries public, are required for Cerro in these times, when through the excellent management of Mr. Quiros in superintending the drainage of the mines and general interests of the place, rich ore shows itself more and more abundantly. The concourse of people being increased, a greater number of interests clash together, and civil as well as criminal suits crowd into court.

NATIONAL MILITIA.

By the articles of the political constitution of Peru, there are supposed to be in every province bodies of national militia as the guarantee of the internal order of the state; but, by the same constitution, the armed force of the nation has no power of political deliberation, as it is declared to be *essentially obedient*. Happy, indeed, might the state be, if its army of the

line and naval squadron were always obedient to the laws, and were proof against the influence of corruption, the wily leaders of faction, and the evils of frequent insurrection!

But in the greater part of the provinces a national militia can hardly be said to exist, except in name; though men titled captains and colonels of such corps are scattered about the country, and strut with their insignia of military importance in hamlets and villages.

In a neighbourhood where the writer resided for several years in the department of Junin, there was a villager of no small local pretension, who held at one time, in his own person, the offices of governor and captain of militia of his district, and, if we remember well, of alcalde also (he being alcalde on the death of the governor whom he succeeded); and in this way he became invested with all the authority of a petty dictator. The province was that of Huanuco, where, through the praiseworthy zeal of Colonel Lucar and Don Pepe (now Colonel) Echegoyen, a troop of cavalry was always kept up in some sort of military order: and the workmen of the different larger estates and little farms around, were called upon to assemble on Sundays under their respective captains; and, at assigned places, to go through some of the simplest military evolutions, using, however, no arms or particular uniform.

These Sunday exercises were generally ill attended; and, of ten or twelve young men on an agricultural estate, it would be usually enough if two or three appeared at one time in the ranks. Upon one occasion, however, when the captain of local militia in the village of Ambo had the honour of having the additional appointment of governor conferred on him, he called upon the writer when indisposed and in bed, and, with great appearance of sympathy and confidential cordiality, congratulated himself upon his promotion, because it would afford him the power, as he had the will, to serve his neighbour. With many such smooth expressions and assurances of kind and honest intentions, calculated to put even a misanthrope off his guard, he ended his visit by requesting that, as it was most desirable to keep up the military spirit of the district, he would expect of the writer that he should use his influence in persuading the young men on his hacienda to attend regularly at the military exercises in the adjacent village; a proposition to which he readily acceded, as it was agreeable to the established laws of the country. On the first or second Sunday following,

six fine young men went to attend the exercises at Ambo; and were seized and put into prison, with many others, under strong guard, to be marched off next day as recruits for the line.

The provincial prisons of Peru are in general very bad and insecure, and they are less frequently used than they should be for any better purpose than that already mentioned, viz. confining the useful and industrious husbandman—thus diminishing a race already too scanty, and yet a race on which the prosperity of the country mainly depends:

“Princes and lords may flourish or may fade,
A breath can make them, as a breath has made.
But a *bold peasantry*, their *country's pride*,
When once destroyed can never be supplied.”

Upon this occasion, we are glad to say, that the new governor's deceitful conduct towards us did not serve his turn as he wished.

The writer galloped to the capital of the department, where he found Colonel Lucar reviewing and selecting the recruits to be sent off from Huanuco to fill up the vacancies in the army of the line; and he must ever feel obliged to the politeness of the colonel, who instantly despatched a peremptory order to the said captain and governor to put our men of Andaguaylla at liberty, and to replace them from the list of idle vagrants, and not of useful husbandmen, within the term of a very few hours,—an order more easily given than executed, as by this time the rumour of imprisonment and seizure for the army had gone abroad, and young men, alarmed for their fate, fled to their woods and lurking-places.

Thus it appears that the real use of this mock militia is not to guarantee the internal order of the department, (which would be best secured by the absence of all troops, as the Indian population are never so well managed as by their own local magistrates of Indian family,) but to serve as a mask, under which to facilitate the means of raising soldiers for the general service. The governor's wily attempt to deceive us under the assurances of friendship is not peculiar, for such unworthy conduct does not disgrace one of these petty tyrants in the eyes of his countrymen.

CHAPTER IV.

Missionary College of Ocopa.—Its foundation, utility, downfall, and decree for its restoration.—Introduction of Christianity along the rivers Marañon, Huallaga, and Ucayali, &c. by the Jesuits and Franciscans.—Letter from Friar Manuel Plaza, the last great missionary of Ocopa, to the prefect of Junin.

Could the department of Junin boast of no other advantages than those which arise from the quantity of precious metal which it annually furnishes, it would be sufficient to substantiate its superior claim to the attention, not of the Peruvian government alone, but to that of all other countries in friendly and commercial relations with Peru.

Higher sympathies, however, than such as emanate from mere pecuniary considerations, must be awakened when it is remembered, that from a corner of this department the voice of Christianity has penetrated into vast regions of heathen and savage tribes, and reached the unsettled wanderers among the thickest entanglements of the woods, which occupy a great portion of the widely extended missionary territory of Peru. From Ocopa issued forth those zealous, persevering, self-denying, and enduring men, the great object of whose lives it has been, in the midst of danger, and in the name of the Saviour, to add to the faith of the church—and to civilized society—beings whose spirits were as dark and uncultivated as the woods they occupied from the confines of the rivers Mantaro and Apurimac on the south, to the river Marañon or Amazons on the north, and from the frontier provinces of the department of Junin on the west, to the great river Ucayali on the east. The missionary college of Ocopa is situated in latitude 12° 2′ south, in the province of Jauja, and at the distance of about twelve leagues to the south-east of Tarma. It was founded in the year 1725 by the commissary of the missionaries, Frater Francisco de San Joseph, with the intention of establishing missions for the conversion of the Indians, who ranged the wild frontier-land we have just alluded to. In the years 1757-8, it was erected by a bull of his Holiness Clement XIII. and schedule of his Majesty Ferdinand VI. into a college *De Propagandâ Fide*.^[17]

This college has attached to it a church built of stone; and we are told that great numbers flocked there in former days, when its altars were decorated with rich donations, and its ecclesiastics celebrated for their saintly character. The missionaries of this college had subordinate religious settlements, or asylums, in other provinces of the department; as, for

example, Huaylas, Huanuco, and also Tarma, at a place called Vitoc, at the entrance of the Montaña. The college was originally constructed to accommodate forty monks, and towards the close of the eighteenth century, when it was under the guardianship of R. P. Fr. Manuel Sobreviela, their number was eighty-four; part of them being distributed in the different settlements, and also in the villages of the neophytes among the wilds beyond the eastern summits of the Andes. The seminary, being under royal protection, was allowed from government six thousand dollars a-year as a charity. The great revolution, which wrested the country from the hands of the Spaniards, also deprived the college of its best support. The Patriots, in the midst of war, proscriptions, confiscations, and persecutions, spared not even this useful institution; the monks dispersed when deprived of government support, and only a few hoary brethren can now be traced among the number of these fugitive fathers. One of them, barefooted and bareheaded, we sometimes visited in his humble cell at San Francisco in Lima. His thoughts, abstracted from the scenes around him, usually dwelt among the tribes of the Huallaga and Ucayali; and with an enthusiasm which brightened up the eye of venerable age, he would point out, in the aisles and cloisters of this great conventual church of his order, the paintings that commemorated the martyrdom of such of his brotherhood as fell victims to the violence of savages whom they piously laboured to turn to Christianity.

The Patriots having at length seen the national loss likely to result from neglecting the territory of the missions, and allowing the half-converted Indians to glide back again into their former savage and independent condition, for the want of officiating priests or zealous monks to continue the work of civilization, in which the Spaniards had engaged with so much spirit and success, it was resolved by the government of Peru, in March 1836, to annul the decree which, in November 1824, was passed to convert this religious college into a common school or academy of general instruction; which, however, was never established on a permanent footing. Besides other reasons of less moment which were assigned in the preamble of the decree for restoring the college to its ancient functions, it was stated that the civilization of the savage tribes of the interior, and their conversion to the holy Catholic faith, was an enterprise worthy of the intellectual light of the age we live in, and acceptable in the sight of the Almighty; that only for this purpose was the college intended at the period of its foundation; that

measures had actually been taken by the government to induce missionaries to come from Europe for the re-establishment of this pious institution, and that therefore it was decreed that the missionary college of Ocopa should be placed precisely on the same footing as before the revolution, or the decree of the 1st of November 1824; that all its rents and property should be restored, and that whatever sums were assigned for the academy alluded to should be transferred to this missionary college; that the Archbishop should appoint a fit person to take charge of the college and receive all its revenues, and to pay from this fund the expenses of repairing the buildings and the passage-money of the expected European monks, whose arrival the Very Reverend Archbishop was required to encourage, while he superintended the necessary repairs of the college, and made such reforms in its regulations and rules as should best harmonize with the republican form of government. Nothing can better prove the decay of the missionary cause, and, we might perhaps add, the decay of practical religion in Peru, (since its own clergy want zeal and enterprise to act as missionaries,) than this document; and though the invitation be more immediately addressed to Spanish ecclesiastics, yet the decree is in that spirit which seems to open the door to any company or association, who, adhering to the Catholic form of religious instruction, may be pleased to extend their benevolence and Christianity to the fertile regions of the Amazons, where they may fulfil their mission far removed from the scenes of political anarchy or misrule, and far beyond the pale of all hostile influence which could impede the exercise of their sacred calling. Experience has long since sufficiently shown that these Indians of eastern Peru are neither incapable of intellectual improvement, nor deficient in those moral elements which form the groundwork of the social edifice; and if ever they should be instructed, and guided, and disciplined in the way of life, according to the Gospel, by active, honest, and enlightened teachers, who know that faith comes by hearing, and hearing by the Word of God, there may yet be exhibited on the banks of Ucayali, and on the fine plain del Sacramento, a great and virtuous people, concerned to know and to do their respective duties, where now cruel barbarism and savage superstition hold their cheerless sway.

To give a full historical account of the transactions of the missionaries of Ocopa, their various expeditions by different routes and with varying success,—or to enter into the interior economy of their college, and details of its discipline,—would be too copious a matter for the narrow space we

have allotted for this subject, which in itself is one of no ordinary interest. And, to detach the history of the missionary exertions of the Jesuits of Quito and San Borja from the labours of the Franciscans of Lima and Ocopa,—to define the precise limits of the conversions by each of these religious orders independently of the other,—would not be free of intricacy; nor does it appear to be necessary to establish the degree of honour due to each, for they both toiled in the same thorny vineyard, and the latter creditably continued what the former had happily commenced. But, to give an idea of the origin of these missions, it may be well to refer back to the discovery of those regions in which they were planted.

The mouth of the Marañon was discovered by Vicente Yanes Pinzon, at the close of the fifteenth century; but Orellana, the Lieutenant-general of Gonzalo Pizarro, Governor of Quito, was the first to sail down its stream, from the point where the Napo joins it, to the ocean, in an armed vessel built at the place of embarkation on the latter river by order of Pizarro, who had himself undergone great hardships, and sacrificed most of his followers on an expedition of inland discovery. Orellana commenced this voyage in the year 1540 or 1541.

Some of the natives were friendly towards him, and others in canoes opposed his progress; and, as the men of one particular tribe were aided by their women in the combat, the Spanish captain gave their female warriors the name of Amazonas,—whence the appellation “Amazons,” which this great river still retains.

Another expedition, under Pedro de Ursua, was undertaken in 1560; but he and most of his followers fell victims to an ambuscade laid for them by the Indians. In 1602, Father Rafael Ferrier, a Jesuit missionary, descended the Marañon to the river Napo, which Orellana had navigated about sixty years before; and on his return to Quito, communicated his discoveries, and his ideas concerning the natives he had seen.

In the year 1616, some Spanish soldiers, stationed on the frontiers of Quito, pursued some Indians into their canoes on the Marañon; in the pursuit they descended this river till they came to the Maynas,—a tribe of Indians who showed such a disposition to amity and to become Christians, that, on the return of the soldiers to the frontier station of Santiago de las Montañas, so favourable a report was made of them to the Viceroy of Peru,

that in 1618 he appointed Don Diego Baca de Vega as Governor of Maynas and Marañon; who was the first to subdue the people of these territories, and subject them to the dominion of Spain.

In 1638, according to Alcedo, the Jesuit fathers, Gaspar Cuxia and Lucas de la Cueva, settled several missions in the country of Maynas on both shores of the Marañon, which continued to flourish until the abolition of the useful society of Jesuits in 1767. ^[18]

At first the capital of Maynas was San Francisco de Borja, which, according to Ulloa, is situated in latitude 4° 28′ south, and 1° 54′ east, of the meridian of Quito. In this city an insurrection of the native Indians took place in the year 1635, which was happily quelled by the indefatigable Jesuits: but afterwards the town of Laguna on the east bank of the Huallaga, in latitude 5° 13′ south, became the principal seat or capital of the missions of Maynas, which extend from St. Borja along both sides of the river Marañon, embracing many villages or settlements, to the frontier possessions of Brazil at Tabatingo. From the Marañon the patriarchal government of the missionaries extended southward along the river Ucayali, and among several of the tribes on its banks, or in the adjacent woods, such as the Cocamas, Piros, and Conibos or Conivos Indians, whom the Jesuits of Quito had in a great measure converted to the faith; but they again revolted, and returned to their original wandering and savage mode of life, having put their pastors to death. After this unfortunate event, several fruitless efforts—especially in the year 1695, and also in 1764,—were made to reconvert these tribes, till at length the Franciscan missionaries of the college of Ocopa succeeded to a great extent in this hazardous undertaking.

But long before the establishment of this college of their order, two Franciscans,—the Fathers Andres de Toledo and Domingo Breda, both bent on making converts to the faith,—left Quito in the year 1636, and, having surmounted the greatest hardships by land and water, arrived at Para. They reported their arrival and discoveries to Santiago Raimundo de Noroña, Governor of San Luis de Marañon, in the united service of Spain and Portugal, for both countries were then under the sovereignty of the crown of Spain. The result of the intelligence thus derived was a further survey up the river, under the command of the Portuguese captain, Texera: and an account of the whole of these proceedings being transmitted from the *Audiencia* at Quito to the Count of Chinchon, Viceroy of Peru, he in the year 1639 sent

back the flotilla of Texera to Para, conveying thither the Fathers Christoval de Acuña, and Andres de Artieda, Jesuits of Quito, and other able men, commissioning them, among other things, to survey minutely the river Marañon and its banks, and, having done so, to embark for Spain, and lay their account before the Council of the Indies; all which they accomplished in a creditable manner.

As early as the year 1631, Franciscan missionaries visited the environs of the river Huallaga, made converts, and entered the country of the Panataguas. Contiguous to Huanuco, and probably within the territorial limits of this ancient tribe, is situated at present the important and civilized Indian town called Panao, which is included in the curacy of Santa Maria del Valle.

From the Panataguas are supposed to have sprung several other tribes of distinct denominations, which had spread over the adjacent country, wherein Christianity had made but slow progress.

From the city of Huanuco the fathers of Ocopa penetrated, by Panao, Muña, and Pozuzo, to the port on the river Mayro, where they formed one of their earliest settlements: from this place they appear to have descended in canoes to the rivers Pachitea and Ucayali. This course is well marked on the map of those parts, published in Lima, in the year 1791, by the literary society entitled “Sociedad de Amantes del Pais de Lima.” This excellent map of the territory of the missions in Peru was dedicated to his Catholic Majesty, Charles the Fourth, Emperor of the Indies, by the said society, and the reverend fathers of the missionary college of Ocopa; whose superior or guardian, Fr. Manuel Sobreviela, enriched it with a plan of the rivers Huallaga and Ucayali, and of the pampa del Sacramento. In the above route from Huanuco, the Franciscans from the time they left their last christianized settlement, Pozuzu, (some years ago depopulated by the small-pox,) had to contend with the Amajes, Carapachos, Callisecas, and other savage tribes, occupying the territory between Pozuzo and the mouth of the Pachitea. From this spot, namely, where the Pachitea joins the Ucayali, to the river Sarayacu, which enters the Ucayali in latitude 6° 45′ south, several streams descend from the plains of Sacramento to join the Ucayali,—such as the Aguaytia, De Sipivos, and Manoa, the environs of which are inhabited or frequented by various tribes of Indians known by the names Sipivos, Conibos, Manoas, and Serebos, &c. Among these they made some

converts; but the principal nation are the Panos, who inhabit the neighbourhood of the Sarayacu, and form a great part of the population of the town of the same name, which is the superior, or rather, at present, the only seat of the Franciscan missions on the Ucayali. This mission, founded by the Franciscan, Father Girbal, in 1791, was visited in February 1835 by Lieutenant W. Smyth and Mr. F. Lowe, in their journey “undertaken with a view of ascertaining the practicability of a navigable communication with the Atlantic by the rivers Pachitea, Ucayali, and Amazon.” They found it under the guardianship of the venerable Father Manuel Plaza, whose account of these parts forms an interesting document in the *Mercurio Peruano*. The climate of Sarayacu is described by this excellent missionary as more free from ague and dysentery than the settlements in low, sultry, and humid situations on the banks of the river Marañon; and Lieutenant Smyth and Mr. Lowe, who give an interesting account of the actual state of the mission, observe that “the climate seems very much like that of the island of Madeira;” and, like the city of Huanuco on the Huallaga, it is refreshed in the dry season by breezes that blow along the river.

All the other missionary settlements in these parts having been abandoned since the downfall of the college of Ocopa, the consequence has been, that the Indians of those settlements have collected round their only remaining spiritual father and friend, Padre Plaza, at Sarayacu, where the population has thus been swelled to the number of about two thousand; and these semi-barbarous tribes honour their faithful pastor, and are very attentive to the service of their church, which is performed partly in the Latin and partly in the Pano tongue. Lieutenant Smyth and Mr. Lowe not being able to realize the object of their expedition by entering the Montaña at Pozuzo, and descending by the Mayro, returned to Huanuco, and descended by the river Huallaga, till they arrived at the river Chipurana, in the province of Maynas, which, according to the missionaries, enters the Huallaga in latitude 6° 30′ south. They ascended the Chipurana as far as they found it navigable; and thence, partly by land and partly by water, they proceeded to Sarayacu, in expectation of being able, through the guidance of Padre Plaza, to effect their expedition up the Pachitea,—an undertaking in which they unfortunately did not succeed, on account of the inundation of the rivers during the wet season, which lasts from November to May, and also for want of a sufficient supply of effects to exchange for the provisions necessary for the support of an escort of the mission Indians, without which

the enterprise is not safe, nor indeed practicable, at any season. Nevertheless, Padre Plaza, before the visit of Lieutenant Smyth and Mr. Lowe, wrote to advise the government of his opinion regarding the communication with the district of his mission by the port of Mayro; and his letter on this subject was published at Cerro Pasco, after accounts had been received of the failure of the expedition undertaken by the gentlemen now mentioned, in company with the Peruvians, Major Beltran and Lieutenant Azcarate. It is to be feared that this reverend monk is too much stricken in years to be much longer able to preserve his usefulness; and, what is yet more to be lamented, at his death it is probable that all the labours of himself and his predecessors in the same field of conversion and civilization will have been thrown away. Friar Manuel Plaza is not likely to have a successor from the school of Ocopa, notwithstanding the decree respecting its restoration, while Peru continues to exhaust its best resources in war, either civil or defensive, against neighbouring states. To enable that country to consolidate its internal strength, and attend to the practical improvements of civil and religious institutions, it must have what the majority of its citizens sincerely long for,—an interval of quiet. Until domestic peace be acquired, the peace of the Gospel is not likely to be sent forth afresh to subdue the turbulent spirit of the Cashivo, or to replant and renew the settlements and friendships that were formerly established by the emissaries from Ocopa; friendships now for the most part forgotten, and settlements no longer to be traced, except on the map of their wide-spreading mission-land, already alluded to as dedicated by their order and the literati of Lima to the King of Spain. But since then the dynasty of kings has been destroyed; and zeal for the missionary cause, except in name and speculation, has almost vanished from the land, where it would appear that patriotism can only thrive on the ruins of all the best institutions of former days: and when the writer of the following letter shall be no more, the name of king and Saviour, if not also of friend^[19] and patriot, may soon cease to be heard or honoured among the woods and glades in the now isolated and forlorn mission of the Ucayali.

“Peruvian Republic.—Mission of the Ucayali.

”Sarayacu, 14th December 1834.

“To the Sub-prefect of Huanuco.

“On the 20th of November last, I replied to the prefect, D. Francisco Quiros, regarding the project of the Supreme Government, which was sent to me by you, 24th September of this present year; and I answered, with the least possible delay, by the Moyobamba post. But, lest that letter should have been in any way mislaid or lost, I think it advisable, as it related to an affair of so much importance, to forward a duplicate of the same, which is as follows:—

“RESPECTED SIR,

“This very day came to my hands your note of the 18th of September of the present year; and having carefully perused it, I have to inform your honour, (Vuesa Señoria,) with the greatest sincerity, that the project adopted by the Supreme Government, of penetrating to the river of Pachitea by the port of Mayro, is the best and safest plan, because of the advantages that would accrue to the republic from opening the navigation of that river; for, from its junction with the Ucayali, up the stream to Mayro, is only a passage of seven or eight days; and from the latter place to Pozuzo, by land, is but an intermediate distance of fourteen leagues.^[20] But there is one obstacle which, as long as it exists, will almost certainly interfere with the enjoyment of a safe traffic on the river Pachitea; namely, that on its banks are situated the pagan Cashivos, cruel cannibals who live on human flesh,—sometimes availing themselves of much cunning and artifice to deceive passengers; and at other times, with all the fierceness of the wild beasts of the forest, fearlessly attacking them, as was proved in two expeditions undertaken from this place by Father Girbal, who the first time only advanced to the nearest huts, when he was compelled to return on account of the scarcity of arms, and the small escort given him by the government. He afterwards advanced to their last encampments (*rancherias*), whence he returned without having realized his purpose of striking the Mayro, where people waited his arrival with provisions and whatever else was required: and since this last expedition,

which was made in the year 1797, no further active measures have been attempted.

“The neighbouring nations of Conivos and Sipivos, who reside by the inland streams of the Ucayali, though they constantly endeavour to drive away these cruel enemies, have never succeeded; for so far is it otherwise, that they suddenly break into the houses, and, not satisfied with putting their inmates to death, carry off the dead bodies to celebrate their banquets with, for the Cashivos have an innate appetite for human flesh.^[21] The project of entering by the Mayro is the most attainable of any other, because, in descending the water, the vessels keep the centre of the river, so that they cannot be reached by the arrows from the banks at point-blank shot: besides, by merely discharging a few fire-arms, they disperse; and as, happily, they do not use canoes, they cannot intercept the passage, or do us material injury. And further, the descent to this point is accomplished in two days only; for which reason it is very necessary that I should have seasonable advice, the time being as nearly as possible fixed, to prevent any disappointment as to our meeting; when, according to the plan proposed by the commissioners, an expedition may be made with every precaution from this point, for the purpose of clearing the passage of so destructive and indomitable a people; and in this way the frontier towns may be able to proceed in extracting from the Montaña its precious productions.

“Actuated by this desire, and that of rendering happy the inhabitants of the Ucayali, I have now, for the space of thirty-four years, felt it my duty to live in these missions; and God grant that my eyes may yet see the prosperity of these regions, since my expedition to the Pangoa failed of producing the advantages expected from it.^[22]

“This expedition I undertook merely to please the fathers of Ocopa; but that the intercourse thus commenced would be of short duration, it was easy to conjecture from the great distance which separated the mission from the college; the

difficult and dangerous navigation of the head-streams—*cabezeras*—of the Ucayali; and lastly, the discordant opinions of the European fathers.

“But the day has now arrived when my wishes will be verified through the skilful arrangements of the Supreme Government; and to the best of my power I will contribute to the success of the enterprise, not only by assisting the commissioners, but also by accompanying them on the expedition, old as I am.

“All the above considerations I submit to your notice for your information and government.—God protect your Honour!

“As the letter which I have alluded to may possibly have miscarried, it will give me great satisfaction to know that this has come to your hands.

“God protect you!

“FRAY MANUEL PLAZA.”

CHAPTER V.

Christianized Indians of the Interior.—Their condition and character.—Hardships imposed on them.
—Desire of revenge.

Tangur, in the curacy of Caina, and department of Junin, is one of those villages so common on the elevated slopes which overlook temperate valleys in the interior of Peru. In this small village, as we are informed by a gentleman who for several years visited it in the character of curate, there are two distinct municipalities, each possessing its separate church and magistrates.

These separate people, who speak the same Quichua language, do not associate together, nor do they even hold their religious festivals on the same day. The origin of this separation of interests, tradition informs us, is as far back as the time of the Incas, when some convicts, ordered from Quito, settled at this place, and formed a distinct family; which has here subsisted since that remote period, without ever mingling its blood with that of its neighbours, or entering into communion or alliance with any other people. This is the more remarkable, as it is the ordinary practice in other remote villages of the interior for the whole body of men to co-operate in any great work, such as constructing bridges for their common good, or building houses for the convenience of individuals; on which occasions one party conducts stones and turf, another builds the walls, a third conveys timber from the distant woods,^[23] and a fourth cuts and lays on the thatch &c. The unanimity in this case, and the want of it in that of Tangur, are equally characteristic of that love which the Indian entertains for the usages of his predecessors in all things. In nothing does he approve of innovation; in his condition he has not yet known any durable or real amelioration, and in nothing does he desire change. In his local prejudices, habits, and daily pursuits, he only thinks, feels, and acts just as others did before him. If the general revolution has been in any degree useful to the poor uninformed Indian of Peru, who has already sunk from the short-lived excitement of patriotic enthusiasm into the dejection of a military despotism,—if it has really improved his prospects, it has been by rousing him, for a while at least, from his wonted apathy to the general concerns and conveniences of life; opening to his view a wider range of imitation and desire, and thus

breaking in upon the hereditary routine of his customs and habits, to which, till now, he has adhered with the unvarying constancy of mere instinct.

The christianized Indians of the Inca dynasty, whose native tongue is Quichua,—for we do not at present speak of the savage, or half-christianized Pano, and other yet unsettled tribes of the Montaña,—are said to be an indolent race; but we have had the opportunity of knowing that their exertions will increase as the prospect of bettering their condition expands, and that in general their labour is only conducted in a slothful manner when it is compulsory, or to themselves unproductive. We have had ample opportunity to know, that when they labour by “*tarea*” or piece-work, and are sure of their wages, they work remarkably well. On their own little farms they are truly hard labourers; and, if they were not so often pounced upon by enemies to industry, the fruit of it would be seen in their growing prosperity. It is those who tyrannize over them who accuse them of laziness, duplicity, and natural perverseness of disposition. Of such persons we may be allowed to ask, Have they ever afforded the Indian any rational encouragement to honesty and industry? Have they ever, by fair dealing, persevered in the experiment of deserving the confidence, of conciliating the affections, or of calling forth the kindly sympathies of these humbler sons of the soil? What virtue, except patience, were they permitted to disclose under Spanish oppression—(would it were mitigated under the patriot system!)—when their masters supplied them with the necessaries of life just on what terms they pleased, and when the Indians could realize no property, however much they redoubled their toil, for in general the fruit of their labour was not their own?

The Indians for the most part are an agricultural people, for more live by tillage than mere pasturage or any other occupation. Many of the modern villages in the temperate climate of the interior were, not many years ago, large farms, possessed by Europeans or their creole descendants; but the labourers, set free at the revolution in consequence of the confiscation of the goods and property of their fugitive or ruined masters, have continued to cultivate the land for their own maintenance, till by degrees their families have swelled into villages, and at length assumed the important character of municipalities. With a few years of undisturbed peace, and exemption from undue exactions, small villages may thus arise and become considerable towns, wherever the locality happens to afford sufficient scope for

cultivation. But as it often occurs that the Indian hamlet is erected on a pinnacle, or on the brow of a hill, around which there is but little suitable soil for the spread of agricultural industry, the consequence is that the father divides and subdivides the same piece of ground among the rising members of his progeny, till at length the means of subsistence become too scanty for the support of the whole family, and, the supernumeraries must seek employment in the mines or elsewhere, as they best may.

The mechanic arts are little needed by Indians who construct their own huts, and, with the exception of their coarse felt-hats, shape their own dress, which in warm situations consists of sandals of raw hide, wide trowsers or breeches open at the knee, a shirt, vest, and sometimes a jacket, and over all a poncho. In cold exposed localities, as Cerro Pasco, they always wear warm woollen stockings and a jacket; not omitting the poncho, which is the indispensable covering by day as well as by night throughout the Sierra. Besides such drawbacks on the growth of Indian population as arise from the want of efficient medical assistance, and the occasionally destructive effects of epidemic diseases, other causes have been frequently assigned, and especially an excessive passion for intoxicating liquors. This propensity operates strongly with the miner, stationed, as he is, in high and frigid localities, where he is much exposed to wet under ground, and to nocturnal frost or snow above it. Here the action of intoxicating drink, particularly when indulged in by those not born in very elevated regions, superadded to the usual effects of a highly rarefied atmosphere, and other causes of a less general character, tend greatly to shorten human life. But in the warm and temperate valleys which intervene between the coast and the Cordilleras this vice is by no means so prevalent as at the mines, where money circulates freely, and all manner of temptation is to be encountered. For though *agua-ardiente*, *guarapo*, and *chicha* usually abound in such places, yet it must not be forgotten that the *peones* or day-labourers of these favoured climates seldom have reals to spend; and that, when they have no money, their credit does not usually extend so far as to enable them to be often intoxicated. For about three years we had seldom less than a score, and frequently as many as fifty or sixty of these Indians, working under the eye of a major-domo; and, except upon some saint-day or festival, we do not recollect having any complaint made to us on the score of drunkenness. Licentiousness is usually stated as a further source of depopulation among all classes and castes in Peru: but, whatever be the true explanation of the

fact, we think that evils springing from such fountains of impurity show themselves comparatively little in the Indian constitution; and though strict regularity of conduct cannot be claimed on the part of the Indian family, yet the modesty of their ancient mamaconas is still remembered among them; and it is a characteristic which to this day honourably distinguishes the Indians from their more cultivated masters, that with them conjugal infidelity is discouraged, punished, and felt to be a crime.

Incessant warfare and intestine broils, by keeping up continued agitation, are at this day as destructive and desolating to the aborigines of Peru, and to the general industry and prosperity of the country, as was formerly the compulsory system of working in the mines, and manufactories or “obrages,” under the Spanish conquerors. The factious and seditious spirit which has gone abroad in the republic is an excuse for a standing army, which in its turn becomes the fertile nursery, or, at least, frequent instrument of faction; and, what is worse, military licence is rapidly pervading all classes of society, and destroying the only true sources of population, which are domestic virtues, domestic habits, and simplicity of life. We shall dwell no longer on such causes, already too well known as principal sources of depopulation in Peru; but, before we quit the subject, it occurs to us to mention that during a litigation, in which the question at issue was to be partly determined by the evidence of tradition and ancient usage, a number of witnesses of the most advanced age in the Vale of Huanuco were called in to give their testimony, of whom several were from seventy to ninety years of age, and, with only one exception, in which the blindness of age came on in an European by birth, all of them were hale old people, and generally of Indian race.

We would remark of these Indians, that although for centuries they have endured oppression with the mute meekness of the lamb bound for sacrifice, they are by no means dead to feelings of domestic tenderness, or insensible to the natural ties of kindred or of country, whence they are violently torn when led away as recruits. Wanderers from their native soil, wherever the public service or the will of an usurper leads, they brood over the loss of the genial freedom, simple habits, and peaceful enjoyments that once were their own, when they herded their flocks or cultivated their corn and pumpkins. In an hospital, on the coast, we have seen some of these poor fellows unable to speak a sentence of Spanish to the physician who

prescribed for their relief; and, in a few extreme instances, despair sunk the powers of life, and a hopeless love of home exhausted their spirits. We have seen one very young lad thus affected who refused food and medicine, until in silent sorrow he expired, a victim to nostalgia, or a love of home, and a broken heart. These hapless beings, whose devotedness to early attachments and associations bespeaks the warmth and fidelity of their affections, though cherished under a cold and apparently a passionless exterior, we found to be indeed reserved, but sagacious; and, when not under any unusual excitement, their minds, though not cheerful, were serene. Their exterior mien always struck us as solemn, and even sad; but this may be partly the effect of the awfully grand and sublime scenery so familiar to their view, which imparts a solemn and contemplative turn to the thoughts of the mountaineer, and influences his moral feelings in such a manner as stamps a certain air of mental gravity on his general deportment and expression. As an individual, the Indian is timid, and he will sooner take a cuff than give one; but, when they assemble for mutual support, then indeed they are seen to fight most valiantly, and like tame oxen, when the blood of one of their number is shed, they all become fearfully courageous. Bold and bloody battles we have seen between strong parties of the native miners in Cerro Pasco, armed for the combat with slings, stones, and clubs. At festivals, too, when roused by drink or enraged by jealousy, they lacerate and maul each other: and the meek-looking, dumpy, Indian woman becomes equally exasperated and vehement if in her quarrels any one should cut away a tress of her long and coarse black hair; for the cutting of these tresses is an odious mark of female dishonour, to which women of every caste in the land,—except the woolly-headed blacks and mulattas, on whom nature has not bestowed these ornaments,—are most acutely and painfully sensitive.^[24]

From the beagle-courage of the Indian, who, like these gentle animals, fights better in company with others than singly, his military character stands very high; and a regiment of Indians when conducted by gallant officers, as was the case during the war of Peruvian independence, are sure to prove indomitably brave and hardy. The dark and zambo soldier of the coast, when urged forward on a rapid Sierra march, is very apt to sink under the pressure of fatigue, conjoined with pinching cold and inevitable privations, to which he has been little accustomed in the warm and humid “potreros” or enclosures near the sea. But the Indian foot-soldier is superior

to such obstacles; and with only the support of a pouch of coca, and a bag of toasted maize, he will continue his march wherever the llama can keep its footing over ledges of rock, and rugged recesses so wild and so land-locked, that, as the dwellers in these solitudes say, it would seem difficult for the birds of the air to escape from them. But though, when engaged on long and forced marches over savage mountains and glens, one of these all-enduring and active natives hardly ever falls behind from mere fatigue, yet he has not so far forgotten the shade of his fig-tree in the bosom of the vale, or his airy home on some distant eyrie, from whence he was dragged in bonds, as not secretly to pant for his native nest; and, on long inland marches, the general or commander who is not singularly vigilant, or uncommonly beloved, has more to fear that the Indian may desert him on the journey than when engaged with the enemy.

In every village of the intermediate valleys, the white vagabond and roguish mestizo have “padrinos,” or protecting friends of their own caste in petty authority, holding the commission of captain of volunteers, governor, or alcalde, or something more subordinate still; but the more industrious Indian, who tills his own piece of ground, and peaceably labours to rear his little group of helpless children, is constantly the victim of oppression. This useful citizen, who happens not to owe a dollar to any man of influence whose interest it may therefore be to interpose with the colonel or sub-prefect of the district in his behalf, has, to use his own pathetic expression, no “arrimo” or powerful support, neither friend nor protector to plead his cause with the local authorities, who, though they are enjoined by the government to enlist none but idle and vicious characters, are daily seen to sacrifice, with insolence and impunity, justice and duty to malice and caprice. The native inhabitants are therefore searched out and dragged from their houses, or from the caves and fastnesses where they have sought concealment. Torn from their forlorn and destitute families, carried away as recruits on every new levy of conscripts, they are bound like galley-slaves, and then driven along, a spiritless crew, hopeless and helpless, from the recesses and glens of the interior to the coast, or elsewhere, as circumstances may require, there to die of ague or dysentery, or, if they survive the usual effects of great changes of climate and diet, to be harshly trained for the exercise of war.

It is a law of the country, contained in article 6th of the Constitution, that the common rights of citizenship be suspended towards the notoriously vagrant, the gambler, the drunkard, and the married man who without cause abandons his wife, or who is divorced on account of his own misconduct. The rich and influential can, when they please, easily evade such laws as these; but, among the peculiar hardships imposed upon the Indian of the interior, it is not the least that he may be seized for a soldier on the alleged ground of his being “mal-casado,” or habitually cohabiting with a woman to whom he has not been previously bound by the bonds of regular marriage. It is not impartial justice that he should be punished in this manner for a delinquency which is almost authorized by the practice of his superiors. These poor people pair together naturally, and at an early age; and would, we think, frequently render their union more binding by marriage, if they could afford it. That this may be understood, it is necessary to say that the curate’s fees for the performance of matrimonial and burial service vary in amount with the caste and complexion of the parties. The fee for marrying an Indian is lower than that assigned for the marriage of a mestizo, and the white man pays more than either. One consequence of this arrangement is, that it is often difficult to ascertain the class of the proposed bridegroom; and the curate may sometimes be induced to raise the beardless Indian to the rank of the scanty-bearded mestizo, and the latter has in his own person great ambition to be thought an “hombre blanco” or white man. The poor agricultural Indian of the Sierra has commonly enough to do to provide himself with his coca,^[25] a hoe, and a chopping-knife, the tools that he usually works with; and seldom indeed has he got on hand as many dollars as would enable him to pay even the lowest rate of marriage fees. Now, he who cannot pay without difficulty the priest for marrying him in a Christian manner, thinks it can be no great harm to imitate others around him whose example ought to be worthy of imitation; and, ignorant of the language which Scripture addresses to his conscience, he contracts a marriage sanctioned by custom, though not by religion.

Another hardship in the Indian’s situation is, that he has often great trouble to pay the customary tribute or capitation tax, from which even the superannuated are not always exempt, though the Treasury professes not to receive contributions from the aged. When we resided in the Vale of Huanuco, men have come to us from the distant province of Conchucos, imploring work, to be paid for not in produce but in money, which is scarce

in Conchucos, that they might be enabled to return with a few dollars to satisfy the collector of revenue, not less inexorable in his demands than the corregidor used to be in exacting the royal tribute, of which this vestige yet remains. On the coast, a day-labourer's wages may be six reals or a dollar, according to circumstances; but in remote parts of the interior, as in the province to which we have referred, wages are very low,—for example, a real or sixpence per day. In Huanuco, wages are nominally three reals a day: but here the native planters usually run accounts with their workmen, whom they supply with such articles as clothes, spirits, maize, coca, and perhaps tobacco; though the cigar is more used by the natives of the coast, and such as use not the coca, than by the agricultural Indian. By this mode of management the poor man is commonly precipitated, before he is aware of it, into his employer's debt, and very often remains involved, and virtually a slave, for the remainder of his life; while his sons after him are made to take upon themselves the burden under which the father sank to his grave. But the same grinding system involves even the better sort of men who are looked upon with respect in their own humble sphere, and permitted to prefix Don as a shining handle to their name,—they also are victims to ruinous custom and superstitious rites; for they are called upon in their turn to bear the expense of being the major-domos of the feasts which are celebrated in honour of the tutelar saint of the village to which they belong.

To defray the expense of these public entertainments, the major-domos have in most instances not only to spend all their savings, but to borrow, and to run up their credit with the sellers of fruit and preserves, the butcher, the baker, the distiller, and *chicheras*, or women that make the country beer, and sell the malt, called *jorra*, made of maize. In short, a major-domo of a festival in a village of any consideration gets well off if one hundred and fifty or two or three hundred dollars, pay his share of the anniversary feast and procession. To sustain the pageantry of one day of drunken and profligate religious enthusiasm in honour of a favourite saint, these men foolishly entangle themselves and families in the miseries of debt and embarrassments that destroy both ease and independence, and lead to a multitude of evils naturally arising out of such degraded circumstances. We have ourselves employed in weeding our cane-fields an honest and industrious family thus reduced to great privation; from which the children could never expect to emerge, after the death of an industrious father,

except by the utmost prudence, perseverance, and industry on their own part, and friendly support on the part of their employer.

To conclude these remarks on the Indian's condition, we have shown that evil example in high places, religious abuses, the exactions of the collector of revenue, and also of the priest, (whom, by the by, the state should deliver from this degrading necessity, by giving him an adequate income out of the tithes which it has appropriated to itself,) the arbitrariness of petty governors, alcaldes, and village captains, together with the restless and overweening ambition of military despots that allow their country no repose,—we have shown that these causes, collectively, tend to render the Indian race—which forms the bulk of the Peruvian nation—insecure in their persons and property, distrustful and cringing in their character, degraded in their morals, and heirs direct to civil and religious bondage.

The curates who reside in the mountain glens and deep corries feel assured, from the well-known feelings cherished by their flocks, that when the day arrives when these uneducated men of the hills shall understand what are their own political rights and physical strength, and shall be commanded by bold and sagacious leaders of their own blood and kind, they will fearfully and cruelly avenge their wrongs on all “advenedizos,” all exotics!—on their white oppressors and sable interlopers!^[26]

CHAPTER VI.

War of Independence.—Unsettled state of the country at the close of 1835 and early in 1836.—Gamarra's Government.—Insurrections.—Guerilla and Freebooters.—Foreign Marines.—Lima invaded from the castles of Callao, under command of Solar.—Orbegoso enters Lima.—Castles of Callao taken by assault.—Battle of Socabaya.—Salaverry taken prisoner.—Execution.—Public tranquillity hoped for under the protection of Santa-Cruz.

Having in the preceding chapters attempted to give a correct idea of the general aspect of Peru, and of the social condition of its inhabitants, we will subjoin a brief sketch of the anarchy into which it fell about the close of the year 1835 and beginning of 1836.

From the year 1810, when first the Patriot flag was triumphantly carried into upper Peru by the spirited Buenos-Ayreans, the natives of lower Peru, or that which is now called the Peruvian Republic, had the path to freedom boldly pointed out to their view. But in Lima, where Spanish influence and loyalty were strongly concentrated, it was not until 1819, when Lord Cochrane appeared with a liberating squadron on the shores of Peru, and the Chilean and Buenos-Ayrean forces in the year following landed on the same coast under the command of General San Martin, that the national spirit of the Peruvian people declared itself in that joyful welcome, and effective support of their proposed deliverers and fellow patriots, which struck dismay into the councils, and confusion into the operations of their proud oppressors. Then, indeed, were kindled all the horrors of civil strife and warfare in this once opulent and peaceful country; and these sanguinary struggles never ceased, until, aided by the Colombian troops, and the directing mind of the great Liberator Bolivar, the Peruvians were at length enabled to throw off the chains of despotism which for more than three centuries they had submissively worn,—chains of which they will long bear evident marks on their national and domestic character; for the battle of Ayacucho, which was gloriously fought and won by the Patriots on the 9th day of December 1824, and which terminated the great liberating campaigns of Peru, has not yet secured prosperity to that distracted country. But the Peruvians, having thrown off the tameness of bondage, and assumed the name of freemen, have yet harder work before them than the expulsion of the Spaniards: they have to finish their own work of regeneration; to surmount all the intestine difficulties and reconcile all the discordant elements which originate among themselves; they have to free

their community of noisy demagogues that poison the public press, and discontented agitators who, affecting the purest zeal for the commonwealth, have an eye only to their own interest, and whose object is change—they care not what, so long as it may benefit themselves. How far they are yet from having realized the final advantages they proposed to derive from their great, and, so far, successful revolutionary struggle, is plainly discovered in times of public disturbance, when all classes suffer more or less severely from grinding contributions and wanton exactions.

The wealth of the “*hacendado*,” or landed proprietor, is dissipated in every turmoil; and the less affluent farmer, or “*chacarero*,” is arrested in his labour, and has his arm paralysed by indigence and violence. Predatory troops, as well as government hirelings, seize and drive off his cattle, lead astray his slaves, press or frighten away his free labourers, destroy his crops, and pillage his granaries; and should the spoliated countryman, or country gentleman, be able to rally his spirits and renew his exertions so as to recover the shock of one year’s depredation, the repetition of the like violence, or an aggression yet more destructive, on the following year, consigns him to hopeless beggary. The miners likewise, though a greatly privileged corporation, are for the most part destitute of real capital; yet, in these times, misnamed patriotic, they are a prey to unjust collectors of tribute, who, on fixing any particular miner’s contribution at a certain sum in current money, which he is unable to pay, take care to recover the amount in *piña*. Now, the extortioner asserts that the miner’s *piña*, though truly excellent, is very badly purified,—that it is of “*mucha merma*,” or sustains great loss of weight in fusion; and, under this false pretence, again comes upon the miner, and obliges him to make up the alleged deficiency: this, in fact, is a surplus, over and above the demands of the government, which swaggering colonels, or other such commissioned pilferers, appropriate without scruple to their own unworthy purposes. Such acts of violence and villany in times of petty revolutions lead to the worst consequences; for they not only occasion great private distress, but create such a general distrust in the government and its rapacious agents, as frequently prevents the miners from remitting silver bars to Lima, when otherwise it might be their interest to do so; and this oppression causes a contraband trade, for which, indeed, the open coasts of the country afford all imaginable facility. Nor should it be here overlooked that, as a common consequence of the frequent public broils in this republic, the small merchant or retail dealer

often feigns, on the convenient plea of bad times, an utter incapacity to pay the wholesale foreigner in Lima who credits him with goods. Should a person of this character once get into the interior with a respectable stock on hand, he is sure to play the part of a gentleman in feasting and dancing, &c.; and it may take some trouble not only to get him to render fair accounts, but to hunt him out of the town or village where, under the pretext of transacting business, he is pleased to locate himself. In short, so great is the disorder in every department of the social and political system in Peru, that, to express the sentiments of a friend of ours, and a distinguished Peruvian statesman, "In Peru there cannot for a long time to come be any other than a military government; every state pretends to regulate itself by a moral government; but, as we have little or no morality in our land, the bayonet must inevitably direct us. Here we have no industry; there is not more than one man in ten that labours for his bread: and putting out of the question the 'empleados,' or those who fill public stations under government, and who are supported at the cost of the state, there is not one in thirty of those mannikins who are daily seen loitering about the streets that live by their own proper industry. Give to the Indian, in whose arm rests our physical strength, an idea of his wants; let him know the conveniences of civilized life; in short, enlighten the mass of our people so as to let them understand something at least of the nature and end of government, and then we shall not have daily revolutions. But, situated as we are at present, we have neither capital, industry, nor private security. All is insecure; all is loose and common, unhinged, unprotected, and without order. Good men have nothing to hope for: the few individuals who have access to our rulers are guided by none but the most sordid motives. It is the ruin of my lacerated country, that no man looks beyond his personal interest; that no one attaches himself to the government with sound intentions, or with any view except that of plunder."

The mountain Indian in particular, whose knowledge and usefulness our patriotic friend would fain improve and enlarge, long inured to servitude, and only acquainted with the rudest arts of life, has never arrived at a correct idea of the extent of his privations, or of the nature of those primary political rights which, by skilful combination and promises, (though himself too ignorant to reason on the merits of the cause,) he was at length goaded forward to assert, and for some time to sustain with manly energy. Thus have the meekest and most submissive of men, through vigorous exertions

on the part of the few who conceived and originated the plan of their independence, been stirred up to despise their unjust rulers, and trained to the use of fire-arms, at the very sight of which they formerly trembled. The effect of such education must for some time be productive of disorder; for, to apply a homely illustration, the fire that is lit by the husbandman to destroy overgrown and noxious weeds, cannot always be checked before it scorches the cane or the corn field; and so it is with hostile passions, which when once excited, though for a good and patriotic purpose, cannot always be quelled at pleasure, or at once restrained within the limits of perfect order and rational liberty. Of this the recent history of Peru affords ample evidence, for, since the close of General Gamarra's troubled government, there has scarcely been a lull of temporary peace in that ill-fated country.^[27] It is asserted by those who best knew this influential man's counsels, that, during his four years' rule, he crushed no fewer than fourteen conspiracies more or less matured against his person or government: yet at the expiration of his lawful term of presidentship, to which dignity his artful schemes conducted him on the ruin of his predecessor, the beloved General La Mar, he had scarcely relinquished his high office in January 1834, when he was seen to rear the standard of rebellion, and hasten the downfall of his country by authorizing insurrection with his example. Although frustrated in this shameful revolt, in the year following he was again at the head of an armed faction, and in open and sanguinary rebellion. But finally, after the disastrous battle of Yanacocha, and total dispersion of his surviving partisans, he came to Lima for refuge against the united and victorious force of his legitimate foes, the Peruvian president Orbegoso, with his Bolivian ally, Santa-Cruz. And here in the capital, while receiving the condolence of his mortified friends, and mourning the loss of his heroic wife, the renowned Panchita,^[28] whose heart in his utmost adversity was presented to him by a confidential female friend, enclosed in a glass,—he had little leisure to weep over it. Ere his awakened sorrow was soothed, he was arrested on a rough military warrant, and, in company with several of his friends or adherents, once more hurried away into banishment by the stern orders of the impetuous General Salaverry, a rival leader, less artful and wary, but more active and daring than himself.

Lest anything should be wanting to crown the accumulated miseries of a distracted and afflicted people, his excellency the provisional president of

the republic, Don Jose Louis Orbegoso, in his address to the Peruvians, dated Tarma, January 4th, 1836, and published in the Redactor of Lima on the 9th day of the same month, solemnly affirmed and promulgated that “the very laws, dictated with the pure intention of securing happiness to the commonwealth, had concentrated within themselves the elements of her destruction. These laws had proved a safeguard to the seditious, and had been the bulwarks of rebellion. Through their operation the executive had been forced to feel the volcano at its feet, though unable to prevent an eruption. Yes, under the overseeing eye of the government, the revolutions had been hatched and brought forth, reared and strengthened into maturity.”

This acknowledgment, from a president duly invested with extraordinary or dictatorial powers, renounced every rational idea of government, and virtually declared the incapacity of the supreme authority to protect the person, property, or rights of the citizen, or to sustain the necessary subordination of society. By this government, which so frankly declared its own imbecility, men either faithless or inept were, perhaps for want of better, appointed to fill offices of high trust and power; and in this way was kindled the train of that sanguinary revolution, which, in the year 1835, burst forth like the flaming combustibles and poisonous eruptions of an overwhelming volcano; spreading consternation, outrage, and desolation over the wide range of its fearful sweep. But, during the whole of this tumultuous period, the Limenian mob, made up, though it be, of mixed and most variegated castes, illustrated by their example how slow the mind is to cast off early and deeply rooted habits; for, after the lapse of so many years of civil dissension, they showed that, as a whole, they still retained the feelings of public subjection (unfortunately not turned to account by any steady government) to which, in olden times, they were habituated under the jurisdiction of the Spaniards. For several days during this period there was no sort of police in the capital. The government and garrison had abandoned it, and shut themselves up within the fortress and castles of Callao; but yet the populace showed a singular measure of forbearance, and the instances of outrage and pillage committed in the streets were exceedingly few.

At this conjuncture of danger and uncertainty, foreign property in the capital was guarded by marines, English, French, and American, from their respective vessels of war on the station: but, for several months previously

to these days of general panic and dismay, the capital had been the theatre of daily broils; the banditti and soldiery being engaged in ceaseless though irregular contest for the mastery both within and without the walls. The inhabitants were affected with a sort of nervous infirmity, or morbid susceptibility of impression, proceeding from the unsubdued feeling of impending danger.

E l'aspettar del male è mal peggiore
Forse, che non parrebbe il mal presente:
Pende, ad ogn'aura incerta di romore,
Ogni orecchia sospesa ed ogni mente;
E un confuso bisbiglio entro e di fuore
Trascorre i campi e la città dolente.

TASSO.

A pillar of dust rising in the distance, or the smoke of burning weeds in the neighbouring farms, were sure to be attributed by the anxious spectator in the city to the less harmless fire of musketry and skirmishers. On the appearance of any such sign, notice was immediately given from the lofty steeple of La Merced, or the arcade of the bridge opposite the palace balconies. If a playful black boy was seen to gallop on his donkey by the trees of the old Alameda, or suburbs of Malambo, then some mercachifle or pregonero^[29] would instantly give the alarm, which was conveyed by the vocal brotherhood with the rapidity of lightning—and “Hay viene el negro Escobar y los ladrones!” (Here comes the negro Escobar and the robbers!) was soon ringing through all parts of the city—whereupon in every direction would follow the running tumult of “Cierra puertas!”—shut doors!—and then the creaking and heavy clash of massy doors, and the jarring of chains and bolts, as every street and area entrance were closed and barricaded. During these moments of self-imprisonment, suspense, and anxiety, the streets were entirely abandoned by the unarmed populace; and the noise from the pavement, caused by the gently progressive motion of an ambling hack, was exaggerated in fancy, so as to imitate the clang and tread of a hundred horses. It produced the same startling effect in the over-excited imagination of those within, (who, to see what passed without, hardly ventured to peep through a key-hole, or from the corner of a latticed balcony,) as the unwelcome rattling of a wheeled carriage or the dull Pantheon car, on the morning succeeding a desolating earthquake, never

fails to produce on sensitive frames while under the still abiding influence of recent alarm. Under such circumstances of general consternation it was that the timely arrival of irregular troops, "*montonera*," under the command of a Patriot general, Vidal, delivered Lima out of the hands of a formidable band of freebooters under the celebrated negro Escobar, who had already begun the work of depredation, and whose sanguinary disposition, if excited by drink or excess, threatened to realize the worst anticipations of the dismayed citizens. In this very condition of infuriated exultation and inebriety, being in the act of plundering a house in open day, he was surprised, and in less than an hour afterwards shot in the plaza; where, only the day before, he had showed off very proudly under the balconies of the archbishop's palace, mounted on a magnificent black steed, which he had taken by force from the prelate's own stable. But now in his last moments his only intelligible prayer was said to be that he might receive forgiveness from the archbishop, whose sacred dignity he had so recently insulted; and, probably, of all the unhappy Peruvians who are brought to suffer death at the "banquillo," there falls not one but shows some mysterious respect for the church; and the greatest criminal among them is never, perhaps, entirely forgetful of his tutelar saint. Whatever their career of life may have been, their faith, well or ill founded, yields them hope at the last hour; and it is allowed by those who witness their tragic end, that they generally die the death of the wicked with the composure of martyrs.

On the day that General Vidal, with his orderly *montonera*, entered at the invitation of the municipality—"cabildo,"—for the protection of the terrified city, it was interesting to observe the contrast presented by the negro Cimarones, when arrayed in the cathedral square of the capital by the side of the freemen of Huamantanga, and the poor but independent Indians of Yuyos, who, of all their tribe and fellow aborigines, are the least passive under political oppression. In the laughing negroes, the perpetual motion of their long and dangling limbs, never at rest in the saddle, betokened an exuberance and locomotive waste of nervous energy; while, on the other hand, the contemplative-looking and compact little Indian, mounted on his hardy nag, just emerged from the solitary and rugged wilds of the mountains, though surrounded with the novelty and excitement of a great city in confusion, never for a moment lost the composure and serenity of his countenance and demeanour.^[30] These highland bands, together with a few other brave but undisciplined volunteers, inspired the lower orders of the

Limenians with that transient enthusiasm to which, on extraordinary occasions, they have more than once shown themselves capable of being raised; and simultaneously they rushed to arms as the bells from every spire tolling the solemn “llamada a fuego,” or the alarm of conflagration, summoned them to the defence of their beloved Lima, which was menaced and ultimately attacked by a formidable sortie from the castles of Callao. The assailants were led on by Solar the governor, and cousin to the spurious president, Salaverry, whose illegitimate cause, now on the eve of being lost for ever, his less energetic relative but faintly sustained. It is worthy of remark, that even on this momentous occasion, the spirit-stirring 6th of January 1836, the patrician youth—“los hijos de familia”—took no active part. Educated with the utmost tenderness of indulgence, they are more inclined to love than arms. In short, the business of their life is pleasure.

Until the last memorable rally and sanguinary struggle at Socabaya, near Arequipa, under that Limenian *lusus naturæ*, General Felipe Santiago Salaverry, the military name of the Patriot officers of Peru had been rapidly sinking into utter contempt. By far the greater number of their spirited and intelligent country-women decried the turncoat fraternity, and regretted that they themselves were not born to carry arms, that they might redeem the fallen honour of their country. These degenerate officers seemed to take pleasure in calling every now and then the attention of the public to their vile “pronunciamientos,” or open abjuration of honourable allegiance to those placed in just authority over them. Such vain and faithless vaunters, whose proudest achievements were but to forsake their duty, bind their chiefs, and desolate their native land, became the objects of public scorn, and were despised even by the softer sex, as being fitter to wield the distaff than the sword.

But Salaverry, a man of vast though ill-directed energy and reckless spirit, made the sky re-echo to his shout of war to the death! And such complete ascendancy did he acquire over the minds of his countrymen, by his almost insane impetuosity and appalling executions,^[31] that he not only constrained them to a state of awe and submission, but, what is more remarkable, inspired them, when he pleased, with martial ardour, and made them emulate the deeds of Zepita, Junin, and Ayacucho. During the gloomy reign of the black banner, and continuance of the revolution of Salaverry, the Limenian women, uneasy beneath the accumulating evils of political

oppression, made their way into the ranks of the insurgents. Disguised in their mysterious “mantos,” they circulated patriotic proclamations, and whispered abroad the low and solemn murmur of public opinion; until at length, on the famous 6th of January 1836, when the populace rushed to the walls, it was shouted aloud from every mouth,—ay, the cannon’s mouth,—to the confusion of rapacious upstarts struggling for ascendancy. And still the women played their part—as they raised the whirlwind, so they rode on it; for, without any metaphor, they were to be seen armed and on horseback amidst the crowd.

Two days after this display of popular feeling so unusual in Lima, the provisional president made his entrance into the city amid loud rejoicings that nothing could exceed. A few weeks after this event, the eminently brave General Moran by a gallant assault forced the castles of Callao, then under the command of the insurgent Solar, to capitulate; and, on the 7th of February, General Salaverry lost the hard-contested battle of Socabaya, also called Altos de la Luna, or Heights of the Moon, a name singularly in character with that high and lunatic excitement which hurried to his doom this enthusiastic child of ambition. He escaped from the field of action with many of his officers, and the remainder of his wearied troops; and, when nearly in sight of their shipping at Islay, they were taken prisoners by our countryman, General Miller, under circumstances which demanded on the part of this very distinguished officer the exercise of that active vigilance, coolness, intrepidity, and self-possession, for which he has been so remarkable throughout his honourable military career.

On Thursday, February 18, 1836, General Salaverry, and eight of his principal officers, were by sentence of court-martial condemned to death; and, accordingly, were publicly shot in the great square of Arequipa. This event, though lamented by a few, was matter of rejoicing to the many, who now looked forwards to the re-organization of the political state of Peru under the protection of General Santa Cruz, the President of Bolivia.

CHAPTER VII.

On Climate and Disease.—Panama, Guayaquil, Peru, and Chile.

For those who propose to cross the Isthmus of Panama, or visit the shores of the Pacific, it may be interesting to be made in some degree acquainted with the influence of particular climates, and the sort of illness which they are most likely to experience at the principal commercial ports, particularly to the south of the line. On this account the author now offers some general hints on these subjects, having it in view to publish as a separate treatise a practical account of the diseases of Peru, described as they occur at different altitudes, in the diversified climate of that country.

The seasons at Panama are divided into wet and dry: the rainy season begins towards the latter end of May, and continues till November; and from November to June, or the latter end of May, is the dry season. At Panama, agues, fevers, bilious and gastric complaints are common in the wet season; but the yellow fever, or "*Vomito negro*," very rarely has been known to pass the mountain barriers which separate the Atlantic from the Pacific. At Cruces the traveller may enjoy a better and safer climate, during the wet and unhealthy months, (when the thermometer never, perhaps, falls below 90°,) than either at Panama or Chagres.

To the north of the Isthmus, along the shores of Central America and Mexico, as far at least as the northern tropic, the climate is considered "malsano," or exceedingly unhealthy; a fact well known to those who trade with Realejo, San Blas, and Mazatlan, where very dangerous remittent fevers prevail.

To the southward of the Isthmus on the shores of Colombia, in about 2° south latitude, we find the port and city of Guayaquil, of well-known commercial importance. Here, the climate is considered unhealthy during the wet season, when the air is sultry and oppressive; but in the dry season Guayaquil is not reckoned particularly sickly. The rain commences in light showers in December, is very heavy in February, and dwindles away in April. From May to December is the dry season.

The wet season, being the hottest, would naturally be considered as summer; but here, as in other places of seasonal or periodical rains, the wet season is called "*invierno*," or winter, and the dry season "*verano*," or

summer; yet the latter is cooler than the former, and allows one to wear warmer clothing than would be agreeable in the rainy months.

In the rainy season the thermometer ascends to 90° or 96° Fahrenheit; but, during the dry season, it ranges from 65° to 85°, being 65° at night, and rarely exceeding 80°, though it sometimes reaches 85° during the day. The rain usually falls in the afternoon or night, seldom in the forenoon, when the sun is often so powerful as nearly to dry up the pools and streets before the evening rain comes on again; however, there are days when no rain falls. The houses being covered with tiles, and furnished with arcades, are sufficiently defended against sun and rain. The plain extending between mountain and sea is, for ten or twelve leagues inland, well wooded, and intersected here and there with smaller rivers which the natives call esteros or lakes, in allusion probably to their appearance during the wet season, when, teeming with alligators, they inundate the beautiful meadows round about; so that the term “river” is only applied, by way of distinction, to the great navigable river of the city, which is so influenced by the tide, at least in the dry season, as to be quite briny to the taste. Here the natives bathe all the year round,—a practice, we believe, which conduces not a little to the general health and fair and stately form of the Guayaquilenian ladies, who are said to be fonder of town, and the ease of their hammocks, than of country air and exercise. The streets of Guayaquil, being steeped in rain, become contaminated for want of police; insects swarm on every side, and vegetable and animal emanations pollute the atmosphere: malaria abounds; and fevers, dysenteries, and various gastric disorders attack the inhabitants, and especially the imprudent stranger, who, trusting in his youth and strength, and not considering that difference of climate demands corresponding difference of life, perseveres in the same habits under every parallel of latitude through which he passes from one temperate zone to the other.

In warm and humid situations, such as Guayaquil, surrounded by rivers, stagnant pools, lagoons, and exuberant vegetation, atmospherical heat may operate in causing disease, not merely by promoting the production of miasmata, but also by increasing the irritability of the organs of the body, so as to predispose to severe attacks of illness. The affection of the skin commonly known under the name of “prickly heat” is very likely to arise from profuse perspiration while in Guayaquil; and all excess in the cuticular

secretion should be avoided by every proper means, such as suitable clothing, temperate living, and moderate bodily exertion, &c. The contrary practice, of encouraging sweat by heating drinks, has a bad tendency, both moral and physical:—physically, it produces, sooner or later, gastric and hepatic diseases;—morally, it furnishes a pretext and excuse for deep potations;—and the end of all is a broken down constitution, and a mind impaired in its noblest powers. In another point of view, without supposing that the fevers which on the shores of the Pacific are termed putrid arise from the want of a due quantity of saline ingredients in the blood, it is not improbable that, when perspiration is excessive and too long continued, it may indeed carry off from the circulation more of these saline portions than can be quite compatible with a state of perfect health. We have sometimes observed horses, when hard pressed on a hot day along the sandy plains of Peru, lie down exhausted and overcome by excessive sweat and muscular exertion; and, on being unsaddled and allowed to cool, the poor animals on such occasions would appear as if covered with hoar-frost, from the quantity of saline matter left behind from the fluids perspired and evaporated.

Moderate transpiration, however, is a cooling process, and a necessary one to the natural condition of the system, when the circulation of the blood is much increased, as is the case under high atmospherical temperature, though at the same time muscular vigour usually becomes much diminished under such circumstances. The functions of the stomach often grow languid as the relaxation of the skin has been great and long continued; but, while the appetite is thus diminished, the flow of bile is apt to be increased, and the bowels often become irregular,—sometimes too lax and irritable, at other times torpid and costive.

In one we may observe that, when the bowels are lax from an overflow of bile, the skin is dry, and that for months together; while in another, exposed to the same changes of climate, the skin is always soft, while the secretion from the kidneys is scanty, and the intestines appear to lack their wonted moisture, and become sluggish, as if deprived of their muscular power of healthy action. But it more usually occurs, on being transported from a cold to a warm and humid climate, that a very notable alteration and increase is observed in both secretions—the biliary and cutaneous, of the liver and of the skin. The state of the bowels therefore requires to be attended to very

particularly in all great transitions of climate; because, from undue accumulations in the intestinal passages during warm and sultry weather, irritation and fever may ensue, and a bilious disorder of the bowels, if neglected, or ill-treated, will too readily decline into a fatal dysentery.

Having in the first chapter of the first volume of this work given a sufficiently minute account of the climate of the Peruvian coast, it will now be enough for us to remark that, at its northern extremity, though bordering on the verdant country of the Equatorial Republic, the air of the coast of Peru is less humid than it is at its southern limit, where it joins the desert of Atacama.

The peculiar dryness of the province of Piura is not explained by the fact that in this part of the coast the Andes retire farther inland than in many others; for, from Piura, we have only to pass the river Tumbez, when, as formerly mentioned, the face of nature is quite changed, and the plains of Guayaquil, though at their lower and more maritime parts far distant from the inland piles of mountains, are nevertheless deluged in rain during the wet season; whereas Payta, the sea-port of Piura, has (as we have been informed by a native of those parts, our enlightened and public-spirited friend, Don Santiago Tabara,) not unfrequently, for years in succession,—sometimes as many as ten or twelve years,—not a shower to give life to a single blade of grass.

At Truxillo, again, the capital of a Peruvian province, situated on the coast in lat. $8^{\circ} 8'$ south, the air is much drier than at Lima or Callao in $12^{\circ} 2'$ of south latitude: yet Truxillo is in the vicinity of lofty mountains which run parallel to the coast; and Huanchaco, its sea-port, is situated at the foot of the lofty Bell Mountains. But, to enumerate no more particulars, we think it will be found true as a general proposition that, from the desert of Atacama to the landing-place of Pizarro on the banks of the Tumbez,—from the southern tropic to close upon the line,—there is a progressive diminution of atmospherical humidity.

The difference thus marked in the state of the air appears to influence very materially the character of several diseases, as intermittent fevers or *tercianas*, which on the northern coast of Peru, or what is called *costa de abajo*, and more particularly in the eminently dry province of Piura, are of

milder type than along the shores of the southern and maritime departments of Peru, known under the name of *los intermedios*.

The Indian population of Piura are a hardy and healthy race of people, naturally inclined to corpulency; and, indeed, the Indians of Peru in general are constitutionally disposed to a sleek rotundity of form, which it would only require ease and good generous diet to call into full developement, so as to render the bulk of this race *as fat as Caciques*. Most of the chronic diseases of the Piuranos are said to result from leaving all to nature in the earlier stages of their complaints; and, among these northern provincialists, phthisis, dysentery, tercianas or agues, and typhus mitior,^[32] are endemic. The same sort of complaints, varying however in the intensity of attendant symptoms, are met with all along the maritime valleys of the coast; and in the list of prevalent diseases at Lima and elsewhere, visceral obstructions, intestinal hæmorrhage, disorders of the heart, and asthma, deserve particular notice. There are also a variety of cutaneous eruptions and nervous diseases of frequent occurrence, upon the nature and cure of which it is not at present our purpose to enlarge.

In consumption, which, in all its various forms, is a common disease on the coast of Peru, a portion of the lungs becoming by degrees ulcerated and destroyed, there is consequently an interruption to the proper discharge of the pulmonary functions, accompanied with nocturnal increase of fever and excessive perspiration. But, even in this advanced stage of the disease, changing the air of the coast for that of the mountains or temperate valleys of the Sierra, is found to produce great relief and prolongation of life.

Spitting of blood from the lungs seems in most instances to depend on the presence of tubercular phthisis, or on an inherent constitutional tendency to this disease; and any accidental excitement, as that from cold or undue exposure to atmospherical vicissitudes, may hurry on cases of pulmonary hæmorrhage to a fatal termination. Suckling, in particular, is known to be apt to occasion spitting of blood, which, if not cured in time, usually ends in those symptoms which characterise consumption of the lungs.

It is curious to observe that, in the warm climates of the coast, cold is the exciting cause of most of the diseases which present themselves, such as catarrh, phthisis, bowel complaints, rheumatism, and even the intermittent

and remittent fevers; for we believe that the baneful influence of malaria would not be nearly so often experienced, were its operation in the developement of fever not aided by some check to the perspiration, or what the natives call *resfrio*.

It is a subject of remark on the sugar estates of Cañete, and other parts of the coast, that the slave population, though they work in the humid cane-fields, are yet by no means so liable to ague as either the white man or Indian. One reason for this difference appears to be that the sebaceous glands of the dark races, and especially the negro, keep their skin smooth and soft with a supply of unctuous or oily matter, of rather offensive odour, but admirably fitted to guard against the evil effects of atmospherical vicissitudes.

As black surfaces radiate heat better than those of lighter tints, it might be expected that the body of the negro would be excessively chilled when exposed to the night air: but the negroes of the coast of Peru often sleep in the open air, without interrupting the healthy action of the dermal system. This is a fact which we are disposed to refer to the preservative effect of the unctuous exudation, because all oily matters being bad conductors prevent the excessive radiation of internal, and the too rapid communication of external heat; and therefore, by this natural inunction of the negro and zambo skin, nature provides a remedy against the extremes of cold or heat under ordinary circumstances.

We thus learn that flannels or woollens, being bad conductors, are, when worn next the skin, very valuable as preservatives of an equal circulation, and therefore of general health, particularly for the European of finer and unanointed skin when subjected to the influence of tropical climates.

During the hot months of January and February, on the coast of Peru, the irritability of the whole system is increased, and particularly of the mucous membrane of the alimentary passages; and cholera morbus thus becomes an exceedingly common disease, for which the standard remedy is ice, or iced water.

The privilege of selling ice in the capital of Peru belongs to the government, who usually let it out for a term of years to the highest bidder. The empresario, or lessee, conveys the ice on mules from the nearest snow-clad mountains at the back of Lima; and is bound to have always on hand a

sufficient quantity for the supply of the capital, and be ready to deliver it at all hours of the day and night. In form of frescos, or cooling drinks, every one uses ice in warm summer weather; and it is considered not merely a luxury or a remedy, but a necessary of life, indispensable for the due preservation of the public health.

The facility of procuring ice renders cholera morbus a disease of easy cure, according to the popular practice of the natives. In the first stage of this malady they administer diluents, such as warm water, linseed or mallow water, with or without a little seasoning of cream of tartar or tamarinds; and these simple drinks they continue to give until they consider that the patient has vomited and voided enough, that is, until all undigested matters be thrown off, and the bowels well unloaded; and then they administer iced water, which produces a powerfully sedative effect.

The death-like coldness of the patient deters neither the vulgar nor the regular practitioner (who sometimes conjoins opiates and iced drinks) from giving this remedy with confidence; and the general consequence of the seasonable use of ice and iced water in this fearful disorder is, that the stage of external coldness is shortened by the early removal of internal heat; and thus the exhausting career of the disease is quickly arrested.

Under this vulgar but satisfactory and long-established treatment of cholera morbus in Lima, where the disease is endemic, though more prevalent in the hot months, vomiting, hiccup, and cramps disappear; reaction is so mild and favourable as never to require the lancet: yet recovery is almost always certain, though cases appear from time to time so intense as to assume the aspect of what is called Asiatic cholera, during which, as a native physician expresses it, the patient is a horrid image of death.

At Ica and various other points to the southward, where vineyards abound, it is observed in vintage-time that to eat freely of the grape on an empty stomach, or without eating bread with the fruit, is one of the most frequent causes of dysentery, which disease is more appalling and fatal on the shores of the Pacific than the cholera morbus to which we have just alluded. It is, however, gratifying to know that, in the form of dysentery which commonly prevails, the calomel and opium plan of treatment, when

discreetly conducted, is assuredly the safest and best yet adopted, whether in Lima or in the interior of the country.

Moquegua, which lies a considerable way inwards towards the mountains behind the sea-port of Ilo, is not less famous for its wine and its grapes, than for its dysenteries and violent agues; but Tacna, on the other hand, about seven leagues inward from the port of Arica, is so healthy as to be a place of resort to the people of the port during the *terciana*, or aguish season, which, over all the coast, is about the vernal and more particularly the autumnal equinox.

The salubrity of climate for which Tacna is distinguished is considered to be partly owing to its vicinity to the cold of the mountains, (for the snowy pass of the Cordillera, which leads to upper Peru, is within four hours' ride of this town,) and still more to a fine dry plain between it and the sea, which only wants water to become rich in agricultural produce.^[33] But in its present state it is free from that malaria which the humidity attendant on irrigation would not fail to engender here as well as in other parts of the coast. In its environs cotton grows spontaneously; and the native women collect it, and make thread from it by means of the *spindle*, just as we have often seen done in some of the warmer inland valleys, where the cotton is indigenous. It is a fact, not perhaps undeserving of notice, that a bud cut off a cotton-tree in the neighbourhood of Arica or Tacna was hung up in the cabin of an English merchant-ship, preserved its vitality in the navigation round Cape Horn, and opened when about half-way between Peru and England.

The whole coast of upper Peru—now called Bolivia—is arid and desert; so much so, that the celebrated president Santa Cruz—who, much to the prejudice of Arica,^[34] made Cobija a free port for the introduction of merchandise,—found that he could not, by sinking pits in the deep sands of Cobija, come at a supply of good water.^[35] For want of water and lucern, mules from the interior of Bolivia often die at the sea-port of Cobija; for there is no vegetation within a great distance from this place. The little water that is obtained at Cobija is brackish, like that in the pozo or well of the great castle at Callao,^[36] which has invariably been observed to give disorder of the bowels to the soldiers, who, during the sieges which that fortress has sustained, were obliged to drink of it. The same has been

observed at Cobija, and therefore there are boats kept there for the purpose of conveying water to it from Paquiza and other distant parts, which makes it an expensive necessary of life.

On several parts of the coast of Peru, water, even for domestic uses, is very scarce; and in the dry season wells are often dug in the beds of dried-up rivers, or in other places in the neighbourhood of irrigated lands. At Port Bermejo and Casma, between nine and ten degrees of south latitude, we are told by the Spanish coasting pilots that, dig where you will, at ten or twelve paces from the sea, you are sure of finding water at the depth of half a fathom that is not very brackish. Wells or pits, however, thus opened in different parts of the coast, are often found to dry up as they do in Lima (where they are common enough) during the dry season, which is the time when they are most required.

In northern Peru the practice of digging pits for water in the beds of rivers is very common; and such is the scarcity of fresh water at the sea-port of Payta, that it is carried to the city on mules, from the distance of several leagues. But on the contrary, at the sea-port of Arica, in southern Peru, good water is found wherever a pit is dug for it; and within two leagues of this port is the fine vale of Asapa, abounding in vines, olives, lucern, corn, &c. and affording a more convenient and copious supply of fresh provisions for shipping than either Payta or Cobija. These facts are of value not only in an economical but medical point of view; since on the quality of the water, as well as of the condition of the atmosphere, in any particular situation, must greatly depend the health of its inhabitants. Thus, in Arequipa, of which Quilca was the old, and Islay the present sea-port, the river-water is said to contain some salts in solution, which render it unwholesome until it is boiled; and this is known to be one of the causes of dysentery, which is a prevalent disease in that city.

The peasantry, who travel with asses between Bolivia and Chile across the deserts of Atacama, pitch their tents by day, to avoid the extreme heat of the sun reflected from the burning sand, and proceed on their journey by night; carrying with them all the water and provisions necessary for the journey. And it may be remarked that the soldiers, sent by order of General Salaverry to invade Cobija, had to march from the landing-place at Iquique over desert sands like these, when under their gallant leader Quirroga they took by surprise the port of Bolivia. These coast marches usually fall to the

lot of the Indian infantry, and these hardy natives of the mountains generally prefer performing them between sun-down and sun-rise; for not being constituted, like the sable races, to live in very warm climates, they are more liable to fever when posting over sandy plains during the noon-tide heat; and, if they do but meet with musk or water-melons on their way, they devour them so greedily that they are sure to fall victims to sundry disorders—as intermittents, remittents, dysentery, &c.

In Chile, Nature puts on a different appearance from what she wears in Peru and Bolivia; there, however, as in these countries, the year is divided into wet and dry—the winter and summer. But in Chile it rains, as in Colombia and the Equatorial Republic, at the same season on the mountains and coast; in which respect it differs altogether from Peru and Bolivia. In the southern extremity of this republic, at about 40° of south latitude, the rains are heavier, and of longer continuance, than in the northernmost part, where it joins the great desert of Atacama. On the coast of Chile very severe gales are experienced, when the coast of Peru is only refreshed by light and gentle breezes.

During summer, the sun at noon is felt very powerfully at the capital of Chile, and it is requisite to guard against the risks of insolation; just as happens in Lima in the month of May, when the mornings and evenings are cool and cloudy, but the mid-day so excessively hot that it has become proverbial, and children and others are at this season warned by the older and more wary Limenians to keep out of the sun in these words, “Quitese de este sol que madura duraznos!”—Get out of this sun, hot enough to ripen peaches! an expression probably used in reference to the mode of ripening fruit of various sorts in Lima, by having it stoved. We understand that this is done chiefly to prevent the birds from eating the fruit, which they would not fail to do if it were left to ripen naturally on the tree. The cheremoya is the fruit most commonly stoved.

In July and August the snow sometimes falls around Santiago, when the native of Lima who visits this Chilean capital is peculiarly struck with the novel appearance of the orange-trees in the “*patios*” or court-yards of the houses, bending under the double weight of fruit and congealed snow; the green leaves forming a remarkable contrast with the sparkling crystals, like the jewel garden of the Incas.^[37] It rarely snows in the valleys; but in the winter of 1834, as we were told, a postman and his horse perished in the

snow on the road between Santiago and its sea-port Valparaiso, ^[38] where in the months of June, July, August, and September, it rains a great deal. But during the dry season, though sometimes foggy in the morning, the sky upon the whole is clear, and the climate healthy and agreeable.

In giving an account of the climate and progress of vegetation on the coast of the middle provinces of Chile, it is stated, on good authority, that “the rainy season, as already mentioned, begins in May, and continues to October; the heaviest rains are in June and July. After a few days of rain, there is an interval of fine weather for at least one or two weeks; and the quantity that falls during the season is small, varying from twelve to sixteen inches. In summer the atmosphere is excessively arid, and there is little or no dew. The temperature at noon, in the middle of the rainy season, is generally about 60°; at night, seldom under 40°, though there is occasionally a little frost. In summer the thermometer at noon stands between 70° and 75°; but, during the night, in clear weather, it frequently falls more than 20°.

“During the latter part of summer, vegetation is almost dormant, and scarcely a plant of any kind is to be seen in flower; but, in a very few weeks after the first rains, every part of the country is clothed with verdure.

“In the south of Chile the heavy rains render the road almost impassable; and, as vegetation does not advance so rapidly there as in the north, he” (*the naturalist*) “can botanise in October, November, December, and January.”^[39]

The following observations on atmospherical vicissitudes and miasmatic matter, with the rationale of their effects in the production of disease among the inhabitants of Santiago of Chile, we have pleasure in being able to offer in the form of a translation from an essay in Spanish, ^[40] published in the year 1828, by Doctor William C. Blest; upon whom, though an Englishman, the Government of Chile conferred the highest professional honours, by nominating him to the protomedical chair, which he fills with credit in that republic.

Dr. Blest, in endeavouring to rouse the attention of those functionaries who preside over the destinies of the republic, to the neglected state of its municipal police, says of Santiago that “The streets, with a few exceptions, have either very bad pavement, or none at all. The canals or water-courses,

(las acequias,) which, without doubt, were originally intended to refresh and purify the city, are at present receptacles of every sort of nuisance; and, not having free exit, they terminate in stagnant pools around the city, which are so many laboratories of putrefaction. The cross streets are left in so shameful a state of neglect, that it is impossible to pass along their narrow foot-paths without being shocked at every step.

“The suburbs, where the poorer and more numerous class of the community reside, are so full of dirt and mud, that even on horseback it is difficult to pass through them. In almost every street there are small and confined apartments, without air or light, except that which enters at the door, and these are occupied by whole families of artisans; so that it is not uncommon to see seven or eight persons crowded together in one wretched abode, where dogs and cats add to the nuisance, and still further crowd the family group.

“Such is a true picture of the police of Santiago; and, to convince the curious reader of its accuracy, we need only refer to the aqueducts which pass through the streets and houses,—to the heaps of putrid matter in the cross streets,—to the deep deposits of mire and marshes,—and to the crowded and unventilated dwellings of the poor and labouring classes....

“It is too well and generally known that at all seasons, and for days together in every week, the aqueducts which pass through the houses are so completely blocked up with the quantity of vegetable matter and dead animals collected in them, that they cannot transmit even the smallest stream of water. The subordinate or cross streets, and many of the principal ones, are not less filthy; and any stranger who visits Santiago will be inclined to believe that, of all the towns of South America, it is the dirtiest.

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“Sad experience, and especially in recent times, has taught, that during the decomposition of organized matter, whether animal or vegetable, under the action of heat and moisture, certain exhalations take place, which possess properties in the highest degree injurious to the health of man. This is a truth which is attested by a multitude of medical authors. It is the good fortune of the inhabitants of Santiago, that the atmosphere in which they breathe does not so readily absorb, or act upon substances undergoing the process of putrefaction, as to engender those dreadful epidemics, which

have carried off millions of lives, and are still reaping their harvest of mortality in various parts of Spain, North America, India, Mexico, Panama, Vera Cruz, ^[42] and many other regions of the Old and New World.

“Were they not thus favoured by the natural salubrity of their atmosphere, the church-bells of the Chilean capital would be daily heard to toll the mournful knell of death, and every home would present the tearful scene of grief and lamentation.

“But although in Santiago the action of the atmospheric air on substances in a state of putrefaction is not so active as to produce such epidemics as those alluded to, yet it cannot be denied that it is capable of acquiring such properties as make it exercise a most baneful influence on the public health: inducing attacks of dysentery, typhus and other fevers, which, from time to time, appear epidemically. In truth, to some general cause of this nature we must attribute those violent and fatal dysenteries which were so very prevalent in the year 1826, and which recurred in the months of March and April of the present year (1828). To a similar cause must be referred that vexatious sort of puerperal fever which in the year 1827 attacked such a number of women; and also those cases of typhus or *chabalongo* which abound, with few exceptions, every year.

“Reasoning on the generally received principle that air at a high temperature occasions a greater degree of exhalation from bodies than cool air does, and from what we know the influence of summer heat to be in other countries, we should suppose that diseases caused by miasmatic matter should be here more common in summer than in winter; but our acquaintance with this climate induces us to think differently on this subject.

“Here, in summer, the air of the atmosphere is uniformly clear and cloudless; and the emanations from the earth’s surface, meeting neither clouds nor mists to impede their ascent, mingle with the other atmospherical ingredients, and diffuse themselves freely through the regions of space. The opposite of this takes place in winter. The heat of the sun is always very considerable, or at least sufficient to disengage from heaps of nastiness and rubbish the noxious vapours which putrefaction has generated in them. At sunset these vapours come in contact with the clouds that gather around us, and soon meet the cold air of the approaching night;

the consequence of which is, that they are precipitated into the lower strata of the atmosphere, and wafted on the nocturnal breeze into the interior of our habitations. Thus a satisfactory and rational explanation is given, why there should be more sickness in winter than in summer; and by associating this view of the matter with the bad ventilation in the houses of the poor, who from their inability to provide themselves with fuel^[43] and warm clothing, are compelled to exclude the free admission of air, we may perceive the reason why at this season the poor are more obnoxious to disease than those whose pecuniary circumstances enable them to protect their homes from the severities of winter by better means than the utter exclusion of the air.

“The generality of persons, overlooking the course of atmospherical changes, imagine that the diminution in the number of cases of sporadic fever observable in summer is owing to the abundant consumption of the fruits of this season. We will not deny that the use of fruit may improve to a considerable extent the health of those persons who in winter and spring have been nourished with strong and stimulating aliment, such as is calculated to disorder the digestive functions; but we are far from thinking that to the use of fruits alone are to be assigned all the good effects which the vulgar fancy they derive from them. We know that in other countries not less bounteously supplied with fruit than Chile, though not favoured with so benign a climate, epidemic diseases prevail more in summer than in winter. For these, and many other reasons which it would be superfluous to detail, we consider ourselves authorized to dissent from public opinion on this subject, and to assign the diminution in the diseases of the character alluded to, during the summer months, to causes more in conformity with medical philosophy: namely, the benign state of the atmosphere in summer; the bodily exercise which the different classes of the community indulge in every summer evening; and the wholesome ventilation which they enjoy at this season, their doors being constantly open, and many of them choosing to sleep even in the open air.”

APPENDIX

ON THE ZOOLOGY OF WESTERN PERU.

Under the head of the zoology of western Peru we beg leave to present to the reader a translation of a chapter on the influence of climate on animals, especially on domestic animals, taken from the work of the late Dr. Don Hipolito Unanue, entitled “Observaciones sobre el Clima de Lima.”

Dr. Unanue was an ornament to society, and honoured by his country, as well under the Spanish as the Patriot government: by the former, as the reader already knows, he was appointed chief of the medical tribunal of Peru; by the latter, president of that republic.

TRANSLATION.

The horrid picture of America which has been drawn by some ultramarine philosophers does not apply to Peru, and can only be viewed as the production of their own excited imaginations. Where else, indeed, could they have found those dark and unfriendly dies which enabled them to depict these happy regions under a repulsive aspect?—as so many dark spots in the creation denied the blessing of Providence,—as the dismal abode of serpents, crocodiles, and venomous monsters. [\[44\]](#)

Quale portentum neque militaris
Daunia in latis alit esculetis;
Nec Jubæ tellus generat, leonum
Arida nutrix.

HOR.

The learned Count Buffon laid down and wished to establish the four following propositions: 1st, that of the animals common to the old and new continents, the breed is larger in the former than in the latter; 2d, that the animals indigenous to the new are less than those of the old hemisphere; 3d,

that the species of domestic animals which have been transplanted from Europe, have degenerated in America; 4th, that this part of the earth furnishes but few races of animals peculiar to itself. But the inaccuracy of these propositions has been demonstrated by the illustrious President Jefferson^[45] in comparative tables of the existing animals on both continents. This, however, seems to be certain, that, as animals depend for their support on the productions of the vegetable kingdom, their number and growth will be in proportion to the luxuriance of vegetation; and therefore, there being in the one hemisphere as well as in the other very extensive plains covered with abundant pastures, and likewise poor and sterile regions that yield little or no nutriment, either hemisphere will exceed the other in the size, number, and beauty of its animals, wherever it happens to exceed the other in the fertility and extent of its woods and meadows.

Peru is by no means well fitted to maintain the numerous species of indigenous animals which inhabit the forests of North America, nor calculated for the multiplication of those which may be transplanted from Europe, to that prodigious extent which is observed on the broad plains and exuberant pastures of Chile and Parana. Still, however, this country, in its coast, its mountain ranges, and Montaña, comprises upon the whole a vast and beautiful variety, with which the pages of natural history will be one day enriched; but it is only our object at present to offer observations on some of the most remarkable of these, as they appear to be influenced by our climate.

INDIGENOUS QUADRUPEDS.

Of the families of quadrupeds found in Peru at the period of its discovery and conquest, the following are the chief.

Paco.^[46]—*Camelus Peruvianus*, Linn. Syst. Nat.;—Molina, Histor. de Chile, Part I.

Alco.—*Canis Americanus*, Linn.; *Kiltho*,—*Thegua*, Mol.

Puma.—*Felis puma*, Linn.; *Pagi*, Mol.

Uturuncu.—*Felis onsa*, Linn.; *Felis gigna*, Molin.—is found in the west of Peru; and by the same name is designated the yaguar of Azara, plate IX. which inhabits the woods and thickets on the eastern side of the country.

Ucumari.—*Ursus Americanus*, Linn.

Tarúca.—*Elaphus*, Linn. ast corpore minor.

Providence, which has everywhere supplied its rational creatures with means of maintenance, and of executing the labours to which they are by nature destined, conferred on the native of the Andes an inestimable boon in the *paco*; by whose wool he is clothed, by whose flesh he is nourished. The fleet huanaco and the timid vicuña afford him amusement and pastime in the chase; and the llama and alpaca convey his goods with safety through the rough and narrow pathways of his native mountains. The long and upraised neck of these animals, their full and expressive eyes, the *urcu*, or tuft which adorns their foreheads, and the dignified air with which they look around them, as with composed and solemn step they march along in a right line like disciplined troops, form altogether a picture of such peculiar and striking beauty as must always be admired, and never can be forgotten.

The *alco* is the most faithful companion of the Indian: it is of middle size; and its body is commonly covered with black wool, all except at the breast and tail, where it is grey. These dogs are endowed with singularly acute powers of perception, their bark never fails to give notice of anything new that happens about the hut or dwelling, and they also attack strangers with great ferocity. Of this race there is a small breed like our lapdogs, which the Indian women carry on their *quipes*,^[47] and cherish in their bosoms; and, as these pet-dogs are taciturn, this peculiarity has made some persons suppose that the alcas do not bark, and therefore belong not to the dog species.

The pacos and alcas inhabit the Sierra or highlands. Such of them as are domesticated descend with their masters to the coast, where they stay but a short time, and then return; for in the heat of the coast they soon fall victims to the *caracha* or itch,—the consequence of increased excitement and circulation on the surface of the body, and a want of perspiration occasioned by the thickness of the skin. Not more remarkable for their beauty are the eyes of the llamas and tarúcas in the Sierra, than are the aborigines of the same mountains for the smallness of their eyes and their inclination

outwards towards the external angle: an useful structure, for it adjusts their sight to their situation, and, by giving them a side-long view of objects, often prevents them from falling over precipices when crossing the wild passes of the mountains. The same peculiarity of structure also defends them from the bad effects of the sun's reflection from the snow, which in white people and the natives of the coast, whose eyes are full and large, induces *zurumpe*, which is a troublesome ophthalmia.

The *tarúca* or deer, and the *puma* or lion, being fitted to endure the temperature of the Sierra and of the coast, pass backwards and forwards from one climate to the other: the deer go about in flocks; but the little lions wander in solitude, apart from others of their kind. The deer are of a middle size, and have pretty horns.^[48] They are fleet, and afford amusement to those who are fond of coursing. The *otorunco* or tiger, and the *ucumari* or bear, do not frequent this side of the Andes, but they inhabit the regions eastward of these mountains, where there are many other animals of prey.

FOREIGN QUADRUPEDS.

The sheep transported from Europe have increased to an amazing degree on the great commons or pastures of *ichu*^[49] which abound in the heights of the Andes; and in the wide-spreading provinces of Collao the quality of the wool is particularly superior.

In the high and Cordillera ranges, horses, asses, and black cattle are, like man, of small size, because their growth is stunted by cold: they are covered with hair which has the softness, length, and consistence of wool; by means of which Nature protects them against the inclemency of those bleak and frozen wastes, even as she does the flowers of such shrubs^[50] as grow on the same frigid heights.

On the other hand, in the valleys and on the coast, where the heat is sufficient to enable the various members of the body to develop themselves freely, those very quadrupeds are of good size, spirited, and showy. The donkey^[51] is strong, and in Lima the most serviceable of the domestic animals; as he is also at the sugar-works, where he carries on his back a great weight of cane from the field to the mill: the horse is graceful and spirited: the bulls are powerful; and in the valleys of Chincha and

Cañete, where a certain wild breed are carefully reared for the bull-ring, these animals are most ferocious.

The black cattle of the Sierra do not endure the climate of the coast: immediately that they descend from their native mountains, to use the vulgar expression, they become *touched*; that is, they become stupified, and die with amazing rapidity. On examining the entrails of cattle thus cut off, the liver, which has a broiled appearance, is observed to be indurated. I conceive that these animals are affected by transition of climate in the same manner as the human species; for, as soon as bullocks from the high and cold regions of the Andes arrive on the warm coast, the circulation of their blood is unusually accelerated and directed to the surface; but, as the skin which covers them is too thick and unyielding to allow of proper transpiration, the consequence is that there arises an ardent fever which destroys them. In beeves this fever is more violent and burning than it is in the paco or alco, because the skin of the latter, being of thinner texture than that of oxen, offers less resistance to the outlet of the humours; so that in the animals of finer skin there comes out a salutary eruption, which saves them; while in black cattle nothing of this sort occurs, and therefore they perish with incredible celerity.^[52]

The butchers have not yet found out a remedy for this disorder. They only know from experience that the mortality among the cattle is greater in summer than in winter: a fact confirmatory of our conjectures as to the nature of the distemper: and therefore it is during the winter, or misty season on the coast, that cattle are driven down from the mountains to supply the Lima market.

Should we compare the dogs reared in this city with those allowed equal freedom in the cities of Upper Peru, it will be found that the former are most indolent and indifferent to everything, so that any one, though an entire stranger, may step over them without the least molestation; but the latter surly curs it is necessary to approach with caution, because they attack all persons with whom they are not well acquainted and on friendly terms.

These animals are subject, especially in spring, to catarrhal epidemics which are peculiar to themselves; and they are also liable to influenzas by which mankind are affected, it being among them that the fatal epidemic commenced in the Trojan army.

Neither in Peru, nor in the neighbouring sections of South America, were dogs ever known to be attacked by hydrophobia prior to 1803; but about this time the malady broke out, during the heat of summer, in the valleys of the northern coast, from whence it extended southward along the maritime plains; having arrived at the city of Arequipa in the spring of 1807, while in Lima it was observed between the summer and autumn of the same year.

Having collected all the necessary data for disclosing the origin of this disorder, and consulted in writing the physicians and well-informed persons who had witnessed its symptoms, I have clearly learned,—1st. That this disease arose spontaneously from the increased atmospherical temperature of the years 1803 and 1804. It commenced on the northern coast, commonly called *Costa Abajo*, where the air was so heated that Reaumur's thermometer indicated the temperature of 30° in some of the valleys: the calms were extreme, without the lightest breeze that could ripple the surface of the ocean; animals rushed into lakes and pools of still water to relieve themselves from the sensation of excessive heat; so that the season described by Horace was fully realized:

Jam Procyon furit,
Et stella vesani Leonis:
... caretque
Ripa taciturna ventis.

2. This disorder affected every sort of quadruped without distinction: and such was the degree of phrensy excited by it, that some animals in their fury bit and tore themselves to pieces; and, in situations where the heat was extreme, several men fell ill with all the symptoms of hydrophobia without having been bit.

3. The malady attached itself more especially to dogs, and some of them suffered so mild an attack that their bite was not mortal; but the greater number were severely affected, and propagated the infection to their kind, to other quadrupeds, and to man.

The mean and niggardly overseer of a sugar-estate had distributed among his negroes, though advised not to do so, some head of cattle that died rabid; which he did under the impression that they were only *tocado*, or touched with that disease which in hot weather usually affects cattle from

the mountains: and the result was, that of the poor negroes who had partaken of this meat, many died with symptoms of hydrophobia.

4. In the towns of Ica and Arequipa the number of individuals who died, after having been bit by mad dogs, was greater, and their cases less equivocal than the preceding.

In Ica a single rabid bitch bit fourteen persons in one night, of whom eight were in one house; some sleeping *al fresco*, or in the open air; others were variously occupied; and the remaining six were among those who, on hearing the alarm, ran to assist in killing the bitch. The surgeon of the place, Don Mariano Estrada, wished to persuade them to submit to be cured; but they rejected his proposal, saying the will of God should be done; and all died with the exception of two men, the one twenty-eight and the other fifty years of age, who agreed to be placed under medical treatment. The physician cured them, happily, on the safest plan; which consists in applying a blister on the part bitten, with a view to promote suppuration from it, and in exciting salivation by means of mercurial inunction.

In the city of Arequipa it was much disputed whether or not the malady was a legitimate hydrophobia, and very learned papers *pro* and *con* were written by the Doctors Rosas and Salvani. In this paper-war much time was lost that should have been taken advantage of for resisting the progress of the malady. True it is, that in many cases those disorders, which by frightened imaginations were represented to be real examples of hydrophobia, were, in point of fact, no such thing; and the alarming misconceptions thus induced were soothed down and removed by persuasive means. Hence, this circumstance, which was the natural consequence of the general panic existing at the time, led Professor Salvani to think that it was precisely the same in all instances, until at length a succession of melancholy results declared the real nature of the disease. Immediately upon being made acquainted that the epidemic hydrophobia approached the capital, the Viceroy of Peru, Abascal, ordered all the dogs in the place to be killed, ^[53] by means of which he liberated Lima from the impending scourge; for though a very few hydrophobic patients entered, during this period, into the hospitals, they were not inhabitants of the city, but some individuals who had come in from the neighbouring farms and valleys.

5. When this calamitous epidemic commenced in the valleys of *Costa Abajo*, Don Jose Figueroa, Bachelor of Arts, wrote me to say, that “the dogs went about with their tails between their feet; they slavered much; hid themselves from human sight; howled lustily; and presently they fell down and moved no more:—as remedies in these cases, cutting off the ears and giving oil were tried in vain. The cats, with their hair on end, ran about the house-tops. Horses and asses got enraged the one against the other; they threw themselves on the ground, rolled about, and instantly on being dead they swelled and putrefied. Black cattle—roaring and lowing—bounded about, fought with each other, in the contest even broke their horns, and they died quickly.”

6. Professor Estrada confidently stated, that of forty-two individuals who died in the city of Ica, after having been bit by mad dogs, the greater number were cut off from twelve to ninety days after the accident. The symptoms which followed the ingraftment of the poison disclosed themselves in the form of convulsions, oppression at the breast, sighs, sadness, laborious breathing, horror at liquids and shining objects, fury, vomiting of dark bilious matter, and an incessant urgent call on the part of the patients that the assistants should depart from them, because they felt themselves impelled to attack, bite, and tear them to pieces: none in this state survived beyond the term of five days.

Since the year 1808 this terrible epidemic has been disappearing. From time to time, however, a dog may be seen running violently hither and thither, and biting all whom he may happen to meet, in the same way as is done by the really mad dog.—But, in the examples wherein no bad results arise from the bite, they may be considered of the same character with the disorder observed by Mr. Colombier, which attacks dogs, renders them furious, and excites them to bite, but has, nevertheless, nothing at all of hydrophobia in it;—still, however, the safest way is to kill the dogs thus affected, and to implore the Father of mercies that these regions may never again experience so severe a visitation.

Canis ore timendo,
Ore vomit flammam.

GERMAN ARAT.

BIRDS.

The shores of the South Sea are covered with myriads of birds, among which are distinguished, for their incalculable number, the Huanaes; from whose ordure, as some believe, is produced that red-coloured earth or manure (huano^[54]), of a penetrating and alkaline smell, which enriches the land so much as to make it yield triple or quadruple the produce it could do without this dressing: a discovery made by the ancient Indians, who were most skilful agriculturists.

Gulls, herons, ducks, and some other families of the feathered class, descend during autumn from the mountain lakes to the coast, where they remain until the commencement of summer, when they again return to the Sierra.

In undertaking this journey, they take their flight in the morning in large flocks; and, as they soon come in contact with lofty barriers of mountains which oblige them to change their course, they ascend the higher regions in a winding and spiral manner, till, after numberless evolutions and gyrations, they have risen above the loftiest peaks of the Cordillera, and find themselves again at liberty to pursue their journey in a direct line.

The condor^[55] often stations himself in the middle of the spires, ^[56] either as acting the part of a guide, or to boast in proud display the strength of wing by which the most vigorous and powerful of birds can soar above all the rest of the feathered race.

In his outward aspect the male bird bears upon him many marks of dignity which distinguish him from the female: such is the crest which serves him as the emblem of monarchy,—a crown; the blackish and loose skin which in folds covers the head, and gathering up behind, after the manner of curls or frizzled hair, resembles a wig; and the white of the wings, when the bird stands erect, gives his shoulders the appearance of being covered with a mantle or cloak.—See the excellent Memoir of Messrs. Humboldt and Bonpland on the Natural History of the Condor, printed in Paris in the year 1807.

Santiago Cardenas, better known by the name of Santiago *el Volador*, or Flier, for many years watched the flight of the condor, with the intention of imitating him; and he left a quarto volume written on this subject, which I

have deposited in the library of the College of San Fernando. In this work he describes three different kinds of condor.

1. Moromoro with ruff (golilla) and mantle of the colour of ashes.—It has of “*embregadura*,” or length, from the point of one wing to that of the other, from thirteen to fifteen feet.

This, of all the condors, is the strongest; and he takes ostentatious delight in combating against the wind, and balancing himself on extended but flutterless wing in the most imposing and majestic manner. It has been said of the moromoro, that, seizing the newly-born lamb, he throws it over his shoulder, where he keeps it steadily fixed; and, having thus secured his prey, rises on the wing, and betakes himself to flight. [\[57\]](#)

2. Condor of ruff and mantle of the colour of clear coffee.—He has of *embregadura* from eleven to thirteen feet, and he is swift and daring.

3. Condor with white mantle and ruff.—He has expanse of wing, or *embregadura*, from nine to eleven feet; and this is the most abundant and beautiful species. The condor inhabits the steep rocks of the Andes; and, according to the observations of Santiago, he makes every day two journeys to the coast in search of food, which shows his prodigious velocity.

In our dissection of this bird, we met with no air-vessel which could maintain a communication between the lungs and the spongy substance of the clavicles, nor any communication between the crop and windpipe. The internal cavity of the chest is lined by a fine and transparent membrane or pleura, which forms various little cells; the lungs descend as low as the abdomen, and adhere, at their posterior extremity, to the spine and ribs, which have perforations at the points of adhesion, communicating with the interior of their spongy body. The texture of the lungs is porous, so that, as soon as they are blown into and inflated through the windpipe, they freely supply with air all the recesses or concealed crevices, great and small, that are about them; and they also fill with air the cavities of the ribs and sternum.

Condor-grease is considered excellent for resolving and dissipating hard glandular tumours of the breast and other parts of the body; and the Peruvians attribute to it as many other virtues as the Europeans do to the

kid, of which it is said by one of their physicians that *totus est medicamentosus*—all is medicinal.

INSECTS.

It is well known that warm and humid countries are infested with swarms of small insects, as flies, mosquitos, zancudos, fleas, &c.; and a certain traveller has asserted that, on this account, Lima was insupportable as a place of residence: but the statement is erroneous, for such insects do not flourish in the midst of population and cleanliness.

Notwithstanding the mildness of winter in Lima, it is sufficient to annihilate the flies and zancudos: mosquitos are not within doors at any season of the year. The flies and zancudos multiply in summer; and the latter are very annoying, especially at night, for they prevent sleep by the buzzing of their wings. But by taking care that no water be left in the house till it become nearly putrid, this little insect will not be allowed to grow troublesome; for the zancudos are the offspring of the animalcules which are produced in water tending to a state of decomposition: neither do the flies prove troublesome by their numbers in the houses where cleanliness is not neglected.

The pediculus may be said to be sterile on the coast, but most highly prolific in the Sierra: insects of this and the cimex kind persecute man wherever he sojourns, and Lima does not appear to be more infested or overstocked with such vermin than certain European cities. In Paris alone there are seventy-seven species of the cimex. [\[58\]](#)

The most intrusive, the most vexatious insect of the torrid zone, is the *pique* or *chigre*, which in other parts is known by the name of *nigua*. Uncleanly in the extreme, it searches the *corrales*, or pens, where pigs are enclosed, and multiplies infinitely in dirty situations. The heaps of rubbish, or sweepings and refuse from streets and houses, &c. are, as it were, in a state of effervescence with *piques*, which also follow the footsteps of man, pursuing relentlessly those with overgrown nails, and others who neglect cleanliness. Less than the flea, but of the same colour, it contrives to introduce itself inside shoes and stockings, and to lodge in the tenderest part of the foot,—in the sole, or under the nails: there it fixes itself, causing as

much pain as would be occasioned by the point of a needle, and it secures its position so well as to render it very difficult of being detached. In attempting to remove it, the soft parts are often ill-treated by the instrument, which is either a needle or pin, commonly used for its extraction; and when, during the operation, the part acted upon becomes tinged with blood, the end of the matter is, that the *pique*, instead of being removed entire, is lacerated, and, the one-half only being taken away, the other is still left inserted under the skin, and there occasions more pain than at first. For this reason, those persons who are accustomed to *piques* keep very quiet when they observe that one of them has fixed itself under the epidermis or outer skin, and leave it undisturbed for a day or two: here it forms its nest; and is gradually metamorphosed into a white globe, of the appearance of a moderately-sized pearl. It holds on fast to the skin, by its mouth, at the point where it first adhered. Having attained maturity, it is in fact nothing else than a group of innumerable little eggs united by a white glutinous matter, and covered by a common envelope which encloses the whole. While growing, the *pique* scarcely causes inconvenience; but instantly it has acquired its due size, if not extracted, it gives rise to very stinging pain. Two or three days after its introduction it will have attained a sufficient growth for being removed. In the performance of this operation the negroes are most expert, on account of the constant practice they have in operating on themselves. With the point of a pin they carefully separate the epidermis under which the *nigua* is fixed, leaving it still attached by its reddish mouth; and then they thread or transfix it, and extract it in its globular form. Great care should be taken not to burst the bag or envelope of the insect at the time of extracting it, for otherwise several ova, equivalent to so many parasitic insects, are left to infest the foot; and besides, should part of the bag be left behind, pain and inflammation will supervene, followed by suppuration to cast off the foreign body. The hollow left after the *pique* is abstracted, is to be filled with snuff or the ashes of a cigar; as it stops any oozing of blood from the little wound, and assists in promoting the separation, or absorption, of any fragments of the envelope of the insect that may have remained behind; and by this means the pain is avoided which otherwise might arise if these parts were left to themselves, and allowed to slough off.

Without having recourse to the process of extraction, the *piques* may be destroyed by rubbing the spot where they nestle with mercurial ointment, or

with a mixture of soap and oil: in either way they are killed, and consequently fall off in form of crust. Tepid oil applied to the parts injured during the extraction of the *pique* or *nigua* affords relief; and it is requisite that the person who has been operated upon take great care not to put his feet in cold water until the incision made in the skin be entirely healed; for otherwise there would be risk of inducing that fearful disease—locked jaw.

EXTINCT ANIMALS.

Notwithstanding the vast distance from one another at which different nations of the earth have been planted, it may yet be traced in their traditions that one great and glorious object had been seen in common by their forefathers, the image of which had been so impressed on their minds, that, when placed in analogous circumstances, it often recurred to their thoughts, and was always referred to, though under different appellations. Thus, in midst of the solemn and sublime apparatus of thunder and lightning, Jehovah descends to the summit of Sinai to give law to the Hebrews. This august and majestic image of the greatness of Divine power is soon after applied to Jupiter darting thunderbolts from the peak of Ida against the armies of Greece; and the *Great Man*^[59] appears in like manner on the mountains of Ohio to exterminate with his darts a fierce animal, which desolated the fertile plains. So also, in former times, the *Heavenly Angel* came down to the summit of Santa Helena in Southern America, to crush and overwhelm a fierce and polluted race of giants, who, having entered these harbours from some unknown clime, devastated the land.^[60]

The Indians of the one and the other hemisphere corroborate the truth of their traditions by being able to present the great molar teeth, or grinders, which are found under the surface of the earth in the places alluded to. In Peru, these teeth, with other bones of enormous magnitude, are found in the province of Chichas, near the tropic of Capricorn; and in Chile there are not wanting vestiges of the same sort of organic remains.

I have had in my possession four of these molar teeth, of which I yet preserve one in the library of the Medical College of San Fernando. When compared among themselves, I have judged, from their configuration, that they did not belong to the same fossil elephant; but, rather, that three pertained to the mammoth, and that one had belonged to the mastodonton of

Cuvier: from which it is to be inferred that those very bulky animals, which in remote ages lived in Siberia and North America, had penetrated into Southern America, where they have left the natives, in the relics of their destruction, or fossil remains, a memorial of the existence and punishment of antediluvian giants.

The bony fragments which are considered to be parts of this gigantic race, may they not rather consist of earthy petrifications in water impregnated with lime? Between the villages of Chorrillos and Miraflores, in the locality named *Calera*, water impregnated with lime is observed to percolate at the foot of the *barranco*, or broken bluff-land; and it deposits on the stones, over which it drips or passes, certain crusts or laminæ, which have the same appearance with the bony laminæ of the human skull. [\[61\]](#)

GEOGNOSTIC DESCRIPTION OF THE COUNTRY IN THE ENVIRONS OF AREQUIPA, WITH AN ANALYSIS OF THE MINERAL WATERS IN THE VICINITY OF THE SAME CITY.

According to the “Guia Politica, Ecclesiastica, y Militar del Peru,” edited by Dr. Unanue of Lima, the city of Arequipa is situated at 16° 13′ 20″ south latitude; and it stands at the distance of about thirty leagues from its present sea-port, Islay. In making from seaward to the port of Chule, the burning mountain of Arequipa, (which now emits no smoke,) described as resembling a sugar-loaf with the top broken off, used to be taken as a landmark by the Spanish pilots on the South-Sea coasts. The population is estimated at about twenty thousand, among whom there are exceedingly few negroes or pardos. Towards the end of the twelfth century, Maita Capac, fourth sovereign of Peru, colonized the valley of Arequipa with three thousand families chosen from the neighbouring provinces; and some refer the origin of the name Arequipa to this early period, as in the Indian language the word Arequipa means, “Remain as you are, if you like it,”—in allusion to the permission given by the Inca to such of his captains as were inclined to settle in this garden in the midst of the desert.

By order of Don Francisco Pizarro, the city was founded in the year 1540. It suffered severely from the great earthquakes of the years 1582, 1600, 1604, 1687, 1715, and 1784.^[62] All about Arequipa is a volcanic country, of which the natural history is very interesting, as may be learned from the following account, taken from an essay originally written and published in Spanish by Don Mariano Rivero, director-general of the mining department in Peru. This essay is deficient in minute chemical analysis; but it has the recommendation of being the only one, so far as we know, that has been written by a native Peruvian on the subjects which it embraces.

TRANSLATION.

GEOGNOSTIC DESCRIPTION.

All the environs of the city of Arequipa are composed of volcanic products; so much is this the case that its edifices are constructed of a white and very light rock, rough to the touch, which contains pieces of pumice-stone and lava. It is called in the country *piedra sillar*, and, in fact, is a real trachyte (traquito). It is met with in different places, at the distance of several leagues from the city,—as, for example, in the glen through which passes the road of Islay. In the route which leads to the glen of Yura, over and above the loose pieces and sand, we meet the trachyte porphyry, not only in large detached masses, but also forming portions of the soil. The principal mass is compact, of a greyish black colour, with crystals of white vitreous felspar, and spangles of bronzed mica; it passes by decomposition into an ash-coloured trachyte, less hard than the first, and very rough to the touch: the crystals of felspar and mica suffer no decomposition.

The adjacent hills are formed of rocks such as these, without presenting a decided stratification. At various points there appears a rock which, from its grain, its little hardness, the pieces of trachyte it contains, and its reddish colour, would seem to be a conglomerate of aqueous formation.

The waters which run through rocky passes give rise to deep excavations, and with much good reason these parts (along which there is a road) are denominated *narrows*,—for they are only a yard and a half broad. The way is intersected at various points by ravines, (quebradas,) through which

several small streams descend from the snowy mountain called Chachani. About a league before we reach Yura, and on the opposite side of the glen, there is an horizontal stratification which extends towards the west, presenting an interesting contrast with the Cordillera on the east, and the hills on the side of the valley along which we pass to Yura.

It is not less worthy the admiration of a geologist to behold, amid so much arid nakedness, the bottom of the deep glen of Yura clothed with pure green and cheering verdure, which comforts the pilgrim, and serves as a soothing restorative to the sick who, sacrificing their domestic comforts and the pleasures of society, go in search of health to the baths situated in this solitary spot.

The glen of Yura, which stretches in the direction of from east to west, is in many parts extremely narrow, as at the site of the baths: but towards the Calera it opens up; and its inhabitants take advantage of this space to cultivate lucern, and collect the sub-carbonate of soda, improperly called saltpetre, which they use in manufacturing the soap consumed in Arequipa. [\[63\]](#)

Having passed the distance of one thousand yards, the glen again becomes narrow, until it joins with the ravine of the river of Yura, which flows from the north-east to south-west. From this place it becomes deeper and narrower, and immediately follows a westerly direction. A rivulet formed by the junction of the smaller streamlets which arise in the hill called Horqueta irrigates some land in the strath beneath; and, pursuing its course to the baths, passes them at a yard's distance, (now ceasing to be pure water,) to unite itself with the many jets that spring up a little above the baths, as I have seen in a ditch recently opened. This rivulet follows the course of the glen; and its waters, being impregnated with saline ingredients, irrigate the lucern fields, and contribute to increase the supply of the carbonate of soda. It at length joins the river of Yura, which during the periodical rains is in the highest degree dangerous to ford, on account of its strong current, and the many stones it carries along its impetuous stream. In the glen of Yura, as well as in the ravine alluded to, the different sorts of earth are of volcanic and transition formation: the first extend to a hundred paces beyond the ferruginous baths; the prevailing rock is trachyte, of a light ash-colour, with pieces of white felspar half decomposed, pumice-stone, and scales of black mica.

In some detached masses are seen in globules the substance called *perlita*, (little pearl,) and black pieces, which, from their general appearance and concavities, look like lava. Ascending a little higher than the sulphurous baths, porphyry is met with; the principal mass of which is compact and black, its fracture conchoidal, and it contains crystals of white felspar. By decomposition it has partly become an ash-coloured rock, less hard and more asperate, in which is found the conduit which emits sulphurous vapours; and in my opinion it is an ancient crater. On the walls of this crater, sulphur is deposited in well-defined, acute, octahedral crystals; and, in some pieces in my possession, the pure sulphur exists in its massive state. These rocks cover a sandy ground, which from its coarse grain, composition, reddish colour, and the fragments of volcanic rocks which it contains, appears to be a sandy conglomerate; it is sufficiently consistent to admit of being cut, and to serve for architectural purposes: there are certain places, as in the Calera, and near the baths, where it is many yards thick: this earth reposes on the transition series.

The earths, or mineral substances last mentioned, occupy all the parts to the north and west: they are composed of *gres*, (a stone abounding in sand,) semi-compact in some layers, and in others it has a fine grain; its colour is a dirty white, inclining to green, and it passes to a lightish black when it is near to the layers of the black "*esquito hojoso*," (foliated schist,) with which it alternates: it contains small spangles of mica. The natives extract laminae of this *gres* of more than a yard in length, and of a quarter in breadth, which serve to line the upper part of the boilers wherein soap is made. The black "*esquito*" also divides itself into laminae of good size; but, for the most part, it breaks very easily, forming small pieces which fall down to the bottom of the glen. A heavy substance of dark colour, which separates in large pieces, and effervesces when brought into contact with the acids, is found in the "*esquito*," and near to the layers of *gres*: it appears to be a carbonate of iron, (*carbonato de hierro litoideo*,) like that which is met with in the coal-mines of England and France. In the "*esquito*" I have observed impressions of plants, vestiges of coal, crystals, and small plates of gypsum.

The transition formation extends to the north and to the west, at great distances. I am also assured that coal is met with near to the village of Yura. Over the horizontal bed of *gres*, the direction of which is from east to west, with an inclination northward, may be observed the white, compact, fibrous

gypsum of Synchronita, distant six leagues from the glen, which, as I conceive, belongs to the gypsum of the vale of Vitor.^[64]

On the south side of the river Yura, in the locality called Calera, is found in layers or coats of considerable thickness, but of little extent, a limestone of a cellular and porous structure, composed in a great degree of very small and delicate tubes: their colour is a dirty white, and by all their signs and characters they appear to owe their origin to infiltration. From this stone is made the lime which is consumed in Arequipa and its neighbourhood.

ANALYSIS OF THE MINERAL WATERS OF YURA.

Mineral waters are distinguishable from common water by their taste, particular smell, colour, temperature more or less raised, and by their not being applicable to domestic purposes. They are found in different parts of the world in springs and wells: sometimes they are of the same temperature with the soil through which they pass, and at other times their temperature rises to the boiling point of water,—and then they are called thermal waters. In the countries where these waters appear, they had attracted the attention of the inhabitants since very remote times, and were medicinally employed internally as well as externally; but, since their component parts were but imperfectly known, they were often applied injudiciously, and they did not always obtain the reputation they merited, for their effects were sometimes contrary to those which the physicians desired to produce.

At the close of the seventeenth century chemists began to discover the substances to which mineral waters owed their peculiar properties; and, since this happy era, they have made such rapid progress in science, that in the present day we are acquainted with many of these substances. This knowledge we owe to simple and more exact methods of analysis. Nature appears to have favoured in an especial manner the environs of Arequipa with thermal springs to cure those maladies to which its inhabitants are subject. This is, however, no more than might be expected, considering the variety of medicinal ingredients with which the waters become impregnated as they slowly percolate through beds of lava, or issue from the deep recesses of burning mountains.

The baths of Yura are situated in a small and narrow glen, several leagues to the N.N.W. of the city, and only one league from the village of the same name,—which, according to my measurement, is one hundred and seventy Spanish yards (varas) above the square of Arequipa.^[65]

The road to the baths is very bad, and, above all, the declivities are so; for the number of stones and narrow windings render it in the highest degree disagreeable, to which the dull uniformity of the landscape also contributes. To the right, all that presents itself to the traveller's view is the lofty volcano, the contiguous hills being denuded of every blade of vegetation except the cactus Peruvianus, of melancholy aspect, seen here and there along the surrounding slopes; and, if he turn his eye to the left, he looks upon sterile plains cut up by mountain torrents, or a group of hills perfectly arid, of greater or less elevation, and in parts covered with white sand.

The analysis of the waters of Yura was attempted by the celebrated naturalist Haenk in the year '96; but, this philosopher not having ascertained their constituent parts, I have now the honour of presenting to the public the result of my investigations regarding these waters, and several others which are used by the inhabitants of the city. In the narrow glen of Yura there are two situations in which springs of thermal water present themselves, and the one is distant from the other about one hundred and fifty yards. The first, called *agua de hierro*, or ferruginous water, is on the left-hand side of the road as we come from the Calera; and those springs which are higher up are denominated *agua de azufre*, or sulphurous water. I will begin by giving an account of the *agua de hierro*.

THE FERRUGINOUS WATER.

From a little plane covered with grass, distant from the rivulet three yards, and four from the ash-coloured trachytic rock, water bubbles up at various points, forming large globules, as if boiling. Its temperature is 94° of Fahrenheit, that of the air being 68°. In the corner where these jets are found, there are at short distances small wells of equal temperature, except one at 67°, which is found at the distance of a yard from the principal jet; and it is the more worthy of notice, as it happens to be very near to the water which indicates higher temperature. All these little wells render

tribute to the principal one, and to the rivulet; their banks, and the bottom of one of the baths, contain a very fine yellow substance, which is the true oxide of iron. These waters are very transparent, without smell, and with taste half acidulated and astringent; they disengage a gas which, passed through lime-water, or a solution of the acetate of lead, throws down precipitates soluble with effervescence in acetic acid. They redden the tincture of violets and blue paper, which loses its colour on drying, and this proves the existence of a free acid; being agitated, an air is disengaged with noise: all the acids, weak and strong, produce effervescence with these waters. The prussiate of potash, when a little of any acid is added to it, causes in the ferruginous water a blue precipitate, which is the prussiate of iron. Iron-water being boiled, it loses the property of effervescing with the acids, of forming a precipitate with the prussiate of potash, and also its astringency. A bottle of water being evaporated affords, during the operation, a light, white precipitate, and its surface becomes covered with a most delicate film. The operation, if continued to dryness, yields sixteen grains of salts, which I have analyzed. It appeared from the analysis that the iron-water is composed of the following ingredients and proportions.

One bottle, or a pound and a half, of water afforded

Carbonic acid	10½ grains.
Muriatic acid	2
Sulphuric acid	¼

A hundred grains of the salts were composed of

Carbonate of magnesia	26 grains.
Carbonate of lime	6
Muriate of soda	15
Bicarbonate of soda	40
Sulphate of iron	3
Insoluble matter, consisting of silex and sulphate of lime	8
	98

This water greatly resembles, in its contents, that of Selz, Spa, and Carlsbad. The exact quantity of carbonic acid which it disengages could not

be ascertained, for want of proper instruments for the purpose. The carbonic acid of the saline parts is sufficient to saturate the lime, magnesia, and soda.

MEDICINAL VIRTUES OF THE FERRUGINOUS WATER OF YURA.

The ferruginous waters are stated by Mr. Rivero, on the authority of Haenk, and of Dr. Vargas of Arequipa, to be tonic, deobstruent, laxative, diuretic, &c.; and, therefore, well calculated to remove general debility, certain forms of hypochondriasis, dyspepsia, and weakness consequent on debauchery.

THE SULPHUROUS WATER.

The jets (*los ojos*) of this water, as we formerly noticed, are placed above the ferruginous baths, situated in a narrow part formed by the trachyte rock on one hand, and on the opposite side by the *gres*, or sandy soil, which furnishes the carbonate of soda. An oblique fissure, extending to the base of the trachyte rock, serves as a conduit to this thermal water, which unites itself with that which flows from the bottom of the bath named Tigre. A short way from this jet there are others which flow from other clefts, at almost the same degree of temperature.

It is observed that the source of these waters is at some distance in the interior of the rock, and, according to my notions, they hold communication with the *crater*, which emits sulphurous vapours, situated a few paces from the water underneath the very house where the sick repose. With respect to the water, having filled a small well situated at the base of the rock where people drink from, the chief object is to have it conducted to the bathing-pits, or basins. The superfluous water not needed for the baths flows out from them by a small channel; and it goes to join the rivulet, which passes within three yards of the bathing apartment.

The sulphurous water flows out in good quantity, making a peculiar noise, and emitting a smell of sulphuretted hydrogen gas, like that of rotten eggs; which is perceptible at the distance of many paces from the place, when the winds are from the east and west. The disengagement of carbonic acid gas occasions the noise alluded to, through the innumerable bubbles

which rise on the surface; and at the same time it occasions a shower of aqueous particles, mixed with sulphuretted hydrogen and carbonic acid gas; its colour is whitish, transparent; and on the walls and canals of the baths, it leaves a whitish substance, somewhat dirty, very fine, which when examined proves to be impure sulphur; its taste is at the same time sweetish and acidulated, but it leaves on the palate the taste of rotten eggs.

In the bathing apartment there are four large reservoirs, or basins, constructed with stone and lime; they are equal in size, communicate the one with the other, and are named Tigre, Sepultura, Desague, and Vejeto: in these reservoirs, however, the temperature of the water is not equal: the first indicates on the thermometer of Fahrenheit 90° , the second 89° , the third 88° , and the fourth 87° ; the air of the habitation being 70° . In the place where this water was first discovered, it at present indicates as many as 90° : Haenk, in the year in which he instituted his analysis, observed that it was only 86° , which proves that the temperature has since then increased.

The gas which arises from this water, when collected, extinguishes flame; produces in lime-water a precipitate which dissolves in acetic acid with effervescence; and it precipitates the acetate of lead, of a dirty yellow colour. The water reddens blue paper; but, on drying, it recovers its original colour, a circumstance which proves that there is a free acid: turmeric paper it does not change the colour of, unless its volume has been decreased by evaporation. A few drops of any acid produce effervescence. The nitrate of silver gives a violet-colour precipitate, the acetate of lead a dirty yellow, the muriate of barytes a white, but it is necessary to add to it a few drops of acid; the prussiate of potash produces a blue precipitate, using the precaution to reduce the water, and of adding to it some drops of nitric or muriatic acid. Liquid ammonia renders it turbid, which shows that it contains magnesia; corrosive sublimate produces a half obscure precipitate, which afterwards effervesces with an acid; it instantly coagulates milk, renders wine and the water of peaches or pears turbid, giving rise to effervescence with the three last. A piece of clean silver, if placed in it for some minutes, becomes somewhat black; when agitated, it disengages carbonic acid with precipitation, and all the water is filled with bubbles; when boiled, it loses its smell, it disengages all the free acid, and it no longer reddens blue paper.

Four bottles of this water, when evaporated, have given forty-three grains of salts: during the evaporation the surface became coated with a white film, and a light white substance was precipitated, which consisted of the carbonate of magnesia and lime, abandoned by the carbonic acid which had held them in a state of solution.

One hundred grains, obtained by evaporation, yielded

Insoluble matter composed of silica and sulphate of lime	10 grains.
Carbonate of magnesia	28
Muriate of soda	14
Carbonate of lime	7
Sulphate of iron,—indications of Bicarbonate of soda	39-98

The sulphuretted hydrogen gas, and the carbonic acid disengaged, could not be ascertained, for want of proper apparatus.

The water of the four baths fitted up for the sick is of the same quality with the one here analyzed; with the difference that in the three last its temperature is less, and that it does not disengage in such quantity the sulphuretted hydrogen nor the carbonic acid.

The *new water* of Haenk, and that from another jet or source which has been more recently discovered, possess the same qualities with those of the baths; differing, however, in this respect, that they do not disengage sulphuretted hydrogen gas. The water recently discovered, of which the temperature is 91°, contains more iron, but not in such quantity as the waters of the ferruginous bath. It is of a somewhat sweetish acidulated taste, and it leaves a certain asperity on the palate. Reagents act upon these in the same way as they do on the sulphurous water.

PECULIARITIES OBSERVABLE IN THE SULPHUROUS WATER BATHS.

Shortly after having entered the bath, the whole body becomes covered with numberless air-globules of a pearly appearance. Some degree of heat and pungency is felt all over the skin; and, soon after immersion in the water, the smell from it ceases to be perceived. It occasions a slight degree

of uneasiness in respiration, arising from the large quantity of carbonic acid and sulphuretted hydrogen which arises from its surface.

MEDICINAL EFFECTS OF THE SULPHUROUS WATER.

According to Dr. Don Jose Maria Adriasola y Arve of Arequipa, it has been found by experience, that, since times of remote antiquity, the baths of the sulphurous waters of Yura have produced the most salutary and specific effects in a great variety of cutaneous diseases. In various instances of chronic disorder of the bowels, and dysentery attended with intestinal ulceration and wasting of the general system, or what has been improperly termed intestinal consumption, Dr. Vargas found that these waters restored the healthy action of the digestive organs, kept up a proper cuticular discharge, and radically cured such apparently hopeless cases.

This same water is also allowed to be excellent for the cure of chronic rheumatism, certain deep-seated pains, and contracted joints, &c.

Mr. Rivero gives the following method as that by which invalids are to profit by the use of the sulphurous baths of Yura.

The first four or six baths must be taken in the bathing-places named Desague or Sepultura, which emit less gas and are of lower temperature; for by entering the bath called Tigre, which is the most active, the body experiences a very disagreeable sensation, and at the same time the breast is peculiarly affected.

To be in a fit condition for enjoying the advantages of the bath, the individual must have the stomach empty, be free from fatigue, perspiration, as well as mental emotion of every sort. The bath should not be continued above three quarters of an hour, and in the Tigre one should not remain above twenty or thirty minutes. Should the nature of the disease so require, the invalids may bathe twice a day. A purgative of cream of tartar or Epsom salts should be taken as a preparative for bathing in these baths. Strict attention to diet, daily exercise to favour perspiration, and great care to avoid exposure to damp or chills during the time of taking exercise, or coming from the bath, are requisite precautions.

The effects of these waters are slow of manifesting themselves, and, for this reason, their continued use in many cases is necessary; to their

perseverance and constancy in this respect many individuals, now in the enjoyment of perfect health, owe their recovery.

STEAM NAVIGATION.

We have in Vol. I. p. 173, alluded to the prospects of a Steam Navigation along the shores of the Pacific Ocean; and we are now happily able to subjoin a few statements on this subject, for the perusal of such of our readers as may not have seen the report of the British merchants and residents at Lima and Callao, upon the subject of opening through Panama a direct communication between Great Britain and the western coast of South America.^[66]

The first meeting on this interesting subject was held in Lima on the 12th of August 1836; and on the 7th of September, at a public meeting, the report of the British Merchant Committee was unanimously approved, and ordered to be printed and circulated in English and Spanish. From a pamphlet, accompanied with documents which detail the general plan of the intended operations of the Pacific Steam Navigation Company, printed by J. M. Masias, in Lima, we extract the following authentic information regarding the “Statistics of Trade, and the favourable influence of Steam Navigation.”

“It is only since the dynasty of Spain ceased to exist in South America, that the shores of the Pacific have been thrown open to foreign commerce; and, when it is considered how much these countries have suffered from continued revolutionary convulsions, the rapid advance of commerce and trade is somewhat extraordinary. The following statement of imports is from the best data which could be obtained.

British	12,000,000 dollars.
North American	2,500,000
French	1,500,000
Peninsular	1,000,000
Germany, and other places on the continent	1,250,000
China, Bengal, and Manilla	750,000
Brazil and Buenos Ayres	300,000
Total	<u>19,300,000</u> dollars.

“Of which there is consumed

In Chile	4,500,000	dollars.
Peru and Bolivia	7,500,000	
Equador and New Granada	1,500,000	
Central America	2,000,000	
Mexico and California	3,800,000	
Total	<u>19,300,000</u>	dollars.

“The whale fishery of the Pacific may be estimated as follows:

North American	12,500,000	dollars.
British	5,000,000	
French	3,000,000	
Total	<u>20,500,000</u>	dollars.

“The whole interest involved, including the Pacific whale fishery, amounts to nearly forty millions of dollars.

“The beneficial influence of steam navigation along the shores of the Pacific, and the opening a communication with Europe and North America, *viâ* Panama, are subjects of deep interest, not only to those engaged in commerce with the Pacific, but also to the whole commercial world.

“The present state of communication is long and tedious between Peru and Great Britain; it may be averaged at about four months; while, by the proposed route, it will be reduced to little more than one-third that period, viz.

Lima to Panama	6	days.
Pacific to Atlantic	1	
Isthmus to Jamaica, by steam,	3	
Jamaica to England	<u>36</u>	
Total number of days	<u>46</u>	

“By substituting steam navigation between Jamaica and England, the voyage from Great Britain to Peru could be performed in little more than a

month.

“The security which will be given to commerce by this prompt communication, the facilities afforded to merchants in realizing the proceeds of their shipments, the consequent increase of trade, the regularity of advices along the shores of the Pacific, so desirable for British commerce,—are points of universal interest. To the squadrons stationed in those seas an immense advantage will also be gained by the facility of communication and the increased efficiency of their operations. The moral influence to be effected will tend to strengthen and sustain the governments of the respective states against the usurpation of revolutionary demagogues.”

We further beg leave to add, on the same very good authority, the following valuable remarks for the information of those who are unacquainted with the localities of Panama, and the difficulties and facilities to be met with in crossing the Isthmus.

“The seasons are distinguished by rainy season and dry season. From June to November constitutes the former; from November to June the latter. During the rainy season, the journey from the Pacific to the Atlantic can be performed in two days; while, in the dry season, twenty-four hours only are necessary: from the Atlantic to the Pacific, during the rainy season, three days are required; and in the dry season it can be accomplished in two days. This difference is owing to the swelling of the river produced by rains. The journey from Panama to Cruces is performed on mules, being a distance of twenty-one miles, over a bad but not a dangerous road. In Cruces, there are canoes of all sizes always in readiness, in which passengers embark, and descend the river to Chagres, the sea-port of the Isthmus, where they re-embark on board the first vessel which suits their convenience.

“The transit of the Isthmus during the dry season is neither inconvenient nor unpleasant: the canoes are covered; provisions, fruits, &c. are abundant along the banks of the river; the temperature, though warm, is perfectly healthy, and there is always personal security. During the rains you are subject to great exposure and consequent illness; but were a good road once opened, and a steamer on the river, there would be no danger at any season, and the journey from sea to sea could be accomplished in *eight or nine hours*, without the slightest inconvenience.”

ECCLESIASTICAL JUBILEE.

The following authentic document we have carefully translated from the Spanish; and, having already referred to it, (vol. i. p. 132,) we now offer no comment on its contents.

We, Dr. Don Jorje de Benavente, Archbishop Elect of Lima, &c. to our clergy, religious communities, and all the faithful residing in and inhabiting this our diocese.

Forasmuch as our most holy father Gregory XVI, Roman Pontiff, and visible head of the Universal Church, moved by the pastoral vigilance and paternal love becoming a successor of St. Peter, has condescended to grant a general jubilee to the whole Catholic world, with a view that the common penitence and prayer of the faithful may obtain from the Father of Mercies and God of all consolation the cessation of the weighty ills that affect the spouse of J. C.; the supreme government of this republic, always zealous for the exact observance of the holy religion which we profess, has given the corresponding pass to the Brief, which with this object his Holiness has despatched, of which the tenor is as follows:

The Pontiff Gregory XVI. to all the faithful Christians who shall see the present letters, health and apostolical benediction.

After having taken solemn possession of the pontificate in the Basilic of the Lateran, we have written many times to our venerable brothers the patriarchs, archbishops, and bishops, concerning the calamities of the Church; encouraging them to oppose with all their might, like walls of Israel, those snares with which, to our great sorrow, she is beset. For the same purpose, also, we have admonished them that they should raise their eyes and hands to the mountain from whence we expect assistance; certain that, with the favour of Him who governs the winds and the sea, tranquillity will be procured, and that the divine mercy descends when the humble prayer ascends to God.

But seeing that the conspiracy of the wicked everywhere gains ground, and that the tempest even increases, we have resolved that in all the Church prayers be publicly offered; by this means opening up the treasure of celestial gifts, to the end that the consciences being regulated and purified in a holy manner from the uncleanness of transgressions, prayers themselves may be rendered more agreeable and acceptable to God, ascending to his presence as a delicate perfume.

It has been an ancient custom of the Romish Church, observed by our predecessors, not only in the commencement of their supreme pontificate, but always when the Lord chastened his people, to resort to the prayers of the community, and to rouse all to penitence by freely opening for them the sacred treasury of the indulgences; with a view that, detesting their iniquities, and humbly making confession of them, they should with confidence draw near to the Throne of Grace, that is to God, who is magnanimous to pardon, and refuses not his mercies even when he is provoked. With this example before us, we, after calling on the Father of Mercies with constant and fervent prayer, promulgate throughout the whole Catholic world an indulgence in the form of a general jubilee; cheerfully hoping that the Author of all comfort shall shorten the days of tribulation, so that, the storm having entirely ceased, the peace of the Church may be firmly established, and public happiness everywhere restored.

Hence, confiding in the mercy of Almighty God, and in the authority of the blessed Apostles St. Peter and St. Paul, —and exercising the power to bind and unbind, which, though of ourselves unworthy, the Lord conceded to us,— we have resolved to concede, and we do grant by the tenor of these presents, full indulgence (such as it has been customary to concede on the year of jubilee) of all their sins to all and to each of the Christians of both sexes residing in and entering into this our capital (Rome), who in the term of

three weeks current from the fourth Sunday of Advent,—that is, from the 23rd day of this month, December, till the 13th inclusive of the following January, on which falls the first Sunday after the Epiphany,—and on the eighth day of the same, shall have twice visited the Basilics of St. John of Lateran, of the chief of the Apostles, and of St. Mary la Mayor, or one of these at least, and there will have devoutly prayed for some time, and fasted Wednesday, Friday, and Saturday of one of the said weeks; and within the same period, having confessed their sins, will have received the most holy sacrament of the Eucharist, and may have given the poor some charity, as to each individual his devotion may dictate: moreover to all the faithful Christians existing in all parts without our capital we concede plenary indulgence, provided they shall have twice visited all or some one of the churches, which may be assigned to them by the local ordinaries, or their vicars or officers, or others under their orders; and, in their default, by the curates of souls, after these our letters may have reached their notice;—it being further required that they should have practised with devotion the other foresaid works in the space of three weeks, to be fixed upon by the churches.

In like manner we concede that voyagers and travellers may enjoy the same indulgence immediately that, on returning to their homes, they put in practice the above-named works, and visit twice the cathedral church, either the greater or parochial, of the place of their residence. We also grant to the regulars of both sexes who live in perpetual seclusion, and all laical persons who, from being in gaols or prisons, or on account of corporeal infirmity, or any other impediment, shall not be able to practise all or any of the works mentioned, that a confessor of those actually approved by the local ordinaries may commute these works into others of a pious kind, or prorogue them to some other early period, and impose on them those which the penitents themselves can exercise; and that he may also dispense the

communion to children who shall have never communicated.

We further concede licence and authority to all and to each of the faithful Christians, seculars or regulars, of whatever order or institute, though the particular denomination should be specified, that they may choose, for the expressed object, any confessor, secular or regular, of those actually approved by the ordinaries, (as also to the nuns, who may choose any confessor of those approved for them, whether the nun be a novitiate or a profesa,) who can absolve them, and liberate them at the tribunal of conscience, and for this time only, from the censures of excommunication, suspension, and other ecclesiastical penalties, for whatever reason enjoined or imposed, either *jure vel ab homine*; and also from all their sins, excesses, crimes, and delinquencies, however grave and enormous they may be, even when they shall have been especially reserved to the local ordinaries, or to us and the apostolical chair, and whose absolution on the other hand should not be considered as granted in the most ample concession; moreover, they may commute into other pious and salutary works all manner of vows, though these may have been made with an oath, and the power of dispensation regarding them reserved to the apostolic chair,—always excepting the vow of religion and chastity, and the bonds (obligatorios) accepted by a third person, or from the commutation of which an injury would follow to another, as also those penal acts which are called preservatives from sin,—unless the future commutation be presumed to be not less sufficient to separate from sin than the first matter of the vow, imposing salutary penance in the above-said cases, and such other particulars as the confessor may be pleased to enjoin.

But not on this account is it our pleasure to give dispensation by these presents in any other public or private irregularity, nor in any defect, stigma, incapacity, or ineptitude, in whatever way contracted; or to confer

authority of dispensation therein, or to qualify and restore such delinquents to their former state, even in the tribunal of conscience; nor to derogate the constitution, *sacramentum penitentiæ*, published with its corresponding explanation by our predecessor of happy memory, Benedict XIV; nor that these letters can or ought to avail in any way those who by us and the apostolic chair, or by any prelate or ecclesiastical judge, may have been expressly excommunicated, suspended, interdicted, or publicly denounced as liable to spiritual animadversion,—unless within the term of the three specified weeks they give satisfaction to, or come to an understanding with, the parties concerned.

Thus then, by the tenor of these presents, and the ineluctable obligation of holy obedience, we strictly enjoin all our venerable brethren, the patriarchs, archbishops, bishops, and all other prelates in the churches, and all ordinaries wheresoever, their vicars and officers, and failing of these, the curates of souls, to the end that when they receive transcripts of these letters, or printed copies thereof, and moved by circumstances of time and place, find it convenient in the Lord to publish the same, they may so publish and cause them to be published in their churches, dioceses, provinces, cities, towns, territories, and places; and having, as far as in them lies, prepared the minds of the people by the preaching of the divine word, may designate the churches and the time whereat and wherein this jubilee may be obtained; all constitutions and apostolical briefs to the contrary notwithstanding,—especially those in which the power of absolution in the cases therein expressed is reserved to the R. P. for the time being,—so that the concessions of such like indulgences and powers, or of any other kind, may be of use to no one, unless express mention or special derogation be made thereof: notwithstanding, moreover, the rule of not conceding indulgences after the manner of this letter, and the statutes and customs of any order, congregation, or institute whatsoever, although the same should be corroborated by oath and apostolic

confirmation, or any other form of stability whatsoever; notwithstanding, moreover, the privileges and pardons in whatever way conceded, approved, and renewed by apostolic letters in favour of the said orders, congregations, institutes, and individuals thereunto belonging. For as much as touching all and each of the above, (and notwithstanding that express and special mention should be made thereof and the tenor thereof, and not by general expressions to the same effect, or that some other particular form ought to be used with regard to them,) holding their tenor to be sufficiently expressed in these letters, and the prescriptions thereof duly observed, we do for this particular time nominally and expressly annul and derogate them, as well as everything that may appear to the contrary, to the end that the foregoing concession may have its due effect.

And with a view that these letters, which cannot be forwarded to every situation, may yet easily reach the notice of all, we desire that to their transcripts, or printed copies, provided they be signed by some notary public, and sealed with the seal of some person constituted in ecclesiastical dignity, may be given in all parts, and by all, the same credit as to the originals, if these shall be shown to them.

Given at St. Peter's in Rome, sealed with the ring of the Sovereign Pontiff, on the 2nd December 1832, the second year of our pontificate.

For Cardinal ALBANO,

A. PICHONI, Substitute.

Lima, September 5th, 1834.—Examined with the exposition of the council of state on the 3rd instant. If the Metropolitan doubts not the authenticity of this brief, he shall cause it to be published in this diocese and the other bishoprics, with this express qualification, namely—without prejudice of the jurisdiction and innate powers of the episcopate.—To this effect let it be returned, retaining a certified copy.—A rubric of his excellency P. O. and of the secretary of state—Leon.

For the reasons expressed,—satisfied of the authenticity of the said brief, and in fulfilment of what is therein ordained, we order it to be published, and in effect we do publish it with due solemnity, as well in this capital as in all the curacies or parishes of our diocese; and we ordain and command that from the 2nd of October first, till the 7th of December, spiritual exercises shall be successively performed in the churches of Santo Domingo, San Pedro, San Agustin, La Merced, San Francisco, San Lazaro, and Santuario de Cocharcas, the distribution of which will be separately assigned, so that the faithful may be prepared to gain the holy jubilee according to and in the manner expressed in the proclaimed brief.

The jubilee will be opened and commence on the second Sunday of Advent; on which occasion will be said in this holy cathedral church a solemn mass, with a sermon, at which all the clergy and religious community will assist; after the conclusion of which the hymn, *Veni Creator*, will be sung: and from this day to the 27th of December inclusive, will expire the three weeks designated by his Holiness for gaining the jubilee, which shall be concluded with a solemn mass as an act of thanks, which will be said in the same manner as the first; and the same will be practised in the other churches as best may be.

To gain the jubilee in this capital, we fix upon the churches of the Cathedral, Santo Domingo, San Francisco, San Pedro, San Agustin, La Merced, and La Parroquia de San Lazaro, which the faithful shall visit twice, or at least one of the said churches, during the three weeks indicated; at the same time putting in practice whatever else the brief requires.

In the suburbs the jubilee will be published, and it will be gained on the same days as in this capital.

In the other parishes of this archbishopric, the three weeks of the jubilee will run from Passion Sunday to Saturday *in Albis* inclusive, of 1835; in consideration that this is the most proper time, for, during it, all the faithful congregate in the parochial capitals to comply with the annual precept of the Church.

To gain the jubilee in the suburbs and other parishes of the archbishopric, we assign the parochial churches, which the faithful shall visit twice, taking care to do whatever else his Holiness prescribes.

With respect to the religious communities of either sex, the sick, imprisoned, voyagers and travellers, it will be according to the literal tenor of the brief cited. And that it may come to the notice of all the faithful of this archbishopric, this our edict shall be fixed on all the doors of the churches; and the necessary copies shall be remitted to all the prelates and parish priests of the archbishopric.

Given in Lima, sealed with the seal of our office, and countersigned by our secretary, on the 26th September 1834.

Signed,

JORJE (George), Archbishop elect, of Lima.

DR. MANUEL GARATE, Secretary.

LANCE'S ADIEU TO LIMA.

The following lines, with the notes annexed, were written under circumstances of great bodily and mental suffering in the year 1833, when the learned and very accomplished gentleman who penned them proposed to return to his native country. They were inscribed by him to his friends, among whom it has long been the author's happiness to have occupied a favoured place. The notes are not only illustrative of the "Adieu to Lima," but also of several incidental remarks contained in the preceding pages—especially of the interesting ruins of Pachacamac alluded to in vol. i. page 144.

I.

Welcome, thou heaving dark blue Sea!
Thy luckless child am I;
Thine island-child was nurs'd by thee
Beneath the northern sky;

II.

In that proud land—baptiz'd of old
To freedom in thy waves—
Confirm'd in glory—steel-ribb'd hold
Of men who'll ne'er be slaves.

III.

On southern shores I've linger'd long,
My life is waning fast;
And Sorrow, with her forked tongue,
Has struck me down at last.

IV.

I'll fling me on my mother's breast,
The sweeping dark blue Sea:
O bear me to my island-nest,
To die among the free!

V.

Land of the Incas, fare thee well!
Thy valé soon is told:
I would no longer in thee dwell
For Huascar's chain of gold![\[67\]](#)

VI.

Thy sceptre erst was sway'd by kings,—
They could not hold it firm;—
But now thy children crouch to things
More worthless than the worm.

VII.

Away! I will not stop to mourn
Thy sorrows or thy shame;
Nor tell what anguish I have borne
Since to thy shores I came.

VIII.

Let Indian pipe, so meek and low,[\[68\]](#)
Bemoan thine Incas' fall;
And Indian maid, with garb of woe,[\[69\]](#)
Her country's wrongs recall:

IX.

And sadder still, the queen![\[70\]](#)

And, sadder still, the cuckoo—
Complain in ev'ry vale,
When ev'ning with her dewy eye
Brings round the hour of wail.

X.
I'll fling me on my mother's breast,
The sweeping dark blue Sea:
O bear me to my island-nest,
To die among the free!

XI.
Like silver bells, yon snowy peaks^[71]
Are hung upon the clouds!
'Tis sweet to see when morning breaks,
And strips them of their shrouds.

XII.
'Tis sweet to see 'midst orange rows
The spires of Lima wind;
But sweeter far than these or those,
To leave them all behind!

XIII.
Then wherefore does my bosom heave?
Why starts th' unbidden tear?
Some scenes there are 'tis hard to leave,
Some friends my soul holds dear!

XIV.
Thou good-man's eyrie, on the rock!^[72]
From thee I'm loth to part!
With names belov'd in mem'ry's book
Sure thou recorded art!

XV.
There, with my friend, full many an hour
Of cank'ring care I'd cheat;
Or range the ample corridor,
The balmy breeze to greet!

XVI.
Or gaze on Pachacamac's height,^[73]
As peal'd the evening gun;
And see its mystic form dilate
Against the setting sun!

XVII.
And thou, dear Garden, 'neath thy bowers^[74]
How swift the moments flew!
They pass'd the sweetness of thy flowers.

Though steep'd in evening dew!

XVIII.

Like stars that seek each other's light, [\[75\]](#)
To form, as poets sing,
A path for gods, and o'er the night
Their mingled radiance fling:

XIX.

E'en so we cluster'd there, with souls
That friendship made but one:—
But hark! my knell of parting tolls,—
Sweet friends, I must be gone.

XX.

I'll fling me on my mother's breast,
The sweeping dark blue sea:
O bear me to my island-nest,
To die among the free!

FOOTNOTES

[1] Wheat and flour are principally supplied from Jauja, and barley from Tarma; fruit from Huanuco, and sugar from Huanuco and Huaylas.

[2] Plata-piña, or simply piña, is the name given to silver not entirely purified from the mercury which adheres to it in the process of amalgamation. Amalgamation is effected by mixing the ore, after it has been ground, with salt and quicksilver; treading the whole together by men or cattle; then allowing it to repose in *cerco*, or in the enclosure in which it has been trodden, for a month or six weeks. At the expiration of this time the quicksilver is supposed to have combined with all the silver in the mass, and to have formed a perfect amalgam, called *pella*, which is separated by washing away the mud and refuse of the ore. The *pella*, thus procured, is white, and so liquid that, by putting it into a strong bag a considerable quantity of the mercury is made, by pressure, to escape, leaving the amalgam of a solid consistence. It is decomposed by a red heat; and the mercury being distilled, it may again be applied to the same purpose as before. In this process, however, there is usually a great waste of quicksilver on account of the bad apparatus employed; and the fixed metal or silver which remains is what is called piña. This piña is usually sold by the miner in round masses larger than cannon-balls; and these balls of silver are, by the trader who does not venture on smuggling, carried to the government smelter stationed at the mines, (an office for many years back honourably filled at Cerro Pasco by a learned and good man, Don Toribio de Oyorzabal,) by whom they are cast into the foundry, and, being there melted down and sufficiently purified, are now cast into bars, which are stamped as being of a proper *ley* or standard purity: after which they may be conveyed to the mint for coinage.

[3] At Huallianca, Hualliy, and several other parts of the rich department of Junin, smelting is used for the extraction of silver: but in Cerro Pasco smelting is little practised. In the district of Yauricocha, and especially in the great or King's mine, the ores are found to contain a considerable portion of the sulphuret of lead; and, also, of the sulphurets of copper, of iron, and of silver.

Such is the quantity of sulphuric acid distributed among the mines of Cerro Pasco, which are placed among limestone hills, that the water which they contain is observed to corrode the iron machinery exposed to its continued action.

[4] "*Cobos*."—This was a duty of one and a half per cent. on the metals extracted from the mines. Its origin, as we are informed, was a grant to this amount, made by the Spanish government in favour of an individual of the name of Cobo. This became a permanent tax which, like the tithes of the metals, afterwards fell into the hands of the government, until both were abolished a few years ago, as alluded to in the text.

[5] We have lately learned that during the years 1837 and early in 1838, quicksilver became so scarce in Peru, that it cost 200 or 220 dollars per quintal. The consequence has been that a private company, under the auspices of the Protector, Santa Cruz, was formed by the enterprising General Otero and others, to clear out the Socabon, or adit, and re-work the long neglected and abandoned mines of Huancavelica—which are distant from Cerro Pasco sixty-six leagues, by the route of Tarma, Jauja, and Iscuchaca. This company has made some progress in the works; but the quantity of quicksilver yet extracted by them cannot be said to have had any sensible influence on the price of this valuable metal, which, in consequence of the large shipments lately made of it, has fallen to about one-half the above enormous price. During the period referred to,—though the drainage and works at Cerro were considerably improved,—no mines of second or third rate could cover the expense of amalgamation; and, therefore, the metal extracted from them was allowed to accumulate in heaps, (constantly guarded by Indian watchmen, called “tapacos,”) to the estimated value of three millions of dollars.

[6] The product of a marc of silver of standard purity is eight dollars four reals, or 1*l.* 14*s.* sterling.

[7] “Guarapo” is the name for the fermented cane-juice used as drink.

[8] “Coto,” or goitre, is common among the inhabitants of Huanuco; but it is a disease very rarely seen on the western side of the Andes.

[9] From *pexe*, fish; *buey*, ox.

[10] Though Mr. Mathews omits to mention its botanical character, it is probable from the name *huasca* (which means rope) that this, like the former plant, is a pliable bejuco. The bejucos are commonly used in Peru as cordage, for the purpose of constructing bridges and fences.

[11] The Chamber of Deputies is composed of representatives elected by the electoral colleges of provinces and parishes. The parochial electoral colleges are composed of all the citizens resident in the parish, congregated according to law. For every two hundred individuals in a parish an elector is nominated; and in every village whose numbers entitle them to name an elector, or have a parochial college, a municipal body is established with a right to superintend its own local interests, consistently with the laws and public good,—and subject to the approbation of the departmental juntas. The electoral colleges of provinces are composed of parochial electors constituted according to law, and they elect deputies to Congress in the proportion of one for every twenty thousand inhabitants, or for a fractional number which exceeds ten thousand. But the province in which the whole population does not come up to ten thousand inhabitants, will nevertheless name a deputy.

[12] See “La Constitucion Politica de la Republica Peruana,” published in 1828.

[13] Cajamarca lies to the eastward of the city of Truxillo, in northern Peru, and, by the post-road, about forty-five leagues inland from the shores of the Pacific. It is the principal town in the province of Cajamarca, and is remarkable in the history of Peru as a seat of the Incas; their baths and palace are yet to be seen, though in ruins. Here the magnanimous Prince Atahualpa, who had

purchased his freedom by an immense ransom of gold and silver, fell a victim to the insatiable cupidity and treachery of Pizarro.

[14] In dry weather sweeping the streets can hardly be considered as a serious punishment; but in the rainy season, when it is customary for the inhabitants to walk with wooden clogs, called *zuecos*, the scavenger's task could not fail to be of very difficult performance in the Cerro.

[15] The practice, common among Catholics, of visiting the tombs of their family, and honouring the spot where the remains of their relatives or friends rest, is not an ostentatious ceremony, but an humble act of devotion, in which we must believe that the heart of the suppliant is deeply engaged. In the niches of the Pantheon at Lima, the renewal of flowers week after week, and even year after year, bears witness that filial or conjugal affection is still cherished in the heart long after the object of endearment has been removed from those who only survive to deplore their own loss, and with tender sorrow pray for the spirit of the departed. Whoever has visited the cemetery of *Père la Chaise*, in Paris, must have been struck by the attention of the living to the dead,—the daily decorations of the grave, and the prayer offered up by pious friends; nor can we suppose any person capable of viewing with cold indifference the flowery neatness which surrounds those monumental tombs, which in *Père la Chaise* seem to triumph over the silence of the grave.

[16] See Memoria, por Jose Villa, Ministro de Hacienda; Lima, 1834.

[17] Guia Politica, Ecles. y Militar, 1793.

[18] One of the greatest Jesuit missionaries was Father Samuel Fritz, a German, who, in 1686, preached the Gospel, and converted many tribes in Maynas. He drew a map of the Marañon and its tributary rivers, which was published in Quito in the year 1707.

[19] "Amigo" or friend, is the first word of Spanish which the mission Indian is taught to speak.

[20] The road which formerly existed between Pozuzo and Mayro is now so overgrown with brushwood as to render it impassable without the aid of the chopping-knife, with the use of which the Indians of Huanuco are well acquainted. By this road the journey from Mayro to Pozuzo was usually performed in two days, and the journey from Pozuzo to the city of Huanuco in three: in all, five days from Mayro to Huanuco.

[21] It might be imagined that this custom of carrying away and eating the dead was a good reason for the ancient practice, still in use among the Indians of the Ucayali, of burying their dead in their houses, as affording some protection against this rage of cannibalism; but among the Inca race of Indians the practice also appears to have existed of old, though without reference to so shocking a cause.

[22] As a comment on this part of Friar Plaza's letter, we cannot do better than introduce a passage illustrative of the allusion here made, translated from a paper of his own in the Merc. Per. and cited by Lieutenant Smyth and Mr. Lowe in the Introduction to their Narrative of a Journey from Lima to Para.

"The three entrances to this district (Ucayali) are by Huanuco and the port of Mayro, by Tarma and the river Chanchamayo, and by the Jauja and Andamarca,

taking the direction of Pangoa, which is passable, and has been so since the year 1815, when I crossed from the plains of Sacramento to Pangoa, where I formed a friendship with various nations on the way; and by this route for seven years the mission has received all its supplies. In this expedition I explored all that was remarkable from Sarayacu, which is fifteen days' distance up the river from the Marañon, and ascended from thence as far as the river Pachitea, in twenty days more." It may be remarked, that the communication with Sarayacu by the rivers Huallaga and Chipurana is so circuitous, that Fr. Plaza does not even mention it as one of the routes to the mission; though this was the route pursued by the late expedition in 1834-5, after the attempt to enter by the Mayro had failed. The Sub-prefect's letter, too, took near three months to reach the mission through the country of Maynas.

[23] As trees of sufficient size for the purposes required are not always at hand, we have seen near a hundred men exhaust their strength in dragging a tree by the means of lazos from deep ravines and hollows. This waste of power might be easily avoided by the help of the pulley, with which they are unacquainted; but they show great skill in the application and management of the lazo, and, when arranged for the tug, their efforts are roused by a song of which the chorus is "Huasca runa!"—Men, to the lazo!

[24] It was a punishment which in certain cases the law of Spain inflicted upon female delinquents, to cut off their hair, and sometimes shave their eyebrows. This, we understand, was done by the common executioner,—hence the sense of disgrace.

[25] The coca leaf is to the Indian of the interior a necessary of life, which he uses from time to time, to renovate his energy for renewed muscular exertion; and in the intervals of labour he often sits down to *chaccha* or to refresh himself by masticating coca seasoned with quick-lime, which he always carries about his person in a little gourd. The lime is used in very small quantity at a time, but in a pulverulent and escharotic state. According to the Indian it counteracts the natural tendency of the coca to give rise to visceral obstructions. Used in moderate quantity, the coca, when fresh and good, increases nervous energy, removes drowsiness, enlivens the spirits, and enables the Indian to bear cold, wet, great bodily exertion, and even want of food, to a surprising degree, with apparent ease and impunity. Taken to excess, it is said to occasion tremor in the limbs, and what is worse, a gloomy sort of mania. But such dire effects must be of rare occurrence; since, living for years on the borders of the Montaña, and in constant intercourse with persons accustomed to frequent the coca plantations, and with Indian yanacones or labourers, all of whom, whether old or young, masticated this favourite leaf, we never had an opportunity of witnessing a single instance in which the coca-chewer was affected with mania or tremor.

[26] The whites have already had a foretaste of this retribution in La-paz, where, as we have been informed, every white man was massacred. The Indians are said to indulge in the hope of yet seeing a prince of their own race on the throne; and such has been their well-founded and now habitual mistrust of the whites, that they have never revealed where all their own treasures and those of the Incas, which were buried after the death of Atahualpa, are to be found. This is a secret to every one but a chosen few of the caciques. A few years before the commencement of the war of independence in Peru, a rising took place among the Indians of some of the inland provinces, under a cacique named Pomacagua:

but this insurrection was soon suppressed. The fact of Pomacagua's being acquainted with the hiding place of the regal treasure alluded to, and his offer to reveal it to save his life, was not believed by the unrelenting Ramires, and he was shot.

[27] Gamarra resigned the government into the hands of the National Convention, which, it appears, was not duly authorized to nominate a president. Under all the circumstances, had Gamarra acted boldly and openly—had he said from the first that he would remain in the government until a congress should assemble, before which he would account for his proceedings, he would have acted not only legitimately, but, as good judges and patriots believed, even wisely: since, by so seasonable an exercise of moral courage, he might have saved his country from anarchy. But, having voluntarily left the government, and publicly as well as solemnly acknowledged the authority of the Convention and the Presidentship of Orbegoso, his conduct afterwards, in taking up arms with the insurgent followers of Bermudes, was unfortunate for himself, discreditable to his party, and ruinous to his country.

[28] This lady united with a vigorous constitution a bold and energetic mind. She was feared by her enemies, but sincerely beloved by her friends. In consequence of the rebellion of her husband, and jealousy of the government that succeeded that of Gamarra, who, but for her talents and influence, could not have governed so long as he did, this ex-presidentess, usually called Panchita, was banished to Chile, where she died of a disease of the heart, which on her death-bed she ordered to be sent to Gamarra after her death.

[29] The mercachifle is a licensed pedlar, and the pregonero a news-crier.

[30] Ever since Europeans became acquainted with the Indian race, self-possession has been noticed as one of their most striking characteristics. Atahualpa was unmoved in the midst of every danger: and Santa-Cruz (of Cacique blood) has, in our own day, signally illustrated the same high feature of character in the Inca family. Finding himself for a moment isolated on the field of battle, and on the point of being pierced through by a trooper, he called out in a commanding voice—"Alza esa lanza y sigue me!"—raise that lance and follow me! Thus, his presence of mind saved his life; for the mysterious power of a superior mind triumphed over the hostile arm of the infuriated soldier—who, now, as we are told, occupies a place in the body-guard of Santa-Cruz.

[31] Only three weeks before he made his revolution, he had suppressed another in the castles of Callao, and shot every fifth man engaged in it. His own treason, while successful, he called patriotism: but he was doomed to suffer the punishment of a rebel.

[32] It has been remarked, by those who have happened to be in Payta during rain, that the soil on these occasions emits a suffocating and oppressive smell. This is probably owing to the quantity of animal and vegetable matter which, during a long continuance of dry weather, accumulates and is left to dry in the sun; and is partially dissolved by the rain, and absorbed by the circumambient air. It would be worth ascertaining by accurate observation whether the typhus of Piura ever becomes aggravated in type in rainy seasons. We never heard of its being contagious.

[33] The valley of Nasca, though situate in the midst of an extensive desert, is rendered very productive in vines, &c. by means of subterraneous aqueducts constructed by the aborigines. Thus, the ancient Peruvians had fertilized the most arid plains, and left monuments of agricultural industry on the coast not less remarkable than their terraced gardens in the Sierra.

[34] This was once the port where the silver from the mines of Potosi used to be embarked in Spanish treasure-ships.

[35] At what is considered the watering-place of Cobija, so sparingly does the fresh water percolate from the rock, that we are informed by an intelligent navigator, well acquainted with these coasts, that it takes a whole night to fill a small cask placed under the precious drop, by the favour of which grow two palm-trees, the only vegetable productions to be seen on the coast of Bolivia.

[36] This fortress was lately dismantled by order of General Orbegoso's government.

[37] The Incas had a garden in the neighbourhood of Cuzco, where all the trees were of gold and silver, and the fruits and leaves of precious stones.

[38] Valparaiso is sometimes called the Vale of Paradise; yet there is anything but a look of Paradise in Valparaiso and its immediate environs. It has been said that the Elysian vale of Quillota, a few leagues distant, is the Paradise alluded to in this appellation, which is a corruption of Va-al-Paraiso, *i.e.* This is the road to Paradise—namely, Quillota.

[39] See Letter from Alexander Cruckshanks, Esq. to Professor Hooker, inserted in Part iv.-v. of the Botanical Miscellany for March 1831.

[40] Ensayo sobre las causas de las Enfermedades que se padecen en Santiago de Chile. Por el Doctor Guillermo C. Blest.

[41] This being written in 1828, it is but fair to suppose that, as the general police of Chile has been vastly improved since that time, the evils here alluded to by Dr. Blest may have been removed.

[42] According to Humboldt, the farm of Enciero, near Vera Cruz, 3,043 feet above the level of the sea, is the superior limit of the vomito or yellow fever; and strangers who come by sea, and therefore pass through a gradual change of atmospherical temperature, are observed to be less liable to contract the yellow fever than the whites and mestizoes who inhabit the table-land of Mexico, (of which the mean temperature is about 60° or 62° of Fahrenheit,) when they descend during the wet season to the port of Vera Cruz. The rains begin in May and end in October, when the "*nortes*" or north winds set in; and during the prevalence of these winds the yellow fever or vomito disappears.—Translator's Note.

[43] The translator understands that chimneys and stoves have of late years become common in the houses of the higher classes in Chile; such, however, as still want these conveniencies make use of the old-fashioned brasiers, or pans of live charcoal. Over these though people may toast their legs if they please, still their backs and shoulders are suffering from cold, as the heat of the *brasero* or brasier is not sufficient to support a proper degree of general temperature in the air of the apartment in which it is placed. These pans are very properly

denounced by Dr. Blest as most unfit, and even dangerous, in the close and ill-ventilated dwellings of the poor.

[44] Western Peru, from the peaks of the eastern Cordillera to the shores of the Pacific, has hardly any venomous animal except the scorpion, which exists in the warm intermediate valleys (in some of which a small black and white snake is also found, which is said to be highly venomous); but alligators, as we have seen, abound about Guayaquil; and the coast of central America is famous for its venomous snakes, as well as for the antidote to their poison, or the bejuco guaco, which in infusion makes an agreeable bitter, something like quassia. The same antidote Providence has planted in the neighbourhood of Tarapoto, on the eastern frontiers of Peru, where venomous snakes also abound. The popular story in Peru respecting the discovery of the properties of the guaco is, that an Indian happening to be present when a condor, or some strong hawk of the numerous species which inhabit the Cordillera, was engaged in mortal combat with a tremendous snake, observed that as often as the bird was wounded he retired to a thicket of guaco, broke off the bark with his beak, dressed his wounds and pruned his feathers with the sap, and returned to the fight with confidence and spirit, till at length he killed the snake, and carried it away in triumph. From this the Indian inferred, that in the juice of the guaco resided the property that counteracted the poison of the snake; and it is vulgarly believed that if you rub your hand and arm with the juice of this bejuco, you may grasp the deadliest serpent with impunity. But, however that may be, the fact is never disputed, that the guaco is a quick, powerful, and certain antidote against the poison of the serpent.—Translator's Note.

[45] Notes on State of Virginia, p. 62.

[46] *Paco*: so called (in the Indian language) because its wool is long and of a bright reddish colour. *Alppa-co*, *sheep of the country*, has the wool long and very smooth; and, though coming under the denomination of Peruvian camel, is not very well fitted to carry a burden. *Llama llamsكاني*, or the *working sheep* (of the Indian) has the wool short and rough; and is the tallest, strongest, and best adapted for the cargo.

EXTERIOR PROPORTIONS OF THE LLAMA.

	Feet.	In.	Lines.
From the crown of the head to the extremity of the sacral bone	6	5	0
The coccyx or tail has of length	1	0	0
From the upper lip to the crown of the head measures	1	1	0
Length of the ear	0	6	6
Length of the neck from its first to its last vertebra	2	5	0
Anterior height, measured from the base of the fore-foot to the edge of the shoulder-blade parallel with the spine	5	5	0
Posterior height, measured from the base of the hind-foot to the spine of the sacral bone	3	6	0

[47] *Quipe*: bundle of clothes which they carry on their shoulders.

[48] The antelope, by the account of the Indians, sometimes looks over the tops of the eastern chain of the Cordillera, in the Vale of Huanuco. In this vale the tiger-cat has been seen; the mucamuca, probably a species of skunk, as it emits a most offensive stench, is common; and here too armadilloes may be found among the thickets of the pasture-grounds, and they are considered by the Indians to be good eating.

Rats are as common as guinea-pigs in all the agricultural valleys of the interior: the fox ranges all over the high hills and table-lands of the Sierra; and among the crevices of the rocks, in high situations, the traveller meets the long-tailed bizcacha, which burrows like a rabbit, and is valued chiefly for its fur.—Translator.

[49] Jarava foliis involutis, spica panicul.—*Flor. Per. et Chil.* t. i. p. 5, icon vi. fig. b. As these pasture-grounds are found at twelve or fourteen thousand feet above the level of the sea, they do not admit of the cultivation and population of the lofty plains of Anahuac or Mexico, which are only six or eight thousand feet above the ocean.

[50] The shrubs, as Dr. Unanue remarks in another part of his work, which grow at the altitude of from twelve to fifteen thousand feet above the Pacific, are of a woody fibre, resinous, and covered with firm bark, to enable them to resist the effects of the piercing cold to which they are naturally exposed.—Translator.

[51] Not only is the ass of Lima the useful quadruped here described, but one of the most ungratefully dealt with by the natives, who seem to have forgotten how honoured this animal had been in ancient times. The saddle-ass is goaded on at a nimble pace by the sharp point of a rib torn from some of the numerous skeletons of mules and horses, &c. which are scattered on the mounds of rubbish, or in the lanes around orchards, within and without the city-walls; and the donkey-driver grins his smile of savage complacency as he swings about his heavy lash, and nicely hits some raw and bleeding spot, the effect of former and frequent inflictions of the same sort at the hands of cruel men.

Ayanque, in his picture of Lima, correctly says,

Veras borricos de alfalfa
Y borricos capacheros,
Borricos cargando harina,
Piedra, cal, ladrillo y yeso.
Veras borricos volar
Al son del latigo huyendo.—Translator.

[52] In travelling from the inland country to Lima we have had occasion to observe, that when the horse reared on the cold table-lands, and not accustomed to any warmer climate, was taken from the Sierra to the coast in the hot months, he pined away almost as fast as a common fowl on the high seas when confined in a coop, and exposed to the spray in rough weather. But the pony, thus affected by the climate of the coast, will with surprising readiness recover his spirits and health as he returns, and ascends the tortuous and shelving paths that lead to his native element near the glacial peaks of the Cordillera.—Translator.

[53] The slaughter thus commenced has passed into a custom of annually destroying these confiding companions of man, when the howl or piteous death-cry of the poor animals rings upon the ear, on fine summer mornings, as the watermen are employed in knocking them down with their iron-pointed sticks in all the streets, and even at the very doors or gates where the persecuted creatures seek protection in vain.

To see them dragged along the streets, bound together by the waterman's lazo, leaving a bloody track behind them, and then heaped up in the public squares, where they are often allowed to lie for days, is truly one of the most painful and disgusting sights which Lima presents, and to which the bloody scenes of the bull-ring are comparatively nothing.—Translator.

[54] This earth or "huano," as the translator has been informed, is an article of commerce at the port of Ilo; whence it is conveyed to the neighbourhood of Arica, the Vale of Tambo, and Arequipa, and sold at so much per quintal. When rubbed between the fingers, it emits an insufferable stench. Tithes are paid upon this valuable manure, which are always put to one side in a heap, and, like the rest, carried away on asses. At Huacho, to the north of Lima, birds' ordure abounds, and is, we believe, used as manure. But, in general, the soil in Peru receives no top-dressing; though about Arequipa, in particular, the agricultural industry of the ancient Indians has always been followed by their successors, who, by means of the huano, compel the same piece of ground to yield several crops annually.

[55] Cuntur de Ccuncuni—to *smell ill*—so called because the condor emits an offensive smell. This name and that of "puma" were celebrated among the ancient Peruvians: they were used as appellatives or surnames in many illustrious families, whose descendants yet live, occupying the rank of Indian nobility or caciques. According to the meaning of these words, it appears that there were two orders of superior dignity in the empire of the Incas,—that of the Condor and Lion; and hence the origin of the surnames, Apucuntur, or Great Condor, as if we were to say, Great Eagle; Cunturpusac, or chief of eight Condors; Cuntur-canqui, Condor, by way of excellence, or Great Master of the Order; Colquipuma, or Lord of the Silver Lion. Cuntur-apachecta is the distinguishing epithet applied to the loftiest peaks of the Andes; denoting that these are sites among which only the Condor, of all the tenants of the air, can take up his abode.

[56] Here the translator would beg leave to remark, that the common carrion vulture, or gallinaza, is a tame and useful scavenger, very fond of taking up his station on spires, high walls, and house-tops: but, as for the bold and soaring condor, he never saw him frequent crowded cities, or sit on spires, as if this king of vultures had come in the spirit of pride imagined by Dr. Unanue.

[57] That this belief in the moromoro's strength and courage is founded on fact, is not very improbable; and that the condor was believed in ancient times, before sheep or lambs were known on the Andes, to carry off young infants, appears from the small drinking-cups which are sometimes dug from guacas, in which the stone is so cut out as to represent the condor carrying off an infant in its talons. The pieces of silver usually found in guacas are representations of natural objects.—Translator.

[58] See Letter of Iturre to Mr. Muños: zancudos, flies, and mosquitos are most troublesome in Andalusia.

N. B. The translator would not venture to decide the question, whether the cimex be more abundant in the metropolis of France or of Peru; but he considers it not unimportant to state, from his knowledge of the fact, that the only effectual means of destroying these insects in Lima, where they are certainly a great nuisance, is to brush over the bed with an infusion of the bruised seeds of the anona (of Lambayeque) in lime or lemon juice. For another set of tormentors, *fleas*, the natives on some occasions use traps, consisting merely of a piece of *bayeta* or baize, which is placed on the part where the enemy is felt to bite; and, as soon as the fleas get into it, they become so entangled in its meshes, that they are caught and executed at once,—for even the fairest hand can show them no mercy. It is curious to observe that, when one is affected with a paroxysm of ague, no fleas come near him: either the aguish blood or perspiration offends them.

The locust is one of the insects sometimes seen in multitudes on the aroma trees of the warm valleys, which they strip of every leaf in a very short time; just as the cauliflowers are devoured by caterpillars and swarms of butterflies of great beauty. The glow-worm often shines among the groves and avenues in a warm and dark night; and at Tarma, celebrated for the fine texture and beautiful tints of its *ponchos*, the cochineal insect is reared on beds of cacti, planted for the purpose, all round the town.—Translator.

[59] The Indians of North America called God the Great Man. See Jefferson's note on Virginia, page 56.

[60] See Garcilaso, t. i. page 313.

[61] It is evident, from the concluding query and remark of Dr. Unanue, that he suspected some speculators in the science of geology of no small share of credulity; and it also appears that he had not himself examined the bony fragments to which he alludes. Had these come to his hands, it is probable that he might have been able to ascertain such specific and distinct characters as should have served to satisfy him that the teeth in his possession were not only by report, but in fact, parts of those skeletons from among which they appear to have been picked up. We may believe that they were conveyed to Lima chiefly on account of their more portable size; while the other more unwieldy bones would have been considered too heavy for being removed so far, by persons who may not have known their scientific value to the geologist.—Translator.

[62] The periods of the great earthquakes of Peru are thus recorded by Dr. Hipolito Unanue.

	Arequipa.	Lima.	Quito.
In the year	1582	1586	1587
	1604	1630	1645
	1687	1687	1698
	1715	1746	1757
	1784	1806	1797

The same author also mentions the following epochs of volcanic explosions.

In Quito.
Cotopaxi, 1534, 1742, 1744.
Pichincha, 1539, 1566, 1577, 1660.
In Arequipa.
Quinistacas, 1600.

[63] Each "*topo*," that is, an extent of 5,000 square yards of this soil, is valued at 1,000 dollars; and every six weeks a harvest of "*salitre*," or the sub-carbonate of soda, is reaped by the owners.

[64] Vitor, here alluded to by Mr. Rivero, is one of the chief valleys in the vicinity of Arequipa. It extends from inland, in a north-west direction, to the large and well-watered valley of Quilca on the coast, and to the north of Islay: on the other hand, to the south-west, is the extensive, rich, and populous valley of Tambo.

Between the vales of Vitor and Tambo there is a sandy, hot desert, (intensely cold at midnight,) with a gradual ascent, through which passes the road from Islay to Arequipa; and on the scorched plain, great numbers of wearied and exhausted cattle are let loose to perish for want of water and pasture; so that along the way-side are to be seen the skeletons and hides of animals sun-dried, and in different grotesque attitudes. Travellers have remarked that along this arid plain, which extends about twenty leagues inland, there are numerous moveable sandhills of regular figure like a half-moon, with the convex side always looking to the sea.—Translator.

[65] Arequipa is above the level of the sea, according to Mr. Rivero's observations, 2704 yards; but he considers that Mr. Pentland has been more exact in estimating its altitude, with the barometer of Fortin, at 2697 yards.

[66] Some months ago, the attention of the public had been called to this subject by the Hon. P. Campbell Scarlett, in a work entitled South America and The Pacific; and, only a few weeks since, a prospectus of a new steam-packet company, under the denomination of Pacific Steam Navigation Company, has been circulated in London. Mr. William Wheelwright has, we believe, the merit of being the zealous projector of this very important undertaking, which now promises to be crowned with success.

[67] "*Huascar's chain of gold*."—See Garcilaso de la Vega. Huascar, in Quichua, signifies 'chain;' and that Inca was so called from an immense chain of gold which was made in his honour. If I remember well, Garcilaso tells us that it required eight hundred men to support the weight of it. It remains buried to this day in a lake not far from Cuzco.

[68] "*Let Indian pipe*."—The Indians of Alto Peru mourn the Incas in "tristes," which they play upon a kind of pipe. In the time of the Spaniards, at one time they were forbidden to tune these mournful airs, from political motives.

[69] "*And Indian maid*."—Many of the Indian women wear a dark drapery suspended from the left shoulder, and falling down to the mid-leg, as mourning for their Incas.

[70] Who is insensible to the sad wild note of the cuculi, the nightingale of this country?

[71] Let any one see the snow-capt mountains at the back of Lima, and that city of spires lying among the dark orange groves at their feet, at daybreak from Callao, and he will say the sight is worth a stave at least; and yet I wish I had never seen it.

[72] Buena Vista, the seat of John Thomas, Esq.

[73] The famous Temple of Pachacamac, whose mighty ruins form a beautiful object from Buena Vista. Pachacamac, like the Temple of Cholula on the plains of Mexico, is a sort of made mountain or vast terraced pyramid of earth. It would be difficult to produce any evidence more conclusive to the benignity of the climate than that which is exhibited on the interior walls of this temple, whereof the mud plaster, though exposed for centuries to the action of the atmosphere, remains to this day with its rude paintings of red and yellow ochre as inviolate and fresh as if it were the work of yesterday. By the bye, it may not be impertinent to mention, that, among these paintings, we find what is called the Grecian Scroll, which, if I am not mistaken, the Grecians borrowed from the Egyptians. This may serve to throw some light upon the origin of Pachacamac. Like that of Mexico,—nay, with still more emphatic gesture,—the gigantic architecture of Peru points to the Cyclopiian family, the founders of the Temple of Babel and of the Egyptian Pyramids. I believe (see Garcilaso) that the Temple of Pachacamac was standing when that part of the coast was conquered by the Incas, so that there is no knowing its age.

[74] Old Green's Nonpareil, where the Hearts of Oak meet.

[75] "*Like stars,*" &c.—The milky way, which, by the bye, is far grander in the southern than in the northern hemisphere, seems to have been formed by the mutual gravitation of myriads of stars. The dark spots which follow the course of that magnificent nebula, on each side of it, are probably the spaces which the stars have left vacant and lustreless.—See Herschel on Nebulæ.

THE END.

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